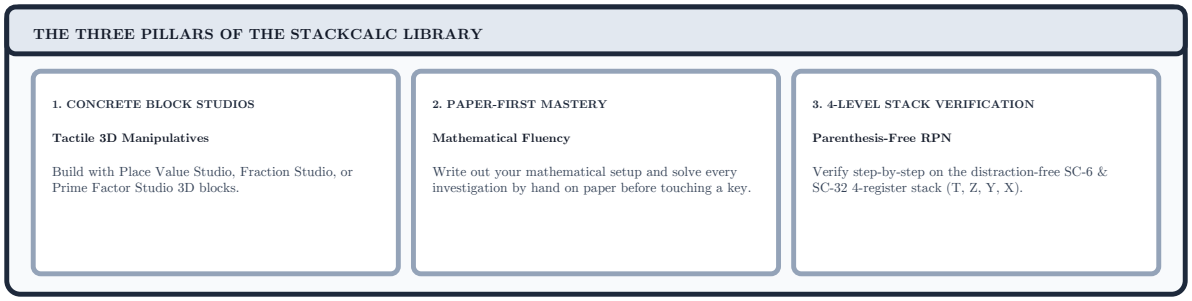
THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 4

HIGH SCHOOL & STEM LAB

Algebra, Trigonometry, Vectors, Logarithms & Digital Logic on the Stack (Grades 9–12 & AP)

A self-contained, rigorous Grades 9–12 & AP STEM student exploration and practice workbook for homeschooling families and classrooms. Every 11-page unit connects Prime Factor and Binary Place Value Block Studios and paper-and-pencil algebraic derivations with Core & Extension Concept Labs, Misconception Spotters, 4-Problem Lab Investigations, six leveled Worked-Example & Practice Worksheets (Practice A through F), a 100-Point Unit Mastery Check, an Applied STEM Lab Case, and 4 Cumulative Mastery Exams (840 active problems), verified step-by-step on the 32-button scientific StackCalc SC-32 calculator.



StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Solving Equations from the Inside Out (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Grade-Band Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Facilitation & Coaching Notes
- Book 2A — Early Elementary Math Lab: Grades K–2 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 2B — Upper Elementary Math Lab: Grades 3–5 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method SC-6 Student Workbook (10 Units • 1,000 Problems • 128 Pages)
- Book 4 — High School & STEM Lab: Grades 9–12 & AP SC-32 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 5 — Stack Speed & Fluency: Full-Year Speed Lessons, Playbooks & 180 Progressive Practice Tests (3,690 Problems • 128 Pages)
- Complete Online Answer Keys & Keystroke Verification Portal: https://stackcalc.io/curriculum/answers

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio: 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio: 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

Automated Unit-Test Verification & 36-Week Open-Education Pacing Architecture

- 100% Unit-Tested Problems: Every mathematical solution and RPN keystroke sequence across the curriculum is verified by automated Python and Swift RPNCore unit tests.
- Flexible Pacing: Compatible with Singapore Math, Saxon, RightStart, Beast Academy/AoPS, Eureka, Illustrative Math, Big Ideas, and Bridges.

COPYRIGHT, STATE STANDARDS ALIGNMENT & OPEN-EDUCATION LICENSE

HIGH SCHOOL & STEM LAB: Algebra, Trigonometry, Vectors, Logarithms & Digital Logic on the Stack (Grades 9–12 & AP)

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

Open-Education License (CC BY-NC-SA 4.0) & Perpetual Classroom Reproduction Permission:

Classroom teachers, math coaches, interventionists, homeschool families, and schools are expressly permitted to photocopy or print student lab worksheets, speed trials, reference cards, and parent welcome letters for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

Trademark Notice & Print Colophon:

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. All curriculum and textbook names are trademarks of their respective owners, none of whom endorse or sponsor this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

Table of Contents

Check off each 11-page unit as you complete the Labs, Practice A–F, Mastery Quiz, and STEM Case.

Section / Unit & Mathematical Topic	Grades & Standards	11-Page Architecture (80 Probs/Unit)	Pages / Done
Preface: Welcome to the Hands-On Math Lab!	Students & Families	Paper & Blocks First, Zero Crutch Rule	**p. iv [ ]**
RPN Stack Dynamics & Hardware Tour Bootcamp	SC-32 Orientation	CRA Framework, Stack Dynamics & Pre-Test	**pp. 1–4 [ ]**
Unit 01: Linear Equations & Slope Focus: Coordinate Slope Triangle Diagram & 4-Level Stack	**Grade 9 – 10 (Algebra 1)** CCSS.MATH.HSA.CED.A.2	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Jan Łukasiewicz (1924), Descartes's Fly & Highway Surveying	**pp. 5–15 [ ]**
Unit 02: Quadratic Factoring & Area Models Block Studio: Prime Factor Studio (Factor Pairs) & 2×2 Algebraic Area Grid	**Grade 9 – 10 (Algebra 1)** CCSS.MATH.HSA.SSE.B.3	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Al-Khwarizmi's Squares (820 CE) & Galileo's Area Scaling	**pp. 16–26 [ ]**
Unit 03: Quadratic Formula & [POLY] Solver Focus: Parabola Root Diagram & SC-32 [POLY] Solver	**Grade 9 – 11 (Algebra 1 & 2)** CCSS.MATH.HSA.REI.B.4.B	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Galileo's Parabolas, Viète's Roots & Kahan's [POLY] Guard	**pp. 27–37 [ ]**
Unit 04: Polynomial Operations & Division Focus: Nested Polynomial Pipeline Diagram & SC-32 Stack	**Grade 10 – 11 (Algebra 2)** CCSS.MATH.HSA.APR.A.1	**Core/Ext • Inv • Practice A–F • Quiz • STEM** The 1535 Milanese Cubic Duel, Horner (1819) & Romberg Calculus	**pp. 38–48 [ ]**
Unit 05: Exponential Growth & Decay Block Studio: Prime Factor Studio & SC-32 Scientific Stack ([y^x], [e^x])	**Grade 10 – 12 (Algebra 2 & Precalculus)** CCSS.MATH.HSF.LE.A.1	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Euler's Base e, Bernoulli's Compounding & Fibonacci's TVM	**pp. 49–59 [ ]**
Unit 06: Natural Logarithms & Doubling Time Focus: Prime Factor Studio (Logarithmic Height Towers) & SC-32 [LN]	**Grade 11 – 12 (Algebra 2 & Precalc)** CCSS.MATH.HSF.BF.B.5	**Core/Ext • Inv • Practice A–F • Quiz • STEM** John Napier's 20-Year Labor (1614) & Libby's C-14 Clock	**pp. 60–70 [ ]**
Unit 07: Trigonometric Functions [SIN, COS, TAN] Focus: Right-Triangle & Unit-Circle Diagram & SC-32 Trig Keys	**Grade 10 – 12 (Geometry & Precalculus)** CCSS.MATH.HSG.SRT.C.6	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Eratosthenes' Shadow (240 BCE) & The Unit Circle	**pp. 71–81 [ ]**
Unit 08: 2D Vectors & Polar Coordinates Focus: 2D Vector Right-Triangle Diagram & SC-32 [$\rightarrow\theta,r$]	**Grade 11 – 12 (Precalc & Physics)** CCSS.MATH.HSN.VM.A.1	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Steinmetz's AC Phasor Vectors & The Wind Triangle	**pp. 82–92 [ ]**
Unit 09: Combinatorics & Probability [nPr, nCr] Focus: Decision Slot Diagram, Statistical Plot & SC-32 [nPr]/[nCr]/[$\Sigma+$]	**Grade 11 – 12 & AP (Precalc & AP Stats)** CCSS.MATH.HSS.CP.B.9	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Pascal & Fermat (1654) and Gauss's Recovery of Ceres	**pp. 93–103 [ ]**
Unit 10: Digital Logic & Base Conversions Block Studio: Place Value Studio (Binary [0]/[1] Bit Sockets) & SC-32 [BASE]	**Grade 9 – 12 & AP (AP CSP & CS)** CCSS.MATH.HSN.RN.A.1	**Core/Ext • Inv • Practice A–F • Quiz • STEM** Leibniz's Binary (1703), Shannon (1937) & 32,768 Hz Quartz	**pp. 104–114 [ ]**

APPENDICES A–D: CERTIFICATE & TRACKER (PP. 115–116), CUMULATIVE EXAMS (PP. 117–120) & 3D BLOCK STUDIOS (PP. 121–124)

Complete verified answer keys & keystroke traces are hosted online at https://stackcalc.io/curriculum/answers#high-school!

Preface: Welcome to the Hands-On Math Lab!

Why we build with blocks and solve on paper first—and how each 11-page unit builds complete mastery.

---

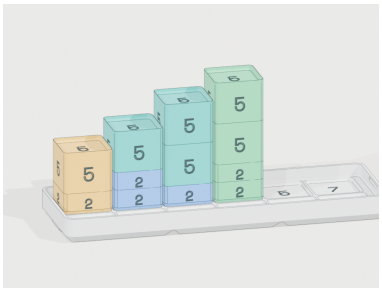
1. PAPER & BLOCKS FIRST: WHY THIS LAB MAKES YOU STRONGER AT MATH (NEVER A CRUTCH!)

Whether you are learning at your homeschool kitchen table or in a classroom, this workbook was built on one simple rule: a calculator should NEVER replace your brain! In every unit, you build Place Value, Fraction, and Prime Factor concepts with physical blocks and solve every problem by hand on paper first. Only after you write your paper prediction do you pick up your distraction-free StackCalc calculator to check your work.

Why does your calculator have an [ENTER] key instead of [=]? Think about how you calculate on paper: first you write your first number on the top line (Waiting Shelf, Line Y). Next you write your second number below it (Workbench, Line X). Only after both quantities are in place do you apply your operation (+, −, ×, ÷) to combine them—just like paper math!

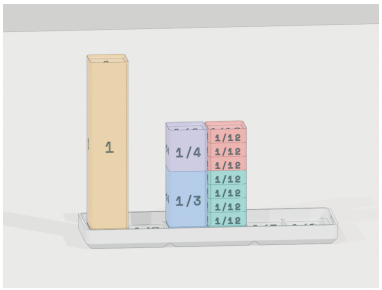
2. HANDS-ON TACTILE MODELS & VERIFICATION FOR HIGH SCHOOL & STEM LAB

1. Logarithmic Prime Towers



Exponential Powers & Log Heights $h(p) = 7.0 \cdot \log2(p)$ mm

2. Proportional Ratio Towers



Trigonometric & Rational Scaling Models

3. Scientific 4-Stack Mat



Zero Parentheses & Automatic Stack Lift

3. 5-DAY PACING & 3-TIER PROGRESSION ACROSS EVERY 11-PAGE UNIT

Every unit in High School & STEM Lab follows a structured 11-page progression (ideal for a 5-day rhythm or 3-tier differentiation):
- Day 1 (Pages 1–3): Core Concept Lab, Extension & Misconception Spotter, and 4-Problem Hands-On Lab Investigation.
- Day 2 (Pages 4–5 • Tier 1 Foundation): Practice A & B — Physical block pairing, step-by-step paper setups, and core 2-step fluency.
- Day 3 (Pages 6–7 • Tier 2 Grade-Level Core): Practice C & D — Word problems, mental math shortcuts, and error diagnostics.
- Day 4 (Pages 8–9 • Tier 3 Challenge): Practice E & F — Multi-step synthesis and fast-finisher championship problems.
- Day 5 (Pages 10–11): 100-Point Unit Mastery Check Assessment & Applied Real-World STEM Engineering Case Study.

• **Online Answer Key & Keystroke Portal:** https://stackcalc.io/curriculum/answers#high-school

Recommended Kit for Book 4 (Grades 9–12) & Try-Before-You-Buy Simulator (stackcalc.io)

• Core Kit for Book 4: StackCalc SC-32 Scientific Calculator + Prime Factor & Place Value Studios + Volume II: Thinking in the Stack.
• Waiting for your physical SC-32 to arrive? Practice on our free interactive web simulator or StackCalc 32 iOS/watchOS app at https://stackcalc.io!

Hands-On Starter: Touch the Math Before the Keys

Why tactile models make arithmetic, fractions, and algebra click before using a calculator.

1. THE CONCRETE-REPRESENTATIONAL-ABSTRACT (CRA) PATHWAY

Mathematics is not a collection of abstract symbols to memorize—it is the study of real patterns and quantities. In this workbook, you always follow the three-stage CRA learning sequence before touching a calculator key:

STAGE 1: CONCRETE

Build with 3D Block Studios

Snap Digit Blocks, 84 mm Fraction Towers, or Logarithmic Prime Blocks into the studio tray.

STAGE 2: REPRESENTATIONAL

Solve by Hand on Paper

Write your mathematical setup and paper prediction step-by-step.

STAGE 3: ABSTRACT

Verify on SC-32 Stack

Push quantities onto Registers Y and X to confirm your paper math matches!

2. YOUR 3D MANIPULATIVE BLOCK STUDIOS & PAPER-FIRST PROBLEM SOLVING

- Algebraic Area Grids & Prime Factor Studio: Factor constant c with Logarithmic Prime Blocks to find binomial dimensions (x + p)(x + q) on a 2×2 area grid before verifying roots with [POLY]!
- 2D Vector Polar Triangles & Slope Diagrams: Sketch vertical Rise Δy (Register Y) and horizontal Run Δx (Register X) to resolve polar magnitude r and angle θ.
- Logarithmic Block Scaling & Binary Trays: Prime Factor block heights $h(p) = 7.0 \cdot \log_2(p)$ mm mirror [LN] and [e^x], while Place Value [0]/[1] sockets model 8-bit binary logic.

TACTILE & VISUAL BRIDGE: Coefficients [a, b, c] → [POLY] Roots • Shelf Y (Δy) & Workbench X (Δx) → [→θ,r] Polar • Log Prime Towers → [LN] & [e^x]

Quadratic Roots & [POLY]

Quadratic Roots & [POLY]: $x^2 - 5x + 6 = 0$

Discriminant $\Delta = b^2 - 4ac = (-5)^2 - 4(1)(6)$

$\Delta = 25 - 24 = 1 > 0$ (2 Real Roots)

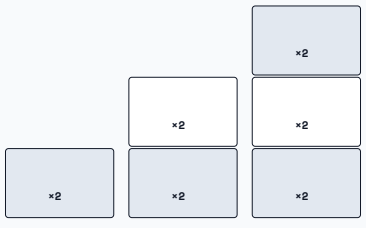
$x = (5 \pm \sqrt{1}) / 2 \rightarrow r_1 = 3,\ r_2 = 2$

Vieta Check: $r_1 + r_2 = 5,\ r_1 \cdot r_2 = 6$

`1 [ENTER] 5 [+/-] [ENTER] 6 [POLY] → 3, 2`

Logarithmic Prime Towers

Logarithmic Prime [2] Towers ($h = n \cdot \log 2$)



Height = Exponent n | $A = P \cdot e^{r \cdot t}$

Right-Triangle Trigonometry

Inverse Growth & Doubling Time: [LN] & [e^x]

Inverse Identity: $\ln(e^x) = x$ & $e^{\ln x} = x$

Continuous Doubling: $e^{r \cdot t} = 2 \rightarrow r \cdot t = \ln(2)$

Doubling Time: $t = \ln(2) \div r \approx 0.6931 \div r$

At $r = 0.07$ (7%): $t = 0.6931 / 0.07 = 9.90$ yr

`2 [LN] 0.07 [÷] → t = 9.90 yr (Rule of 70!)`

Paper Steps First — The Universal Lab Rule

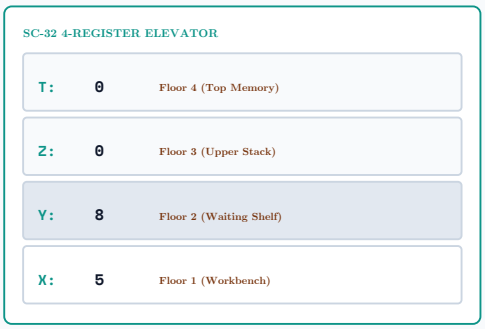
Never touch a calculator key until you have built with blocks or written your paper steps. The StackCalc instrument is your personal laboratory verification tool—not an answer generator!

SC-32 Calculator Dynamics: Calculating Without Parentheses

How the Waiting Shelf [Y] and Workbench [X] make calculations natural and transparent.

1. MEET THE FOUR-LEVEL STACK (T, Z, Y, X)

Ordinary calculators force you to type parentheses or rely on invisible hidden memory. The StackCalc calculator displays up to 4 lines on screen simultaneously (T, Z, Y, X):
- Register X (Floor 1): Active Workbench where you type your current number.
- Register Y (Floor 2): Waiting Shelf holding your first number after [ENTER].
- Registers Z & T (Floors 3 & 4): Upper memory levels that lift & drop automatically.



2. ESSENTIAL SC-32 STACK ELEVATOR DYNAMICS

1. Quadratic Solver ($x^2 - 5x + 6 = 0$):

[1] [ENTER] [5] [+/−] [ENTER] [6] [↷] [POLY]

Roots 2 and 3 appear directly in X and Y without formula drudgery.

2. 2D Vector Conversion (x=3, y=4 → r=5, θ=53.13°):

[4] [ENTER] [3] [↶] [▸θ,r]

Converts rectangular coordinates (y in Y, x in X) to magnitude and angle.

3. Deepest Inner Expression (100 ÷ (3 + 4×5)):

[4] [ENTER] [5] [×] [3] [+] [1] [0] [0] [x⇄y] [÷]

Solves inner 4×5+3=23 first, then swaps [$x \rightleftarrows y$] and divides into 100.

4. Continuous Growth & [LN] ($t = \ln(2) \div 0.05$):

[2] [LN] [0] [•] [0] [5] [÷]

Computes exact continuous doubling time (13.86 yrs) in 4 keystrokes.

Elevator Lift & Drop Rule • Try It Live on Your SC-32 or Web/iOS Simulator

Every binary operation (+, −, ×, ÷) combines Waiting Shelf Y and Workbench X into X while upper registers drop automatically. Key the 4 sequences above on your physical SC-32 calculator (or the free interactive simulator & StackCalc 32 iOS/watchOS app at stackcalc.io) before turning to Page 3!

Hardware Tour: Meet Your SC-32 Instrument

Understanding keycaps, shift layers, LCD annunciators, and error recovery.

1. THE SC-32 KEYBOARD ARCHITECTURE & LAYERS

The StackCalc SC-32 features an authentic 32-button scientific and programming keyboard layout:
- Primary Layer: Direct one-touch keys for numbers, arithmetic, trigonometry ([SIN], [COS], [TAN]), and exponents ([y^x], [$\sqrt{x}$]).
- Gold [f] and Blue [g] Shift Keys: Access secondary scientific functions printed above each mechanical keycap.
- Dedicated Solvers: [POLY] extracts quadratic roots from standard-form coefficients [a, b, c]; [$\rightarrow\theta,r$] converts (x, y) to Polar (r, θ).
- Programmer Mode: [BASE] toggles HEX, DEC, OCT, and BIN to verify 8-bit binary [0]/[1] registers on the Place Value Tray.

ALGEBRAIC & KEYPAD BRIDGE: Coefficients [a, b, c] → [POLY] Roots • Shelf Y (Δy) & Workbench X (Δx) → [$\rightarrow\theta,r$] • Binary [0]/[1] Tray → [BASE]

Scientific & Logs



Dedicated Solvers



Stack & Base



2. ESSENTIAL HARDWARE RECOVERY & STACK MANAGEMENT RULES

- Stack Swap [$x\rightleftarrows y$]: Made a mistake in operand order (for subtraction or division)? Press [$x\rightleftarrows y$] to swap Workbench X and Shelf Y!
- Stack Drop [DROP]: Discards the number in Register X; Register Y drops down to take its place.
- Stack Duplication: Pressing [ENTER] when no new number is typed copies Register X into Register Y (great for doubling and squaring!).
- Division by Zero Protection: Dividing by zero displays 'Error: Div by 0' without clearing your upper stack memory.

2-SECOND STACK AUDIT: Always verify Waiting Shelf Y = First Quantity (Minuend / Dividend) and Workbench X = Second Quantity!

SWAP ORDER [$x\rightleftarrows y$]



Flips Register Y and Register X before pressing [−] or [÷].

ROLL DOWN [DROP]



Removes an unwanted entry in X and pulls Shelf Y down to X.

DUPLICATE [ENTER]



Copies X into Y so [+] doubles or [×] squares instantly.

Simulator Quick-Start

You can practice all hardware key sequences right now in your web browser at stackcalc.io before proceeding to the diagnostic test.

Diagnostic Baseline: Test Your Readiness

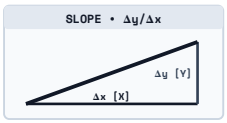
Solve these four baseline problems on paper first, then verify on your calculator.

DIAGNOSTIC BASELINE WARM-UP — SOLVE ON PAPER FIRST, THEN VERIFY ON THE STACK

Complete these 4 diagnostic problems on paper first in the dot-grid workspace, then run the verification keycap sequence on your stack.

Diagnostic 1: Multi-Step Rational Evaluation

Evaluate $(24 - 6) \div 3$ on paper first, then verify with zero parentheses:

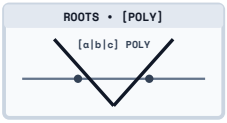


My Paper Prediction: _____ [ ] Verified on SC-32

2 4 ENTER 6 - 3 ÷

Diagnostic 2: Quadratic Discriminant

Evaluate $b^2 - 4ac$ on paper for $a = 1$, $b = 6$, $c = 8$:

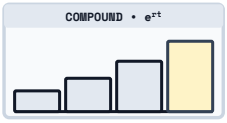


My Paper Prediction: _____ [ ] Verified on SC-32

6 ↶ x² 4 ENTER 1 × 8 × -

Diagnostic 3: Exponential Compounding

Evaluate $100 \times (1.05)^2$ on paper and verify on the stack:

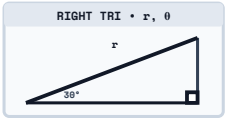


My Paper Prediction: _____ [ ] Verified on SC-32

1 0 0 ENTER 1 • 0 5 ENTER 2 y^x ×

Diagnostic 4: Right-Triangle Trigonometry

Evaluate $10 \times \sin(30°)$ on paper using special triangles (Degree mode):



My Paper Prediction: _____ [ ] Verified on SC-32

1 0 ENTER 3 0 SIN ×

Linear Equations, Rate of Change & Slope Pipelines

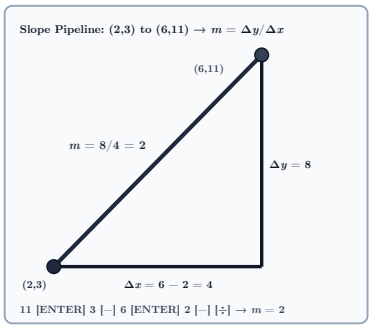
Mathematical Focus: Coordinate Slope Triangle Diagram & 4-Level Stack • Standards: CCSS.MATH.HSA.CED.A.2, CCSS.MATH.HSF.IF.B.6

---

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Evaluating Slope on the Stack: Points (2, 3) and (6, 11)):

Stack registers lift Δy to Shelf Y automatically while you compute Δx in Register X, enabling zero-parenthesis division.

Companion Book (Vol. II: Thinking in the Stack — Prologue, Ch. 2: Inside-Out Thinking & Ch. 12: Descartes's Coordinate Bridge) • Jan Łukasiewicz (1924), Descartes's Fly & Highway Surveying: Polish logician Jan Łukasiewicz proved postfix order eliminates every parenthesis (Prologue & Ch. 2)—and Descartes's coordinate bridge (Ch. 12) unpeels $m = (y_2 - y_1) \div (x_2 - x_1)$ across Z, Y, X!



2. KEY MATHEMATICAL PRINCIPLES: EVALUATING SLOPE ON THE 4-LEVEL STACK

- Slope is the constant rate of change: $m = \Delta y \div \Delta x = (y_2 - y_1) \div (x_2 - x_1)$.
- Inside-Out evaluation: evaluate numerator $(y_2 - y_1)$ first into Register Y, then denominator $(x_2 - x_1)$ into Register X, then press [÷].
- Point-slope form: $y - y_1 = m(x - x_1)$. Evaluate m$(x - x_1)$ on the workbench, then add y_1.
- Horizontal lines have slope $m = 0$; vertical lines have undefined slope (division by zero).

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: Plot Point A (2, 3) and Point B (6, 11) on a coordinate slope triangle diagram, labeling vertical rise Δy and horizontal run Δx. Question: Rise $\Delta y = 11 - 3 = 8$. Run $\Delta x = 6 - 2 = 4$. Slope $m = 8 \div 4 = 2$. (Prediction: 2)

Verification Keystrokes: `1` `1` `ENTER` `3` `-` `6` `ENTER` `2` `-` `÷`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`1` `1`	11	y_2 (11) enters Register X
`ENTER`	11	11 lifts to Register Y
`3`	3	y_1 (3) enters Register X
`-`	8	$\Delta y = 11 - 3 = 8$! Stored in Register X
`6`	6	x_2 (6) enters; 8 lifts to Register Y
`ENTER`	6	6 lifts to Register Y; 8 lifts to Register Z
`2`	2	x_1 (2) enters Register X
`-`	4	$\Delta x = 6 - 2 = 4$! (8 waits in Register Y)
`÷`	2	$\Delta y \div \Delta x = 8 \div 4 = 2$! Slope $m = 2$

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Find the slope passing through (1, 4) and (5, 16) on paper, then check on SC-32.

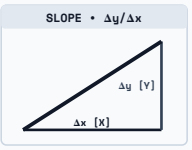
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Evaluating Point-Slope Equations: $y = m(x - x_1) + y_1$

Deepening Concepts • Standards: CCSS.MATH.HSA.CED.A.2

1. EXTENSION CONCEPT & SETUP (Evaluating Point-Slope Equations: $y = m(x - x_1) + y_1$):

Evaluating inside-out mirrors order of operations directly without requiring algebraic rearrangements.

Extension Setup: Sketch the slope triangle starting from reference point ($x_1 = 3$, $y_1 = 5$) with slope $m = 2$ to target input $x = 7$. • Question: Evaluate inner difference first: (7 − 3) = 4. Multiply by slope: 4 × 2 = 8. Add y_1: 8 + 5 = 13. (Prediction: 13)

2. EXTENSION STACK SEQUENCE (SC-32): [7] [ENTER] [3] [-] [2] [×] [5] [+]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[7]	7	x (7) enters Register X
[ENTER]	7	7 lifts to Register Y
[3]	3	x_1 (3) enters Register X
[-]	4	$x - x_1 = 7 - 3 = 4$
[2]	2	m (2) enters Register X; 4 in Register Y
[×]	8	m$(x - x_1) = 4 \times 2 = 8$
[5]	5	y_1 (5) enters Register X; 8 in Register Y
[+]	13	$y = 8 + 5 = 13$! Linear evaluation verified

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1 — Computing Run over Rise Instead of Rise over Run: Student calculates $\Delta x \div \Delta y = 4 \div 8 = 0.5$.

Mathematical & Stack Remediation: Emphasize 'Rise Y over Run X'—Δy is vertical and sits in Waiting Shelf Y first!

Misconception 2 — Subtracting Coordinates in Inconsistent Order: Computing $(y_2 - y_1)$ in numerator but $(x_1 - x_2)$ in denominator, flipping the sign.

Mathematical & Stack Remediation: Always enter Point 2 into Waiting Shelf Y and Point 1 into Workbench X.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [y_2] [ENTER] [y_1] [-] [x_2] [ENTER] [x_1] [-] [÷] = Slope m

Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Given line with $m = 3$ through (2, 4), find y when $x = 8$.

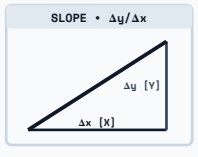
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



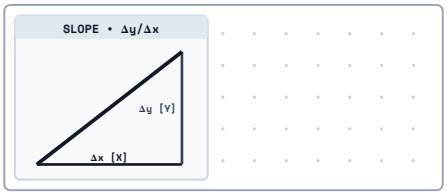
Unit 01 Lab: Linear Equations, Rate of Change & Slope Pipelines

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch your number line or equation balance in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Find slope between (3, 5) and (7, 17): m =

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

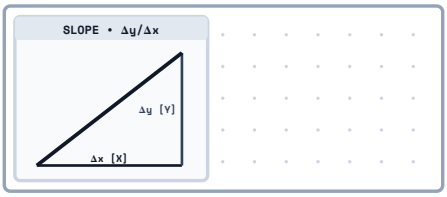
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find slope between (-2, 8) and (4, -4): m =

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

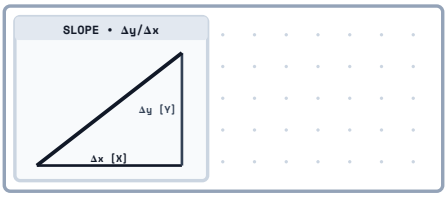
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Line has $m = 4$ through (1, 3). Find y when $x = 6$: y =

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

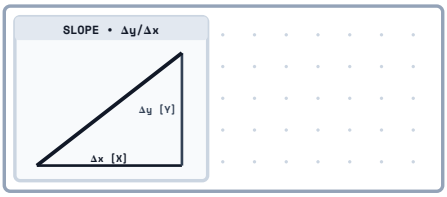
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Find the x-intercept of $3x + 6y = 24$: x =

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 01 Practice A — Slope Between Two Coordinate Points: $m = (y_2 - y_1) \div (x_2 - x_1)$

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find slope through (2, 3) and (6, 11).

$\Delta y = 11-3 = 8$; $\Delta x = 6-2 = 4$; $m = 8 \div 4 = 2$. Key: [11] [ENTER] [3] [-] [6] [ENTER] [2] [÷] [÷].

Verified Answer: 2

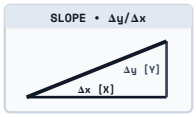
EXAMPLE 2 Find slope through (-2, 8) and (4, -4).

$\Delta y = -4-8 = -12$; $\Delta x = 4-(-2) = 6$; $m = -2$.

Verified Answer: -2

1 Tier 1 • Builder

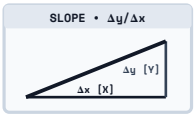
Find slope between (1, 2) and (5, 10): $m =$



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

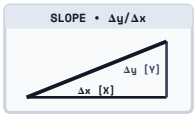
Find slope between (2, 5) and (6, 17): $m =$



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

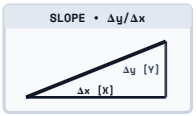
Find slope between (3, 4) and (8, 24): $m =$



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

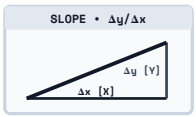
Find slope between (1, 15) and (6, 5): $m =$



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

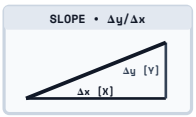
Find slope between (-3, -2) and (2, 13): $m =$



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

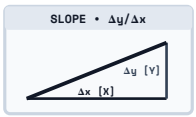
Find slope between (-4, 9) and (2, -3): $m =$



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

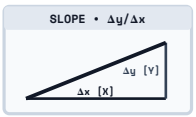
Find slope between (0, -5) and (4, 15): $m =$



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

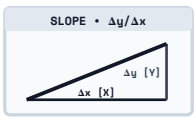
Find slope between (2, 7) and (10, 11): $m =$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

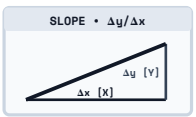
Find slope between (-5, 6) and (3, 6): $m =$



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Find slope between (4, -3) and (8, -15): $m =$



Paper Step: _____ | | SC-32 Ans: _____

Unit 01 Practice B — Slope-Intercept & Point-Slope Evaluations on the Stack

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate $y = 3(x - 2) + 5$ when $x = 7$.

$7 - 2 = 5$; $5 \times 3 = 15$; $15 + 5 = 20$. Key: [7] [ENTER] [2] [-] [3] [×] [5] [+].

Verified Answer: 20

EXAMPLE 2 Evaluate $f(x) = -4x + 18$ when $x = 3$.

$-4 \times 3 + 18 = 6$. Key: [-4] [ENTER] [3] [×] [18] [+].

Verified Answer: 6

1 Tier 1 • Builder

Evaluate slope-intercept $y = 4x - 7$ at $x = 5$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

Evaluate slope-intercept $y = -3x + 20$ at $x = 4$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

Evaluate slope-intercept $y = 2.5x + 6$ at $x = 8$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

Evaluate point-slope $y = 2(x - 3) + 5$ at $x = 9$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

Evaluate point-slope $y = 4(x - 1) + 3$ at $x = 6$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

Evaluate point-slope $y = -2(x - 4) + 10$ at $x = 7$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

Evaluate point-slope $y = 5(x + 2) - 8$ at $x = 3$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

Evaluate point-slope $y = -3(x + 1) + 12$ at $x = 4$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

Evaluate point-slope $y = 1.5(x - 2) + 4$ at $x = 10$: $y =$

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

Find y-intercept b given $y = 19$, $m = 3$, $x = 5$ ($b = y - mx$):

SLOPE • Δy/Δx

Δy [Y]

Δx [X]

Paper Step: _____ [ ] SC-32 Ans: _____

Unit 01 Practice C — Highway Grade, Velocity & Linear Rate Word Problems

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 A highway rises 240 ft over a 4,000 ft horizontal run. Find percent grade.

$(240 \div 4000) \times 100 = 6\%$.

Verified Answer: 6%

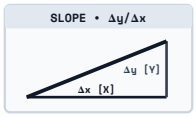
EXAMPLE 2 A train is at mile 45 at $t = 1$ hr and mile 195 at $t = 3.5$ hr. Average speed?

$(195 - 45) \div (3.5 - 1) = 60$ mph.

Verified Answer: 60 mph

1 Tier 2 • Detective

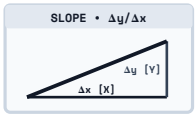
A mountain pass rises 350 m over a 5,000 m horizontal run. Find the percent grade:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

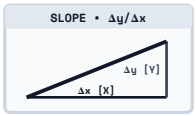
A wheelchair ramp rises 15 in over a 180 in run. Find the slope ratio:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

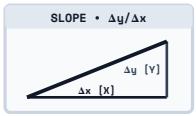
A car travels from mile 30 at $t = 1$ hr to mile 210 at $t = 4$ hr. Speed (mph):



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

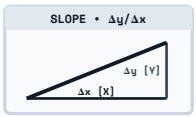
A tank holds 500 L at $t = 0$ and 320 L at $t = 6$ min. Drain rate (L/min):



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

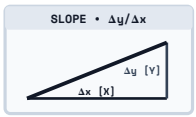
An elevator descends from 420 ft to 60 ft in 12 seconds. Velocity (ft/s):



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

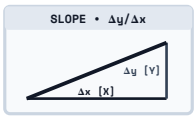
A plumber charges a \$65 call fee plus \$45/hr for 4 hours. Find the total cost:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

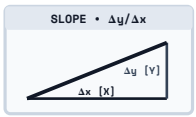
Temperature at sea level is 20°C and drops 6.5°C per km. Temp at 4 km:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

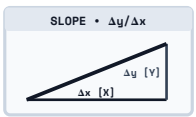
A cloud server costs \$120/mo base + \$0.08 per GB. Cost for 1,500 GB:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

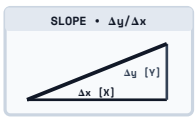
A battery charges from 15% to 95% in 40 minutes. Rate (% per min):



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: Sam computed $\Delta x \div \Delta y = (6-2) \div (14-2) = 0.33$. Find true slope $\Delta y \div \Delta x$:



Paper Step: _____ | | SC-32 Ans: _____

Unit 01 Practice D — Intercepts of Standard Form ($Ax + By = C$) & Linear Equations

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find x-intercept of $3x + 6y = 24$.

Set $y = 0 \rightarrow 3x = 24 \rightarrow x = 24 \div 3 = 8$.

Verified Answer: 8

EXAMPLE 2 Solve $4x - 7 = 21$ on the stack.

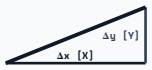
$(21 + 7) \div 4 = 28 \div 4 = 7$. Key: [21] [ENTER] [7] [+] [4] [÷].

Verified Answer: 7

1 Tier 2 • Detective

Find x-intercept of $4x + 5y = 40$: $x =$

SLOPE • Δy/Δx

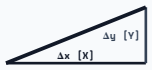


Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 2 • Detective

Find y-intercept of $4x + 5y = 40$: $y =$

SLOPE • Δy/Δx

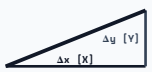


Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 2 • Detective

Find x-intercept of $6x - 8y = 48$: $x =$

SLOPE • Δy/Δx

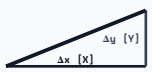


Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 2 • Detective

Find y-intercept of $6x - 8y = 48$: $y =$

SLOPE • Δy/Δx

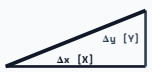


Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 2 • Detective

Solve linear equation $3x + 9 = 36$: $x =$

SLOPE • Δy/Δx

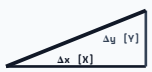


Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 2 • Detective

Solve linear equation $5x - 12 = 43$: $x =$

SLOPE • Δy/Δx

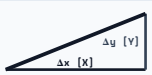


Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 2 • Detective

Solve linear equation $7x + 15 = -20$: $x =$

SLOPE • Δy/Δx

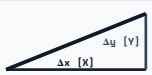


Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 2 • Detective

Solve linear equation $-4x + 18 = 2$: $x =$

SLOPE • Δy/Δx

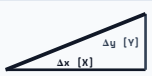


Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 2 • Detective

Solve variables on both sides $6x - 4 = 2x + 20$: $x =$

SLOPE • Δy/Δx

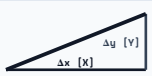


Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 2 • Detective

Solve variables on both sides $9x + 5 = 4x + 40$: $x =$

SLOPE • Δy/Δx



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 01 Practice E — Parallel/Perpendicular Slopes, Midpoints & Collinearity

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 Line L has slope $m = 4$. Find perpendicular slope m_{perp}.

$m_{perp} = -1 \div m = -1 \div 4 = -0.25$.

Verified Answer: -0.25

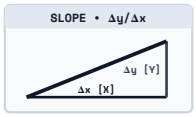
EXAMPLE 2 Find x-midpoint of (-3, 4) and (9, 10).

$(-3 + 9) \div 2 = 3$.

Verified Answer: 3

1 Tier 3 • Challenge

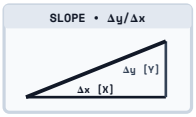
Find the perpendicular slope to $m = 2$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

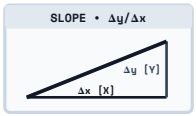
Find the perpendicular slope to $m = -5$:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

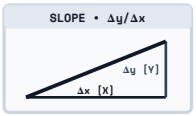
Find the perpendicular slope to $m = 0.4$:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

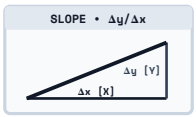
Find x-coordinate of midpoint between (2, 5) and (14, 9):



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

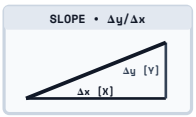
Find y-coordinate of midpoint between (2, 5) and (14, 9):



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

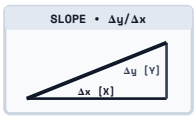
Find x-coordinate of midpoint between (-6, 3) and (10, -7):



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

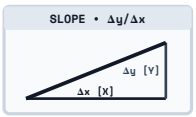
Find y-coordinate of midpoint between (-6, 3) and (10, -7):



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

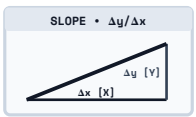
Given slope $m = 3$ through (2, 4) and (6, y2), find y2:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

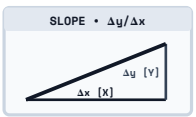
Given slope $m = -2$ through (1, 10) and (5, y2), find y2:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Given slope $m = 4$ through (3, 5) and ($x2$, 25), find x2:



Paper Step: _____ | | SC-32 Ans: _____

Unit 01 Practice F — Linear Intersections, Break-Even Analysis & Challenge

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find intersection x of $y = 5x + 2$ and $y = 2x + 14$.

$x = (14 - 2) \div (5 - 2) = 12 \div 3 = 4$.

Verified Answer: 4

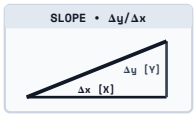
EXAMPLE 2 Company A charges \$100 + \$15/hr; B charges \$40 + \$30/hr. Break-even hours?

$(100 - 40) \div (30 - 15) = 4$ hrs.

Verified Answer: 4

1 Tier 3 • Challenge

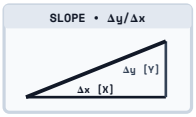
Find intersection x of $y = 4x + 3$ and $y = 2x + 15$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

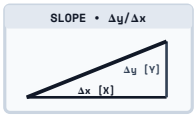
Find intersection x of $y = 7x - 5$ and $y = 3x + 19$:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

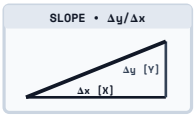
Find intersection x of $y = 3x + 12$ and $y = -2x + 32$:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

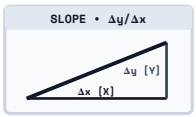
In Problem 3 (where $x = 4$), find the intersection y-coordinate:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

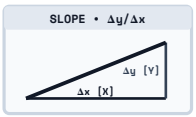
Break-even units: Fixed cost \$1,200, unit cost \$8, selling price \$20:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

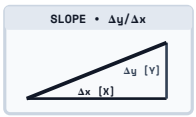
Break-even units: Fixed cost \$2,400, unit cost \$15, selling price \$35:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

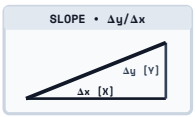
Linear interpolation: at x=0, y=10; at x=10, y=60. Find y at x=4:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

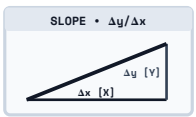
Find the area of the triangle bounded by x=0, y=0, and $3x + 4y = 24$:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

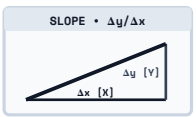
Find the average rate of change of $f(x) = x^2$ from $x = 2$ to $x = 6$:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Break-Even Rate Challenge: Two automated carts start at $y = 5$ m and $y = 25$ m with speeds 8 m/s and 3 m/s. At what time t do they meet?



Paper Step: _____ | | SC-32 Ans: _____

Unit 01 Mastery Check: Linear Equations & Slope

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Find slope between (2, 5) and (6, 21): m =

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Find slope between (-3, 10) and (5, -6): m =

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Evaluate point-slope $y = 3(x - 4) + 7$ at $x = 10$: y =

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Find x-intercept of $5x + 4y = 60$ (set $y = 0$): x =

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Solve $6x - 14 = 34$ on the stack: x =

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Find the perpendicular slope to $m = 4$: m_{perp} =

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

A road rises 180 m over a 3,000 m horizontal run. Find percent grade:

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

Find intersection x of $y = 5x + 4$ and $y = 2x + 19$: x =

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Given slope $m = 4$ through (2, 3) and (7, y2), find y2 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: A student found slope between (1, 3) and (5, 15) as (5−1)÷(15−3) = 0.33. Find true slope:

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 01 STEM: Civil Highway Grade & Mountain Pass Surveying

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: CIVIL HIGHWAY GRADE & MOUNTAIN PASS SURVEYING

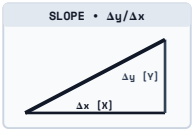
Companion to Vol. II, Ch. 2 & Ch. 12: Civil engineers survey elevation changes Δy (Register Y) over horizontal runs Δx (Register X) across mountain passes.

Companion Book (Vol. II: Thinking in the Stack — Prologue, Ch. 2: Inside-Out Thinking & Ch. 12: Descartes's Coordinate Bridge) • Jan Łukasiewicz (1924), Descartes's Fly & Highway Surveying: Polish logician Jan Łukasiewicz proved postfix order eliminates every parenthesis (Prologue & Ch. 2)—and Descartes's coordinate bridge (Ch. 12) unpeels $m = (y_2 - y_1) \div (x_2 - x_1)$ across Z, Y, X!

Mathematical Focus: Coordinate Slope Triangle Diagram & 4-Level Stack • Stack Shortcut: [y_2] [ENTER] [y_1] [-] [x_2] [ENTER] [x_1] [-] [÷] = Slope m

CASE INVESTIGATION 1 STEM Modeling

West Ascent Ramp: A survey team measures milepost (2, 3) to summit (6, 11). Find the highway slope $m = \Delta y \div \Delta x$:



1. Formula / Stack Plan: _____

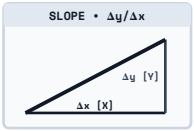
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

East Descent Grade: From ridge (-4, 10) down to valley (2, -8), compute the descent slope $m = \Delta y \div \Delta x$:



1. Formula / Stack Plan: _____

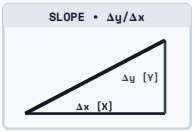
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Viaduct Pier Elevation: Using point-slope profile $y = 3(x - 2) + 5$, find the pier height y at station $x = 7$:



1. Formula / Stack Plan: _____

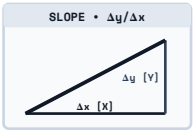
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student calculates $\Delta x \div \Delta y = 4 \div 8 = 0.5$. Correct the mistake.



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

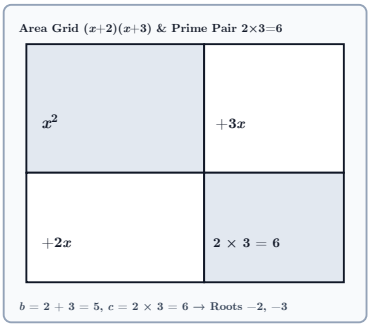
Algebraic Factoring, Area Models & Quadratic Trinomials

Hands-On Block Studio: Prime Factor Studio (Factor Pairs) & 2×2 Algebraic Area Grid • Standards: CCSS.MATH.HSA.SSE.B.3, CCSS.MATH.HSA.REI.B.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks):

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$).

Companion Book (Vol. II: Thinking in the Stack — Ch. 6: Galileo's Square-Cube Area Law & Ch. 7: Viète's Root Identities) • Al-Khwarizmi's Squares (820 CE) & Galileo's Area Scaling: In Baghdad's House of Wisdom, Al-Khwarizmi factored quadratics with geometric squares and rectangles (Ch. 7), while Galileo's Square-Cube Law (Ch. 6) explains why area scales with x^2!



2. HANDS-ON BLOCKS & CONCEPT RULES: HOW PRIME FACTOR BLOCKS & AREA GRIDS MODEL POLYNOMIAL FACTORING

- Quadratic trinomials $x^2 + bx + c$ form a rectangular area with width $(x + p)$ and height $(x + q)$.
- Factoring rule: decompose constant c with Prime Factor Blocks to find p and q where $p \times q = c$ and $p + q = b$.
- 2×2 Area Grid Check: sketch the 4 sub-rectangles x^2, px, qx, and pq to confirm the linear middle term $(p + q)x = bx$.
- On the SC-32: Evaluate (x + p)(x + q) for test values using stack multiplication.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Block Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$. Question: Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots $x = -2$ and $x = -3$.

Verification Keystrokes: 1 ENTER 5 ENTER 6 ↷ POLY

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
1	1	Coefficient $a = 1$ entered into X
ENTER	1	1 lifted to Y
5	5	Coefficient $b = 5$ entered into X
ENTER	5	b in Y, a in Z
6	6	Coefficient $c = 6$ entered into X
↷ POLY	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: $x + 2$), Root 2 = -3 in X (factor: $x + 3$)!

STACK SHORTCUT & BLOCK STUDIO TIP (SC-32):

[x] [ENTER] [2] [+] [x] [ENTER] [3] [+] [×] (Evaluates Trinomial) • Quadratic trinomials $x^2 + bx + c$ form a rectangular area with width $(x + p)$ and height $(x + q)$.

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

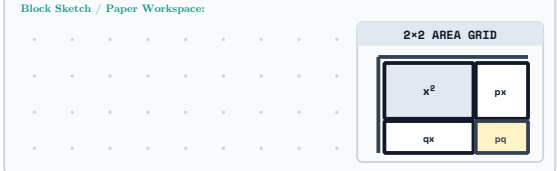
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



Extension: Signed Factor Pairs: Factoring Trinomials with Negative Terms

Deepening Concepts • Standards: CCSS.MATH.HSA.REI.B.4

1. EXTENSION CONCEPT & SETUP (Signed Factor Pairs: Factoring Trinomials with Negative Terms):

When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. You can check both roots on the calculator after factoring on paper.

Hands-On Extension Setup: Factor |c| = 6 into Prime Factor blocks [2] and [3] and sketch a 2×2 algebraic area grid for $x^2 - x - 6$. • Question: Assign signs to the factor pair (2, -3) so their product is -6 and their sum is 2 + (-3) = -1: factors are $(x - 3)(x + 2)$.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [ENTER] [1] [+/−] [ENTER] [6] [+/−] [↷] [POLY]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1]	1	Coefficient $a = 1$
[ENTER]	1	1 lifted to Y
[1]	1	Enter magnitude of b
[+/−]	-1	Coefficient $b = -1$ in X
[ENTER]	-1	b in Y, a in Z
[6]	6	Enter magnitude of c
[+/−]	-6	Coefficient $c = -6$ in X
[↷] [POLY]	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: $x - 3$), Root 2 = -2 in X (factor: $x + 2$)!

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms ($2x$ and $3x$).

Hands-On Block Studio Remediation: Sketch a 2×2 algebraic area grid for $(x + 2)(x + 3)$ and factor 6 into Prime Factor blocks [2] and [3] so students see that $b = 2 + 3 = 5$ and $c = 2 \times 3 = 6$.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block Studio Remediation: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [x] [ENTER] [2] [+] [x] [ENTER] [3] [+] [×] (Evaluates Trinomial)

Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Factor $x^2 - 5x + 6$ into two binomials by finding two negative factors of +6 whose sum is -5, and state both roots.

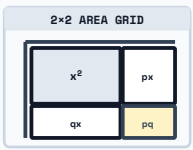
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



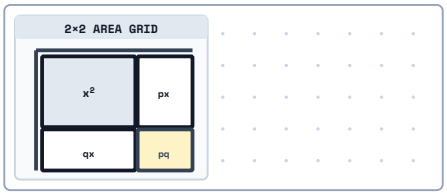
Unit 02 Lab: Algebraic Factoring, Area Models & Quadratic Trinomials

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch your number line or equation balance in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

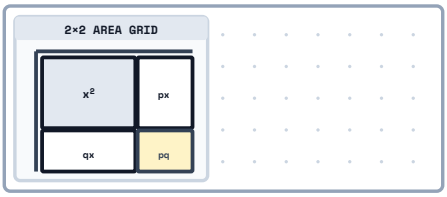
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Factor x^2 - 4 (difference of two squares, where the linear coefficient $b = 0$).

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

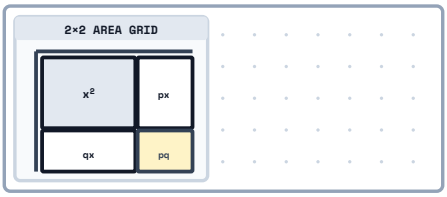
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

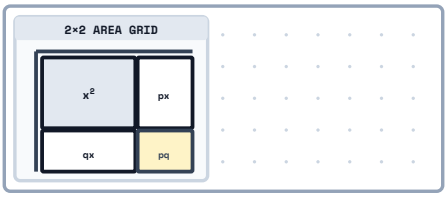
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Factor x^2 - $5x + 6$ using signed factor pairs and state both roots.

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did the Prime Factor or Binary Place Value blocks confirm the structure on the SC-32 stack?

Unit 02 Practice A — 2×2 Area Models, Binomial Expansion & Base-10 Polynomials

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Place (x + 3) along the height and (x + 4) along the width of a 2×2 area grid.**

Interior holds 1 x^2 square, (3+4) = 7 x-rectangles, and 3×4 = 12 unit squares → $x^2 + 7x + 12$.

Verified Answer: $x^2 + 7x + 12$

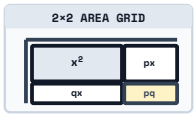
EXAMPLE 2 **Verify $(x + 3)(x + 4) = x^2 + 7x + 12$ at base-10 $x = 10$ on the SC-32.**

At $x = 10$: 13 × 14 = 182, and 100 + 70 + 12 = 182!

Verified Answer: 182

1 Tier 1 • Builder

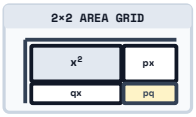
Area Grid Expansion: Find the constant corner product $p \cdot q$ in $(x + 3)(x + 6) = x^2 + 9x + c$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

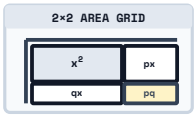
Area Grid Expansion: Find the linear x-term sum (p + q) in (x + 4)(x + 7) = x^2 + bx + 28:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

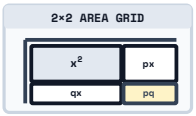
Evaluate the area $(x + 2)(x + 4)$ when $x = 3$:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

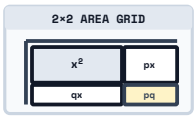
Evaluate $x^2 + 5x + 6$ when $x = 3$:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

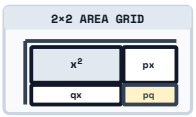
Evaluate x^2 - 4x + 7 when $x = 5$:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

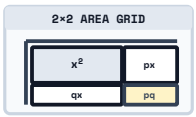
Combine quadratic like terms: $(3x^2 + 5x) + (2x^2 + 4x)$ at $x = 1$ (sum of coefficients):



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

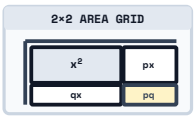
Expand using the Distributive Property: $3x(2x + 5)$ =



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

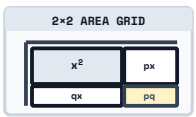
Factor out the GCF from the area: $12x^2 + 18x$ =



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

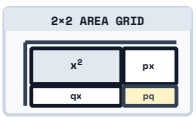
Expand the difference-of-squares area grid: $(x$ - 5$)(x + 5)$ at $x = 10$:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Base-10 Polynomial Check: Evaluate $(x + 2)(x + 3)$ when $x = 10$:



Paper Step: _____ | | SC-32 Ans: _____

Unit 02 Practice B — Quadratic Factoring ($x^2 + bx + c$) & Roots ([POLY])

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Factor $x^2 + 7x + 12$ and find its roots.

Find p, q with $p \cdot q = 12$ and $p + q = 7 \rightarrow 3$ and $4 \rightarrow (x + 3)(x + 4)$; roots $x = -3, -4$.

Verified Answer: (x+3)(x+4); -3, -4

EXAMPLE 2 Solve $x^2 - 5x + 6 = 0$.

$(-2)(-3) = 6$ and $(-2)+(-3) = -5 \rightarrow (x - 2)(x - 3) = 0 \rightarrow x = 2, 3$.

Verified Answer: $x = 2, 3$

1 Tier 1 • Builder

Expand the binomial product: $(x + 2)(x + 5) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Expand the binomial product: $(x - 3)(x + 4) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Expand the difference of squares: $(x - 6)(x + 6) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Factor completely: $x^2 + 6x + 8 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Factor completely: $x^2 + 9x + 20 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Factor completely: $x^2 - 7x + 10 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Factor completely: $x^2 - 2x - 15 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Find both roots of $x^2 - 6x + 8 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Find both roots of $x^2 + 2x - 8 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Vieta's Check: What is the sum and product of the roots of $x^2 - 9x + 14 = 0$?

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

Unit 02 Practice C — Algebraic Modeling, Projectile Roots & Error Analysis

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 A rectangle has length (x + 4) and width (x + 2) with area 35. Find x (>0).

$(x+4)(x+2) = 35 \rightarrow x^2 + 6x - 27 = 0 \rightarrow (x+9)(x-3) = 0 \rightarrow x = 3.$

Verified Answer: $x = 3$

EXAMPLE 2 Spot the Mistake: Sam expanded $(x + 3)^2$ as $x^2 + 9$.

$(x + 3)(x + 3)$ has two $3x$ cross-terms: $x^2 + 3x + 3x + 9 = x^2 + 6x + 9$!

Verified Answer: $x^2 + 6x + 9$

1 Tier 2 • Detective

When $x = 10$, the polynomial $x^2 + 5x + 6$ equals what 3-digit base-10 number?

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

A square garden of side x is extended by 3 m on one side and 5 m on the other. Write its area:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

A ball's height is $h(t) = -t^2 + 6t$. At what time $t > 0$ does it hit the ground ($h = 0$)?

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

Find two positive numbers whose sum is 11 and whose product is 28:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

The sum of two numbers is 20 and their difference is 6. Find both numbers:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Evaluate the discriminant $b^2 - 4ac$ for $2x^2 - 5x + 2 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Solve the system: $y = 2x + 1$ and $x + y = 10$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Find the vertex x-coordinate $x = -b / (2a)$ of the parabola $y = x^2 - 8x + 19$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

To enter -5 on the SC-32, press 5 then [+/-] (CHS). What value is in Register X?

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: Leo said the roots of $(x + 2)(x + 5) = 0$ are $x = 2$ and $x = 5$. Fix the signs!

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ | | SC-32 Ans: _____

Unit 02 Practice D — Factoring Leading Coefficients ($ax^2 + bx + c$) & Rational Expressions

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Factor $2x^2 + 7x + 3$ and find its roots.

$2x^2 + 6x + x + 3 = (2x + 1)(x + 3) \rightarrow$ roots $x = -0.5, -3$.

Verified Answer: $(2x+1)(x+3)$; -0.5, -3

EXAMPLE 2 Simplify the rational expression $(x^2 - 9) \div (x + 3)$ for $x \neq -3$.

$(x - 3)(x + 3) \div (x + 3) = x - 3$.

Verified Answer: x - 3

1 Tier 2 • Detective

Area Grid Expansion: Expand $(2x + 1)(x + 4) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 2 • Detective

Area Grid Expansion: Expand $(3x - 2)(x + 3) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 2 • Detective

Expand perfect square: $(x + 5)^2 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 2 • Detective

Expand perfect square: $(2x - 3)^2 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 2 • Detective

Factor completely: $2x^2 + 5x + 2 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 2 • Detective

Factor completely: $3x^2 - 12 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 2 • Detective

Simplify rational quotient: $(x^2 - 16) \div (x - 4) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 2 • Detective

Simplify rational quotient: $(x^2 + 7x + 10) \div (x + 2) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 2 • Detective

Evaluate $(x^2 - 25) \div (x - 5)$ when $x = 15$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 2 • Detective

Find the positive root of $2x^2 - 7x + 3 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

Unit 02 Practice E — Al-Khwarizmi's Completing the Square & Parabola Vertex Form

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Al-Khwarizmi's Geometric Square: Complete the square for $x^2 + 10x$.**

Split $10x$ into two $5x$ wings on the 2×2 area grid; add the $5 \times 5 = 25$ corner square → $(x + 5)^2$!

Verified Answer: 25 ($(x + 5)^2$)

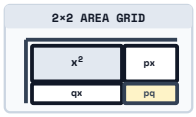
EXAMPLE 2 **Write $y = x^2 - 6x + 14$ in vertex form by completing the square.**

$y = (x^2 - 6x + 9) + 5 = (x - 3)^2 + 5$ → Vertex $(3, 5)$.

Verified Answer: $(x - 3)^2 + 5$; $V(3, 5)$

1 Tier 3 • Challenge

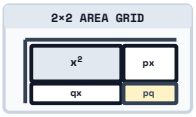
Al-Khwarizmi's Corner Square: Find $c = (b/2)^2$ to complete the square for $x^2 + 6x + c$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

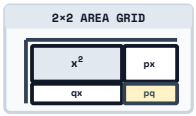
Al-Khwarizmi's Corner Square: Find $c = (b/2)^2$ to complete the square for $x^2 - 12x + c$:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

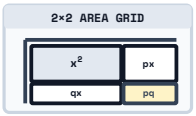
Solve $x^2 + 10x = 39$ by adding 25 to both sides ($(x + 5)^2 = 64$, x > 0):



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

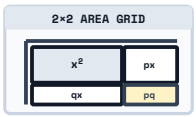
Solve $x^2 + 6x = 16$ by adding 9 to both sides ($(x + 3)^2 = 25$, $x > 0$):



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

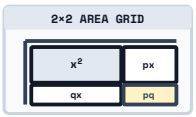
Find the vertex (h, k) of $y = (x - 4)^2 + 7$:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

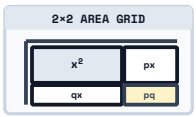
Find the minimum value of $y = x^2 - 10x + 29$:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

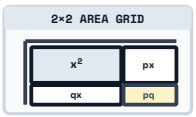
Find the maximum value of $y = -x^2 + 6x + 1$:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

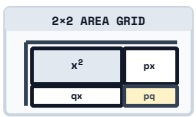
Compute discriminant $\Delta = b^2 - 4ac$ for $x^2 - 6x + 9 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

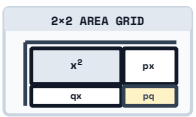
Compute discriminant $\Delta = b^2 - 4ac$ for $x^2 + 2x + 5 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

For what value of c does $x^2 + 8x + c = 0$ have a repeated root?



Paper Step: _____ | | SC-32 Ans: _____

Unit 02 Practice F — Vieta's Root-Coefficient Identities & Difference-of-Squares Mastery

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate $53^2 - 47^2$ mentally using Difference of Squares $(a - b)(a + b)$.

$(53 - 47) \times (53 + 47) = 6 \times 100 = 600$!

Verified Answer: 600

EXAMPLE 2 If r_1 and r_2 are the roots of $x^2 - 6x + 4 = 0$, find $r_1^2 + r_2^2$.

$r_1 + r_2 = 6$ and $r_1 r_2 = 4 \rightarrow r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1 r_2 = 36 - 8 = 28$!

Verified Answer: 28

1 Tier 3 • Challenge

Difference-of-Squares Shortcut: Evaluate $53^2 - 47^2 = (53 - 47)(53 + 47) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 3 • Challenge

Difference-of-Squares Shortcut: Evaluate $65^2 - 35^2 = (65 - 35)(65 + 35) =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 3 • Challenge

Vieta's Sum of Roots: Find $r_1 + r_2$ for $3x^2 - 15x + 7 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 3 • Challenge

Vieta's Product of Roots: Find $r_1 \times r_2$ for $4x^2 + 9x - 20 = 0$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 3 • Challenge

Reciprocal Root Sum: If $r_1 + r_2 = 8$ and $r_1 r_2 = 4$, find $1/r_1 + 1/r_2$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 3 • Challenge

Sum of Squared Roots: For $x^2 - 5x + 3 = 0$, find $r_1^2 + r_2^2$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 3 • Challenge

Difference of Cubes: Evaluate $4^3 - 3^3 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 3 • Challenge

Factor $x^3 - 8 =$

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 3 • Challenge

Find the positive solutions to $x + 6/x = 5$:

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 3 • Challenge

Error Diagnostics: Why does entering $x^2 - 5x + 6$ into [POLY] require [5] [+/-]?

2×2 AREA GRID

x^2	px
qx	pq

Paper Step: _____ [ ] SC-32 Ans: _____

Unit 02 Mastery Check: Quadratic Factoring & Area Models

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Evaluate the quadratic area $2x^2 + 3x - 4$ when $x = 3$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Area Grid Expansion: Expand $(x + 3)(x + 4)$ and evaluate at $x = 2$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Al-Khwarizmi's Square: Find $c = (b/2)^2$ to complete the square for $x^2 + 14x + c$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Factor completely: $x^2 + 8x + 15 =$

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Find both roots of $x^2 - 8x + 12 = 0$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Find both roots of $x^2 - 25 = 0$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

A rectangle has sides (x + 1) and (x + 5) and area 21. Find the positive value of x:

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

Find two numbers whose sum is 13 and product is 40:

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

What is the sum of the roots of $x^2 - 11x + 24 = 0$?

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: Why does $(x + 4)^2$ equal $x^2 + 8x + 16$ instead of $x^2 + 16$?

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 02 STEM: Renaissance Garden Quadrangle Proportions

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: RENAISSANCE GARDEN QUADRANGLE PROPORTIONS

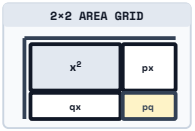
Companion to Vol. II, Ch. 6 & Ch. 7: Renaissance architects proportioned walled courtyards using 2×2 algebraic area grids (x + p)(x + q) and Prime Factor blocks.

Companion Book (Vol. II: Thinking in the Stack — Ch. 6: Galileo's Square-Cube Area Law & Ch. 7: Viète's Root Identities) • Al-Khwarizmi's Squares (820 CE) & Galileo's Area Scaling: In Baghdad's House of Wisdom, Al-Khwarizmi factored quadratics with geometric squares and rectangles (Ch. 7), while Galileo's Square-Cube Law (Ch. 6) explains why area scales with x^2!

Block Studio: Prime Factor Studio (Factor Pairs) & 2×2 Algebraic Area Grid • Stack Shortcut: [x] [ENTER] [2] [+] [x] [ENTER] [3] [+] [×] (Evaluates Trinomial)

CASE INVESTIGATION 1 STEM Modeling

Main Cloister Lawn: Factor the courtyard area $x^2 + 5x + 6$ on the 2×2 area grid and verify the corner constant product:



1. Formula / Stack Plan: _____

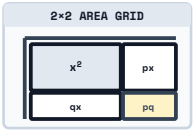
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Fountain Terrace Cutout: Factor the difference-of-squares plaza $x^2 - 16$ and verify the corner square area:



1. Formula / Stack Plan: _____

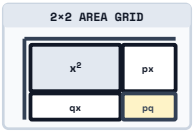
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Pavilion Stone Count: Evaluate the expanded courtyard area (x + 3)(x + 4) when base module $x = 5$ m:



1. Formula / Stack Plan: _____

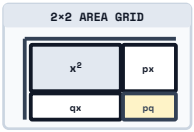
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student claims $(x + 3)^2 = x^2 + 9$ at $x = 2$ (getting 13). Find true $(2 + 3)^2$:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

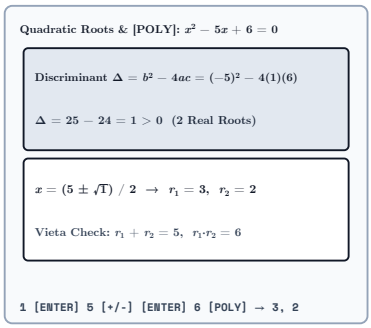
The Quadratic Formula & Automatic [POLY] Root Extraction

Mathematical Focus: Parabola Root Diagram & SC-32 [POLY] Solver • Standards: CCSS.MATH.HSA.REI.B.4.B, CCSS.MATH.HSN.CN.C.7

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Evaluating the Discriminant on the Stack: $b^2 - 4ac$):

Stack lift stores b^2 in Register Y while you compute the $4ac$ product in Register X, evaluating Δ cleanly.

Companion Book (Vol. II: Thinking in the Stack — Ch. 7: Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder ([POLY], [SOLVE])) • Galileo's Parabolas, Viète's Roots & Kahan's [POLY] Guard: Galileo proved in 1638 that projectiles trace parabolas, François Viète linked root sums to $-b/a$, and William Kahan guarded [POLY] against cancellation!



2. KEY MATHEMATICAL PRINCIPLES: MASTERING QUADRATIC ROOTS AND THE DISCRIMINANT

- Discriminant $\Delta = b^2 - 4ac$: If $\Delta > 0$, 2 real roots; if $\Delta = 0$, 1 double root; if $\Delta < 0$, 2 complex conjugate roots.
- The Quadratic Formula: $x = (-b \pm \sqrt{b^2 - 4ac}) \div (2a)$.
- SC-32 Instant [POLY] Solver: Enter coefficients a, b, c and press [POLY] to solve both roots into Registers Y and X!
- Stack key sequence: [a] [ENTER] [b] [ENTER] [c] [POLY] → Register Y holds Root 1, Register X holds Root 2.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: For $x^2 + 5x + 6 = 0$, identify standard-form coefficients $a = 1$, $b = 5$, $c = 6$ on your parabola sketch. Question: $b^2 = 25$. $4ac = 4 \times 1 \times 6 = 24$. Discriminant $\Delta = 25 - 24 = 1 > 0$ (2 distinct real x-intercepts).

Verification Keystrokes: `5` `↶` `x²` `4` `ENTER` `1` `×` `6` `×` `-`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
5	5	b (5) enters Register X
↶ x²	25	$b^2 = 25$ in Register X
4	4	4 enters Register X; 25 lifts to Register Y
ENTER	4	4 lifts to Register Y; 25 lifts to Register Z
1	1	a (1) enters Register X
×	4	$4 \times a = 4$ in Register X
6	6	c (6) enters Register X; 4 in Register Y
×	24	$4ac = 24$ in Register X; 25 waits in Register Y
-	1	$b^2 - 4ac = 25 - 24 = 1$! Discriminant $\Delta = 1$

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Evaluate the discriminant for $2x^2 - 7x + 3 = 0$ on paper and verify on the stack.

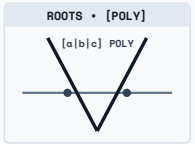
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Instant Quadratic Root Extraction with [POLY]

Deepening Concepts • Standards: CCSS.MATH.HSA.REI.B.4.B

1. EXTENSION CONCEPT & SETUP (Instant Quadratic Root Extraction with [POLY]):

The [POLY] algorithm executes the complete quadratic formula internally, placing both roots onto the stack.

Extension Setup: Write standard-form coefficients $a = 1$, $b = 5$, $c = 6$ on paper and push them into Registers Z, Y, and X. • Question: Roots of $(x + 2)(x + 3) = 0$ are $x = -2$ and $x = -3$.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [ENTER] [5] [ENTER] [6] [↗] [POLY]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1]	1	$a = 1$ enters Register X
[ENTER]	1	1 lifts to Register Y
[5]	5	$b = 5$ enters Register X
[ENTER]	5	5 lifts to Register Y; 1 lifts to Register Z
[6]	6	$c = 6$ enters Register X
[↗] [POLY]	-3	Press [POLY]: Register X = -3, Register Y = -2! Both roots solved

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1: Forgetting the Negative Sign on -b in Manual Calculation

Mathematical & Stack Remediation: Point out that [POLY] avoids sign slip by storing b directly with its true sign.

Misconception 2: Assuming a Negative Discriminant Means 'No Solution'

Mathematical & Stack Remediation: Negative discriminant produces complex conjugate roots (with imaginary unit i).

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [1] [ENTER] [5] [ENTER] [6] [POLY] → Y: -2, X: -3

Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Use [POLY] on the SC-32 to solve $2x^2 - 5x - 3 = 0$.

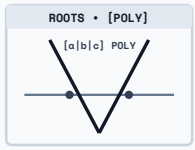
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



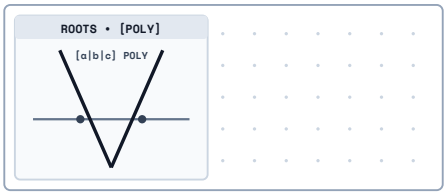
Unit 03 Lab: The Quadratic Formula & Automatic [POLY] Root Extraction

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch your number line or equation balance in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate discriminant for $x^2 + 6x + 9 = 0$: $\Delta =$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

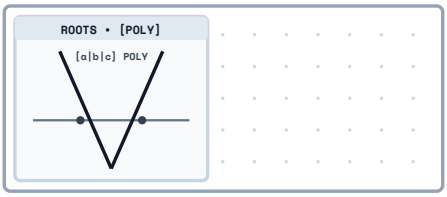
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Solve roots with [POLY]: $x^2 - 7x + 12 = 0 \rightarrow$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

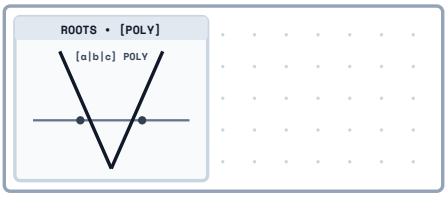
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Solve roots with [POLY]: $x^2 + 2x - 15 = 0 \rightarrow$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

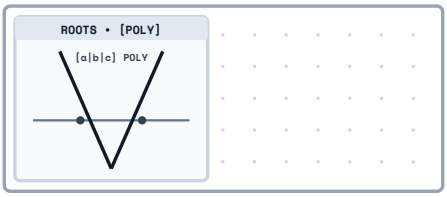
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate discriminant for $x^2 + 4x + 13 = 0$: $\Delta =$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 03 Practice A — Evaluating the Quadratic Discriminant: $\Delta = b^2 - 4ac$

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find Δ for $x^2 + 5x + 6 = 0$.

$5^2 - 4(1)(6) = 25 - 24 = 1$. Key: [5] [$x^2$] [4] [ENTER] [1] [×] [6] [×] [-].

Verified Answer: 1

EXAMPLE 2 Find Δ for $2x^2 - 7x + 3 = 0$.

$(-7)^2 - 4(2)(3) = 49 - 24 = 25$.

Verified Answer: 25

1 Tier 1 • Builder

Discriminant of $x^2 + 7x + 10 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Discriminant of $x^2 - 6x + 9 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Discriminant of $x^2 + 3x - 10 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Discriminant of $2x^2 + 5x + 2 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Discriminant of $3x^2 - 10x + 3 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Discriminant of $4x^2 - 12x + 9 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Discriminant of $x^2 + 4x + 13 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Discriminant of $2x^2 - 3x - 5 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Discriminant of $5x^2 + 2x + 1 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Discriminant of $6x^2 - 11x - 10 = 0$: $\Delta =$

ROOTS • [POLY] [a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Practice B — Instant Quadratic Root Extraction with [POLY] (Monic $a = 1$)

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Solve $x^2 + 5x + 6 = 0$ with [POLY].

Key: [1] [ENTER] [5] [ENTER] [6] [POLY] → $Y = -2$, $X = -3$.

Verified Answer: $x = -2, -3$

EXAMPLE 2 Solve $x^2 - 7x + 12 = 0$ with [POLY].

Key: [1] [ENTER] [-7] [ENTER] [12] [POLY] → $Y = 4$, $X = 3$.

Verified Answer: $x = 4, 3$

1 Tier 1 • Builder

Solve with [POLY]: $x^2 - 5x + 6 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Solve with [POLY]: $x^2 - 8x + 15 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Solve with [POLY]: $x^2 + 2x - 15 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Solve with [POLY]: $x^2 - 3x - 10 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Solve with [POLY]: $x^2 + 7x + 12 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Solve with [POLY]: $x^2 - 9x + 20 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Solve with [POLY]: $x^2 + 4x - 21 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Solve with [POLY]: $x^2 - 16 = 0$ ($b = 0$) → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Solve with [POLY]: $x^2 - 10x + 25 = 0$ → double root =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Solve with [POLY]: $x^2 - 11x + 24 = 0$ → roots =

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Practice C — [POLY] Root Extraction for Non-Monic Quadratics ($a \neq 1$)

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 Solve $2x^2 - 5x - 3 = 0$ with [POLY].

Key: [2] [ENTER] [-5] [ENTER] [-3] [POLY] → $x1 = 3$, x2 = -0.5.

Verified Answer: $x = 3, -0.5$

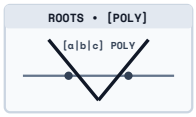
EXAMPLE 2 Solve $4x^2 - 12x + 9 = 0$ with [POLY].

Key: [4] [ENTER] [-12] [ENTER] [9] [POLY] → $x = 1.5$.

Verified Answer: 1.5

1 Tier 2 • Detective

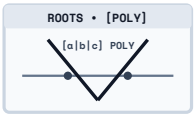
Solve with [POLY]: $2x^2 - 7x + 3 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

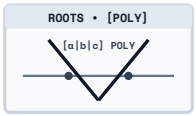
Solve with [POLY]: $2x^2 + 5x + 2 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

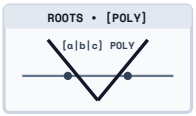
Solve with [POLY]: $2x^2 - 3x - 5 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

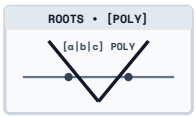
Solve with [POLY]: $4x^2 - 4x - 3 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

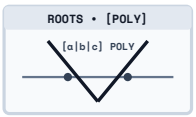
Solve with [POLY]: $5x^2 - 13x - 6 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

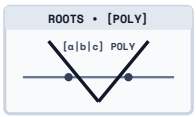
Solve with [POLY]: $2x^2 - 9x + 4 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

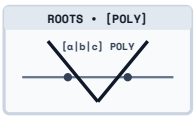
Solve with [POLY]: $4x^2 - 25 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

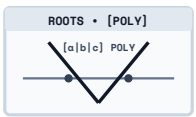
Solve with [POLY]: $5x^2 - 8x + 3 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

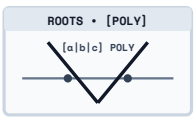
Solve with [POLY]: $8x^2 - 10x + 3 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Solve with [POLY]: $10x^2 - 17x + 3 = 0$ → roots =



Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Practice D — Manual Quadratic Formula Steps & Axis of Symmetry

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find axis of symmetry $x = -b \div (2a)$ for $y = 2x^2 - 12x + 7$.

$x = 12 \div (2 \times 2) = 3$.

Verified Answer: 3

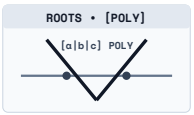
EXAMPLE 2 Compute larger root of $x^2 - 6x + 5 = 0$ via $(6 + \sqrt{16}) \div 2$.

$(6 + 4) \div 2 = 5$.

Verified Answer: 5

1 Tier 2 • Detective

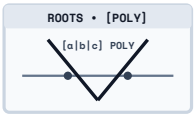
Axis of symmetry $x = -b \div (2a)$ for $y = x^2 - 8x + 5$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

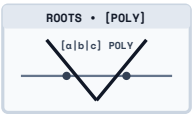
Axis of symmetry $x = -b \div (2a)$ for $y = 2x^2 + 12x - 9$:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

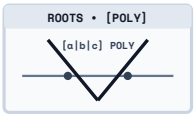
Axis of symmetry $x = -b \div (2a)$ for $y = -3x^2 + 18x + 4$:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

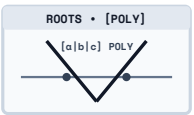
Axis of symmetry $x = -b \div (2a)$ for $y = 4x^2 - 20x + 11$:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

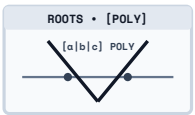
Evaluate $\sqrt{\Delta}$ for $2x^2 - 7x + 3 = 0$ (where $\Delta = 25$):



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

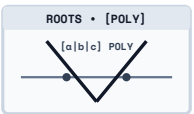
Evaluate $\sqrt{\Delta}$ for $3x^2 - 10x + 3 = 0$ (where $\Delta = 64$):



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

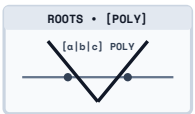
Find the larger root via $(-b + \sqrt{\Delta}) \div (2a)$ for $2x^2 - 7x + 3 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

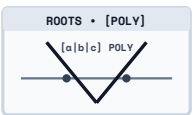
Find the smaller root via $(-b - \sqrt{\Delta}) \div (2a)$ for $2x^2 - 7x + 3 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

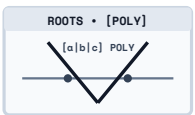
Find the larger root via quadratic formula for $x^2 - 4x - 1 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Find the Golden Ratio positive root of $x^2 - x - 1 = 0$:



Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Practice E — Ballistics Trajectories, Bridge Parabolas & Physics

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 A rocket's height is $h(t) = -16t^2 + 64t + 80$. When does it hit ground ($h = 0$)?

Use [POLY] with a=-16, b=64, c=80 → t = 5 s (and -1 s).

Verified Answer: 5 s

EXAMPLE 2 In Ex 1, find time of peak apogee $t = -b \div (2a)$.

-64 ÷ -32 = 2 seconds.

Verified Answer: 2 s

1 Tier 3 • Challenge

Ballistics: $h(t) = -16t^2 + 64t + 80 = 0$. Find positive landing time t:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

In Problem 1, find the time of maximum height t:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

In Problem 1, find the maximum height at $t = 2$ s:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

A stone dropped from 144 ft has $h(t) = -16t^2 + 144 = 0$. Find landing time t:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Ballistics: $h(t) = -5t^2 + 20t + 25 = 0$. Find positive impact time t:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

In Problem 5, find the peak height at $t = 2$ s:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

A rectangle has length (x + 4) and width x with area 45. Find x > 0:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

A right triangle has legs x and x + 7 and hypotenuse 13. Find x > 0:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Braking distance $d = 0.05v^2 + v = 60$ ft. Find the positive speed v:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Spot the Mistake: For Δ of $x^2 - 6x + 5$, Leo did $-6^2 - 20 = -56$ instead of $(-6)^2 - 20$. True Δ:

ROOTS • [POLY]
[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Practice F — Vieta's Root Theorems, Vertex Minimums & Mastery

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 By Vieta's formulas for $ax^2 + bx + c = 0$, sum of roots $= -b/a$, product $= c/a$.

For $2x^2 - 10x + 12 = 0$: sum $= 5$, product $= 6$.

Verified Answer: 5

EXAMPLE 2 Find minimum value of $f(x) = x^2 - 6x + 14$ at vertex $x = 3$.

$3^2 - 6(3) + 14 = 5$.

Verified Answer: 5

1 Tier 3 • Challenge

Vieta's sum of roots $(-b \div a)$ for $2x^2 - 14x + 9 = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Vieta's product of roots $(c \div a)$ for $2x^2 - 14x + 9 = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

Vieta's sum of roots $(-b \div a)$ for $4x^2 + 20x - 11 = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Vieta's product of roots $(c \div a)$ for $4x^2 + 20x - 11 = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Given roots r1 = 4 and r2 = -7, find $b = -(r1 + r2)$ for $x^2 + bx + c = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Given roots r1 = 4 and r2 = -7, find $c = r1 \times r2$ for $x^2 + bx + c = 0$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Find vertex minimum of $f(x) = x^2 - 8x + 21$ at $x = 4$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Find vertex maximum of $f(x) = -2x^2 + 12x - 10$ at $x = 3$:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Find positive k so $x^2 + kx + 36 = 0$ has a double root ($\Delta = 0$):

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Vieta's Root-Sum Verification: Extract both roots of $2x^2 - 10x + 12 = 0$ with [POLY] and add them on the stack:

ROOTS • [POLY]

[a|b|c] POLY

Paper Step: _____ | | SC-32 Ans: _____

Unit 03 Mastery Check: Quadratic Formula & [POLY] Solver

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Evaluate discriminant $\Delta = b^2 - 4ac$ for $2x^2 - 7x + 3 = 0$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Evaluate discriminant $\Delta = b^2 - 4ac$ for $x^2 + 4x + 13 = 0$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Solve roots with [POLY]: $x^2 - 7x + 12 = 0$ →

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Solve roots with [POLY]: $x^2 + 2x - 15 = 0$ →

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Solve roots with [POLY]: $2x^2 - 5x - 3 = 0$ →

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Find the axis of symmetry $x = -b \div (2a)$ for $y = 3x^2 - 18x + 7$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

A projectile follows $h(t) = -16t^2 + 48t + 64$. Find positive landing time t:

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

Find Vieta's product of roots $(c \div a)$ for $5x^2 - 17x + 30 = 0$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Find vertex minimum value of $f(x) = x^2 - 6x + 13$ at $x = 3$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: A student got $\Delta = -11$ for $x^2 - 3x - 5$ by subtracting $4(1)(-5)$ as $+20$. Find true Δ:

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 03 STEM: Trajectory Ballistics Flight Arc Analysis

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: TRAJECTORY BALLISTICS FLIGHT ARC ANALYSIS

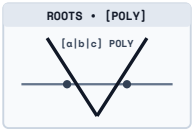
Companion to Vol. II, Ch. 7: Aerospace guidance engineers analyze parabolic flight equations on the 4-level stack using [POLY] and [SOLVE] to find real crossings and impact times.

Companion Book (Vol. II: Thinking in the Stack — Ch. 7: Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder ([POLY], [SOLVE])) • Galileo's Parabolas, Viète's Roots & Kahan's [POLY] Guard: Galileo proved in 1638 that projectiles trace parabolas, François Viète linked root sums to $-b/a$, and William Kahan guarded [POLY] against cancellation!

Mathematical Focus: Parabola Root Diagram & SC-32 [POLY] Solver • Stack Shortcut: [1] [ENTER] [5] [ENTER] [6] [POLY] → *Y*: -2, X: -3

CASE INVESTIGATION 1 STEM Modeling

Radar Target Clearance: Evaluate discriminant $\Delta = b^2 - 4ac$ for flight clearance curve $2x^2 - 7x + 3 = 0$:



1. Formula / Stack Plan: _____

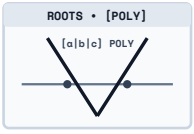
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Wind-Tunnel Calibration Arc: Extract the reference roots of $x^2 + 5x + 6 = 0$ using [POLY]:



1. Formula / Stack Plan: _____

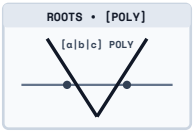
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Downrange Impact Solution: Extract the roots of trajectory profile $2x^2 - 5x - 3 = 0$ using [POLY]:



1. Formula / Stack Plan: _____

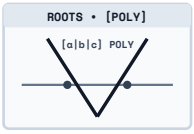
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student says $\Delta < 0$ for $x^2 + 4 = 0$ means no roots. Evaluate Δ:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

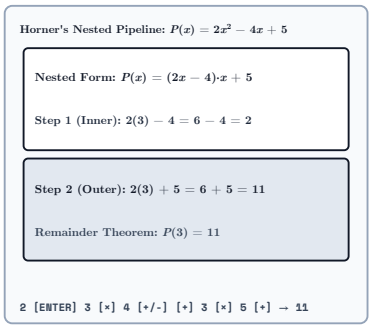
Polynomial Operations, Horner's Rule & Synthetic Division

Mathematical Focus: Nested Polynomial Pipeline Diagram & SC-32 Stack • Standards: CCSS.MATH.HSA.APR.A.1, CCSS.MATH.HSA.APR.B.2

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Evaluating Polynomials via Horner's Rule on the Stack):

Horner's nested method requires only multiplication and addition, avoiding all manual exponentiation.

Companion Book (Vol. II: Thinking in the Stack — Ch. 7: Horner's Nested Scheme, Ch. 8: Romberg Calculus ([図]) & Ch. 14: High-Speed Fluency) • The 1535 Milanese Cubic Duel, Horner (1819) & Romberg Calculus: William George Horner showed that rewriting $ax^3 + bx^2 + cx + d$ as $((ax + b)x + c)x + d$ cuts multiplications in half (Ch. 7 & 14)—the same polynomial foundation used in Romberg quadrature (Ch. 8)!



2. KEY MATHEMATICAL PRINCIPLES: EVALUATING HIGH-DEGREE POLYNOMIALS EFFICIENTLY

- Horner's Rule (Nested Multiplication): $ax^3 + bx^2 + cx + d = ((ax + b)x + c)x + d$.
- Evaluating with Horner's Rule eliminates high powers and executes in linear time on the stack.
- The Remainder Theorem: When $P(x)$ is divided by $(x - c)$, the remainder equals $P(c)$.
- On the SC-32: Write coefficients in descending degree order on paper (inserting 0 for missing powers) and multiply-add across.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: For $P(x) = 2x^2 - 4x + 5$ at $x = 3$, write the nested form $(2x - 4)x + 5$ and list coefficients $[2, -4, 5]$ in order. Question: $P(3) = 2(9) - 4(3) + 5 = 18 - 12 + 5 = 11$. Nested: $(2(3) - 4)(3) + 5 = (6 - 4)(3) + 5 = 2(3) + 5 = 11$.

Verification Keystrokes: 2 ENTER 3 × 4 +/− + 3 × 5 +

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
2	2	Coefficient a (2) enters Register X
ENTER	2	2 lifts to Register Y
3	3	$x = 3$ enters Register X
×	6	$2 \times 3 = 6$ in Register X
4 +/−	-4	Coefficient b (-4) enters Register X
+	2	$6 + (-4) = 2$ in Register X
3	3	$x = 3$ enters Register X; 2 in Register Y
×	6	$2 \times 3 = 6$ in Register X
5	5	Constant c (5) enters Register X
+	11	$6 + 5 = 11$! $P(3) = 11$ verified

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Evaluate $P(x) = 3x^3 - 2x^2 + 4x - 8$ at $x = 2$ using Horner's rule on the stack.

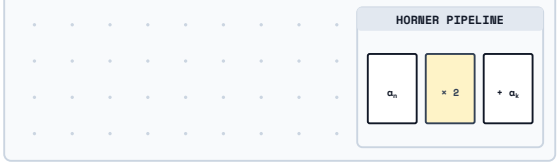
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Synthetic Division on the Stack & The Remainder Theorem

Deepening Concepts • Standards: CCSS.MATH.HSA.APR.B.2

1. EXTENSION CONCEPT & SETUP (Synthetic Division on the Stack & The Remainder Theorem):

The intermediate sum gives the quotient coefficient (-3) and the final sum gives the remainder (0).

Extension Setup: Write coefficients [1, −5, 6] of $P(x) = x^2 - 5x + 6$ in a synthetic division table on paper to divide by $(x - 2)$ at $c = 2$. • Question: If remainder is 0, $(x - 2)$ is a factor. $P(2) = 4 - 10 + 6 = 0$. (Prediction: Remainder = 0)

2. EXTENSION STACK SEQUENCE (SC-32): `1` `ENTER` `2` `×` `5` `+/-` `+` `2` `×` `6` `+`

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
`1`	1	Lead coefficient 1 enters Register X
`ENTER`	1	1 lifts to Register Y
`2`	2	$c = 2$ enters Register X
`×`	2	$1 \times 2 = 2$
`5` `+/-`	-5	$b = -5$ enters Register X
`+`	-3	$2 + (-5) = -3$ (linear quotient coefficient)
`2`	2	$c = 2$ enters Register X; -3 in Register Y
`×`	-6	$-3 \times 2 = -6$
`6`	6	constant 6 enters Register X
`+`	0	$-6 + 6 = 0$! Remainder is 0, $(x - 2)$ is a factor

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1 — Dropping Missing Degree Terms in Synthetic Division: Dividing $x^3 - 7x + 6$ and omitting the $0x^2$ coefficient.

Mathematical & Stack Remediation: Always write a 0 placeholder for any missing degree term in your synthetic division row: 1, 0, -7, 6.

Misconception 2 — Flipping the Sign of Divisor c: Dividing by $(x - 2)$ and using $c = -2$ instead of $c = +2$.

Mathematical & Stack Remediation: The root of $(x - c) = 0$ is $x = +c$.

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Test if $(x - 3)$ is a factor of $P(x) = x^3 - 4x^2 + x + 6$ using synthetic division on stack.

1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



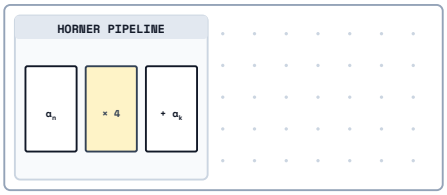
Unit 04 Lab: Polynomial Operations, Horner's Rule & Synthetic Division

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch your number line or equation balance in Box 1. (2) Write your algebraic steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate $P(x) = x^2 - 6x + 8$ at $x = 4$: $P(4) =$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Evaluate $P(x) = 2x^3 + 3x^2 - 5x + 1$ at $x = 2$: $P(2) =$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Find remainder of $(x^3 + 2x^2 - 7x - 2) \div (x - 2)$: $R =$

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate $P(x) = 4x^2 - 12x + 9$ at $x = 1.5$: P(1.5) =

1. Number Line / Equation Balance:



2. Algebraic Setup & Inverse Steps:

Signed / Equation Setup:

Inverse Operation Step(s):

Verified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 04 Practice A — Evaluating Polynomials via Horner's Nested Pipeline

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate $P(x) = 2x^2 - 4x + 5$ at $x = 3$.

$((2 \times 3) - 4) \times 3 + 5 = 11$. Key: [2] [ENTER] [3] [×] [-4] [+] [3] [×] [5] [+].

Verified Answer: 11

EXAMPLE 2 Evaluate $P(x) = 3x^3 - 2x^2 + 4x - 8$ at $x = 2$.

$((3(2) - 2)2 + 4)2 - 8 = 16$.

Verified Answer: 16

1 Tier 1 • Builder

Horner evaluation: $P(x) = x^2 + 3x - 4$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n | × 2 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

Horner evaluation: $P(x) = 2x^2 - 5x + 7$ at $x = 3$: $P(3) =$

HORNER PIPELINE

a_n | × 3 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

Horner evaluation: $P(x) = 3x^2 + 4x - 10$ at $x = 4$: $P(4) =$

HORNER PIPELINE

a_n | × 4 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

Horner evaluation: $P(x) = x^3 + 2x^2 - 3x + 1$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n | × 2 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

Horner evaluation: $P(x) = 2x^3 + 3x^2 - 5x + 1$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n | × 2 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

Horner evaluation: $P(x) = x^3 - 4x^2 + 6x - 2$ at $x = 3$: $P(3) =$

HORNER PIPELINE

a_n | × 3 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

Horner evaluation: $P(x) = 2x^3 - 3x^2 + x + 8$ at $x = -1$: $P(-1) =$

HORNER PIPELINE

a_n | × -1 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

Horner evaluation: $P(x) = 4x^2 - 12x + 9$ at $x = 1.5$: P(1.5) =

HORNER PIPELINE

a_n | × 1 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

Horner evaluation: $P(x) = x^3 + 5x^2 + 2x - 8$ at $x = -2$: $P(-2) =$

HORNER PIPELINE

a_n | × -2 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

Horner evaluation: $P(x) = 3x^3 - 7x^2 + 5x - 1$ at $x = 3$: $P(3) =$

HORNER PIPELINE

a_n | × 3 | + a_k

Paper Step: _____ [ ] SC-32 Ans: _____

Unit 04 Practice B — The Remainder Theorem: Remainder of $P(x) \div (x - c)$ is $P(c)$

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find remainder of $(x^3 - 4x^2 + 2x + 5) \div (x - 2)$.

Evaluate $P(2) = ((1(2) - 4)2 + 2)2 + 5 = 1$.

Verified Answer: 1

EXAMPLE 2 Find remainder of $(2x^3 + x^2 - 6x + 4) \div (x + 2)$.

Evaluate $P(-2) = -12 + 4 + 12 + 4 = 8$.

Verified Answer: 8

1 Tier 1 • Builder

Find the remainder of $(x^2 - 7x + 15) \div (x - 3)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Find the remainder of $(2x^2 + 5x - 3) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Find the remainder of $(x^3 - 2x^2 + 4x - 5) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Find the remainder of $(x^3 + 3x^2 - 4x + 2) \div (x - 1)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Find the remainder of $(2x^3 - 5x^2 + x + 7) \div (x - 3)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Find the remainder of $(x^3 + 4x^2 + x - 6) \div (x + 1)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Find the remainder of $(x^3 - 3x^2 + 2x + 8) \div (x + 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Find the remainder of $(3x^3 - 4x^2 - 5x + 6) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Find the remainder of $(x^3 - 6x^2 + 11x - 6) \div (x - 4)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Find the remainder of $(2x^3 + 3x^2 - 8x + 3) \div (x - 1)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

Unit 04 Practice C — The Factor Theorem & Missing Zero-Coefficient Terms

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 Divide $x^3 - 7x + 6$ by $(x - 2)$ (note $0x^2$ term!).

Coefficients: 1, 0, -7, 6 → $P(2) = ((1(2)+0)2 - 7)2 + 6 = 0$.

Verified Answer: 0

EXAMPLE 2 Divide $x^3 - 8$ by $(x - 2)$ (coefficients 1, 0, 0, -8).

$P(2) = 8 - 8 = 0$.

Verified Answer: 0

1 Tier 2 • Detective

Test factor $(x - 2)$ on $x^3 - 7x + 6$: $P(2) =$

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

Test factor $(x + 3)$ on $x^3 - 7x + 6$: $P(-3) =$

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

Find remainder of $(x^3 - 5x + 2) \div (x - 3)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

Find remainder of $(2x^3 - 9x + 4) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

Find remainder of $(x^3 + 4x^2 - 9) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Find remainder of $(x^3 - 27) \div (x - 3)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Find remainder of $(x^3 + 8) \div (x + 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Test factor $(x - 3)$ on $x^3 - 4x^2 + x + 6$: $P(3) =$

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

Test factor $(x + 1)$ on $x^3 - 4x^2 + x + 6$: $P(-1) =$

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Find the depressed linear coefficient b1 in $(x^2 - 5x + 6) \div (x - 2)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

Unit 04 Practice D — Quartic (4th-Degree) Polynomials & Symmetry

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 5$ at $x = 2$.

$(((1(2)-2)2+3)2-4)2+5 = 9$.

Verified Answer: 9

EXAMPLE 2 Evaluate even quartic $P(x) = x^4 - 5x^2 + 4$ at $x = 2$.

$2^4 - 5(2^2) + 4 = 16 - 20 + 4 = 0$.

Verified Answer: 0

1 Tier 2 • Detective

Quartic Horner: $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 5$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

Quartic Horner: $P(x) = x^4 + x^3 - 2x^2 + 3x - 1$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

Quartic Horner: $P(x) = 2x^4 - 3x^3 + x^2 - 5x + 2$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

Even quartic $P(x) = x^4 - 5x^2 + 4$ at $x = 2$: $P(2) =$

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

Even quartic $P(x) = x^4 - 5x^2 + 4$ at $x = 3$: $P(3) =$

HORNER PIPELINE

a_n	× 3	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Find the remainder of $(x^4 - 16) \div (x - 2)$:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Find the remainder of $(x^4 - 1) \div (x - 3)$:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Find the remainder of $(x^4 + 2x^2 - 8) \div (x - 2)$:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

Evaluate odd cubic $P(x) = x^3 - 4x$ at $x = 3$:

HORNER PIPELINE

a_n	× 3	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Verify odd symmetry $P(-3) = -P(3)$ for $P(x) = x^3 - 4x$ at $x = -3$:

HORNER PIPELINE

a_n	× -3	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

Unit 04 Practice E — Engineering Transfer Functions & Volume Polynomials

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 An open box from a 12×10 sheet has $V(x) = x(12-2x)(10-2x)$. Find $V(2)$.

$2 \times 8 \times 6 = 96$ cm^3.

Verified Answer: 96 cm^3

EXAMPLE 2 Evaluate filter damping $D(s) = s^3 + 2s^2 + 2s + 1$ at $s = 2$.

$8 + 8 + 4 + 1 = 21$.

Verified Answer: 21

1 Tier 3 • Challenge

Box volume $V(x) = x(12 - 2x)(10 - 2x)$ at cut $x = 1$ cm:

HORNER PIPELINE

a_n	× 1	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Box volume $V(x) = x(12 - 2x)(10 - 2x)$ at cut $x = 2$ cm:

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

Box volume $V(x) = x(12 - 2x)(10 - 2x)$ at cut $x = 3$ cm:

HORNER PIPELINE

a_n	× 3	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

3rd-order Butterworth polynomial $B(s) = s^3 + 2s^2 + 2s + 1$ at s = 2:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

3rd-order Butterworth polynomial $B(s) = s^3 + 2s^2 + 2s + 1$ at s = 3:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Beam deflection polynomial $D(x) = x^3 - 6x^2 + 9x$ at $x = 2$ m:

HORNER PIPELINE

a_n	× 2	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Beam deflection polynomial $D(x) = x^3 - 6x^2 + 9x$ at $x = 3$ m:

HORNER PIPELINE

a_n	× 3	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Revenue polynomial $R(x) = -2x^3 + 15x^2 + 100x$ at $x = 5$ units:

HORNER PIPELINE

a_n	× 5	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Thermal heat capacity $C(T) = 2T^2 + 5T + 12$ at $T = 10$ K:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Spot the Mistake: In $(x^3 - 7x + 6) \div (x - 2)$, Sam omitted $0x^2$ and got $(1(2)-7)2+6 = -4$. True $P(2)$:

HORNER PIPELINE

a_n	× x	+ a_k

Paper Step: _____ | | SC-32 Ans: _____

Unit 04 Practice F — Unknown Coefficients, Depressed Equations & Mastery

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find k so $(x - 2)$ is a factor of $x^3 - 3x^2 + kx - 4$.

$P(2) = 8 - 12 + 2k - 4 = 0 \rightarrow 2k = 8 \rightarrow k = 4$.

Verified Answer: 4

EXAMPLE 2 Given $P(x) = (x - 2)(x^2 - 3x - 4)$, find all roots.

$x = 2$, and [POLY] on $x^2 - 3x - 4 = 0$ gives $x = 4, -1$.

Verified Answer: $x = 4, -1$

1 Tier 3 • Challenge

Find k so $(x - 2)$ divides $x^3 - 3x^2 + kx - 4$ evenly:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Find c so $P(x) = 2x^2 - 5x + c$ has remainder 10 when divided by $(x - 3)$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

Factor $x^3 - 6x^2 + 11x - 6$ given root $x = 1$: solve depressed $x^2 - 5x + 6 = 0$ with [POLY]:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Factor $x^3 - 4x^2 + x + 6$ given root $x = 3$: solve depressed $x^2 - x - 2 = 0$ with [POLY]:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Polynomial composition: if $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find g($f(2)$):

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Polynomial composition: if $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find f($g(2)$):

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Find the average rate of change of $P(x) = x^3$ from $x = 1$ to $x = 4$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Find the sum of all coefficients of $P(x) = 3x^4 - 2x^3 + 5x^2 - 7x + 4$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Find the constant term of $P(x) = (2x - 3)^3$:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Factor Theorem Root Test: Evaluate $P(x) = x^3 - 4x^2 + x + 6$ at $x = 3$ using nested Horner's method:

HORNER PIPELINE

Paper Step: _____ | | SC-32 Ans: _____

Unit 04 Mastery Check: Polynomial Operations & Division

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Evaluate $P(x) = 2x^2 - 4x + 5$ at $x = 3$ via Horner's rule:

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Evaluate $P(x) = 3x^3 - 2x^2 + 4x - 8$ at $x = 2$ via Horner's rule:

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Find the remainder of $(x^3 + 2x^2 - 5x + 4) \div (x - 2)$:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

Test if $(x - 2)$ is a factor of $x^3 + 2x^2 - 7x - 2$: $P(2) =$

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Find the remainder of $(x^3 - 7x + 6) \div (x - 3)$:

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Evaluate quartic $P(x) = x^4 - 5x^2 + 4$ at $x = 3$:

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Find box volume $V(x) = x(14 - 2x)(10 - 2x)$ at cut $x = 2$ cm:

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

Given $x = 2$ is a root of $x^3 - 6x^2 + 11x - 6 = 0$, solve depressed $x^2 - 4x + 3 = 0$ with [POLY]:

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Find the sum of all coefficients of $P(x) = 4x^3 - 3x^2 + 6x - 2$:

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Error Analysis: Dividing by $(x - 2)$, a student evaluated $P(-2)$ instead of $P(2)$ on $x^2 - 5x + 6$. Find true $P(2)$:

[ ] +10 pts [ ] SC-32 Ans: _____

Unit 04 STEM: Signal Processing Filter Transfer Functions

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: SIGNAL PROCESSING FILTER TRANSFER FUNCTIONS

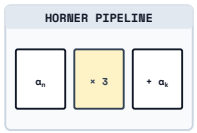
Companion to Vol. II, Ch. 7, Ch. 8 & Ch. 14: Audio engineers evaluate high-order filter transfer polynomials on the stack using Horner's nested pipeline.

Companion Book (Vol. II: Thinking in the Stack — Ch. 7: Horner's Nested Scheme, Ch. 8: Romberg Calculus ([⊠]) & Ch. 14: High-Speed Fluency) • The 1535 Milanese Cubic Duel, Horner (1819) & Romberg Calculus: William George Horner showed that rewriting $ax^3 + bx^2 + cx + d$ as $((ax + b)x + c)x + d$ cuts multiplications in half (Ch. 7 & 14)—the same polynomial foundation used in Romberg quadrature (Ch. 8)!

Mathematical Focus: Nested Polynomial Pipeline Diagram & SC-32 Stack • Stack Shortcut: [a] [ENTER] [x] [×] [b] [+] [x] [×] [c] [+] (Horner's Pipeline)

CASE INVESTIGATION 1 STEM Modeling

Stage-1 Gain Response: Evaluate filter polynomial $P(x) = 2x^2 - 4x + 5$ at frequency ratio $x = 3$ via Horner's rule:



1. Formula / Stack Plan: _____

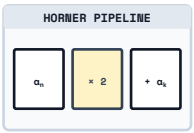
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Notch Filter Null Check: Evaluate cubic transfer function $P(x) = x^3 - 4x^2 + x + 6$ at $x = 2$ via Horner's rule:



1. Formula / Stack Plan: _____

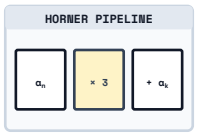
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Remainder Theorem Telemetry: Evaluate filter polynomial $P(x) = 2x^2 - 3x + 5$ at x = 3 to find the remainder when dividing $P(x)$ by $(x - 3)$:



1. Formula / Stack Plan: _____

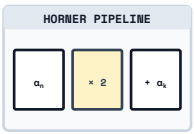
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student omits the $0x^2$ coefficient in $P(x) = x^3 - 7x + 6$ at $x = 2$. Evaluate $P(2)$ with the $0x^2$ term included:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

Exponential Growth, Decay & Continuous Compounding

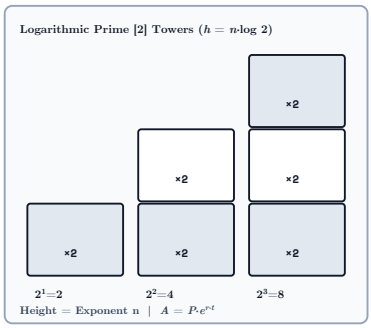
Hands-On Block Studio: Prime Factor Studio & SC-32 Scientific Stack ([y^x], [e^x]) • Standards: CCSS.MATH.HSF.LE.A.1, CCSS.MATH.HSF.LE.A.4

---

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Continuous Compounding & Euler's Constant e):

Exponential growth $P \cdot e^{rt}$ compounds continuously by scaling a base quantity according to an inner rate-time product ($r \times t$). Evaluating from the inside out—computing the exponent r × t first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: 100-Decade Scaling, Ch. 3: Exponential Growth ([e^x]) & Ch. 11: Fibonacci's [TVM]) • Euler's Base e, Bernoulli's Compounding & Fibonacci's TVM: Jacob Bernoulli discovered $e \approx 2.71828$ by compounding interest every instant (Ch. 3), while Fibonacci's 1202 Liber Abaci launched the time-value-of-money cash-flow equations (Ch. 11)!



2. HANDS-ON BLOCKS & CONCEPT RULES: EVALUATING EXPONENTIAL MODELS WITH ZERO PARENTHESES

- Annual Compounding: $A = P(1 + r)^t$. Evaluate inside-out: $(1 + r)$ first, raise to power t with [y^x], then multiply by P.
- Continuous Compounding: $A = Pe^{rt}$. Evaluate exponent product $r \times t$ first, apply [e^x], then multiply by P.
- Exponential Decay: $A = P(1 - r)^t$ or $A = P \times (1/2)^{t/h}$ for radioactive half-life.
- On the SC-32: The [y^x] and [e^x] scientific keys execute directly without requiring equation scratchpads.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Block Setup: Stack Prime Factor [2] blocks (7.0 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).
Question: Invest \$1,000 at 7% continuous interest for 10 years: $A = 1000 \times e^{0.07 \times 10} = 1000 \times e^{0.7}$. Will it double? ($e^{0.7} \approx 2.01$).

Verification Keystrokes: `0` `•` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
`0` `•` `0` `7`	`0.07`	Rate $r = 0.07$ entered in X
`ENTER`	`0.07`	Rate lifted to Y
`1` `0`	`10`	Time $t = 10$ entered in X
`×`	`0.7`	Exponent $r \times t = 0.70$
`e^x`	`2.0138`	$e^{0.70} = 2.01375$ (Growth factor: more than doubled!)
`1` `0` `0` `0`	`1000`	Principal $P = 1000$ (growth factor lifts to Y)
`×`	`2013.75`	Total accumulated amount $A = \$2{,}013.75$!

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: A bacteria culture starts with $P = 500$ cells and triples every hour. Calculate the population after $t = 4$ hours.

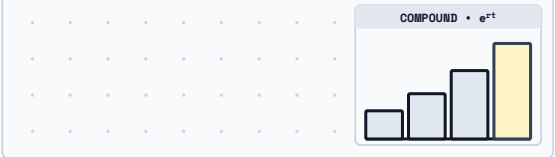
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



Extension: Continuous Compounding with the Natural Exponential [e^x]

Deepening Concepts • Standards: CCSS.MATH.HSF.LE.A.4

1. EXTENSION CONCEPT & SETUP (Continuous Compounding with the Natural Exponential [e^x]):

Continuous compounding represents the mathematical limit of compounding as intervals approach infinity.

Hands-On Extension Setup: Compare discrete power growth on the Prime Factor Studio tower with continuous compounding: Principal P = \$1,000, rate r = 5% (0.05), time t = 10 years. • Question: Exponent $r \times t = 0.5$. $e^{0.5} \approx 1.6487$. $A \approx$ \$1,648.72. (Prediction: \$1,648.72)

2. EXTENSION STACK SEQUENCE (SC-32): [0] [•] [0] [5] [ENTER] [1] [0] [×] [e^x] [1] [0] [0] [0] [×]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[0] [•] [0] [5]	0.05	Rate r (0.05) enters Register X
[ENTER]	0.05	0.05 lifts to Register Y
[1] [0]	10	Time t (10) enters Register X
[×]	0.5	$r \times t = 0.5$ in Register X
[e^x]	1.6487	$e^{0.5} = 1.648721$ in Register X
[1] [0] [0] [0]	1000	$P = 1000$ enters Register X; e^{rt} in Y
[×]	1648.72	Pe^{rt} = \$1,648.72! Continuous compounding verified

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS

Misconception 1: Entering Annual Rate as a Whole Percentage (5 instead of 0.05)

Hands-On Block Studio Remediation: 5% means 5 per 100 = 0.05. Multiplying by 1.5 instead of 1.05 produces catastrophic inflation!

Misconception 2 — Multiplying Principal by Base before Applying the Exponent: Typing [1000] [×] [1.05] then raising to 10.

Hands-On Block Studio Remediation: Order of operations requires exponents before multiplication: $P \times (1+r)^t$.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [1000] [ENTER] [1.05] [ENTER] [10] [y^x] [×] = \$1,628.89

Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: A bacterial colony of 500 cells grows continuously at rate $r = 0.12$ per hour. Find population at $t = 8$ hrs.

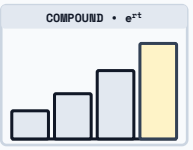
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



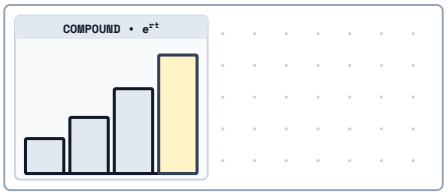
Unit 05 Lab: Exponential Growth, Decay & Continuous Compounding

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the growth/decay curve or timeline in Box 1. (2) Write your exponent/log steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Invest $2,000 at 6% compounded annually for 5 years: A =

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

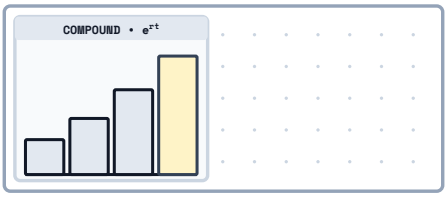
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Continuous compounding: $5,000 at 4% for 8 years: A =

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

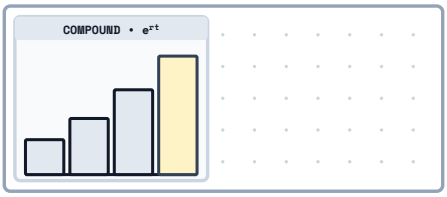
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Depreciation: Car drops 15% per year from $30,000 after 4 years: A =

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

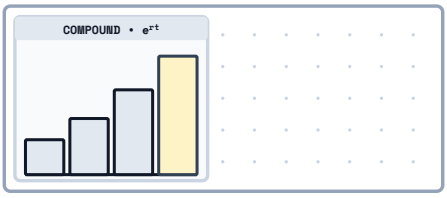
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Half-life: 100g substance with half-life 6 days after 18 days: A =

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did the Prime Factor or Binary Place Value blocks confirm the structure on the SC-32 stack?

Unit 05 Practice A — Discrete Exponential Growth & Compound Interest: $A = P(1 + r)^t$

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Invest $1,000 at 5% annual interest for 10 years.**

$1000 \times (1.05)^{10} = \$1{,}628.89$. Key: [1000] [ENTER] [1.05] [ENTER] [10] [y^x] [×].

Verified Answer: $1,628.89

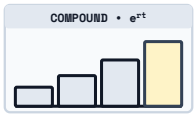
EXAMPLE 2 **A town of 5,000 grows 4% per year for 6 years.**

$5000 \times (1.04)^{6} = 6{,}326.60$.

Verified Answer: 6326.6

1 Tier 1 • Builder

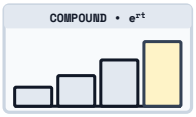
Evaluate exponential power: $2^{10} =$



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

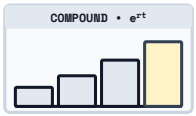
Evaluate exponential power: $3^{6} =$



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

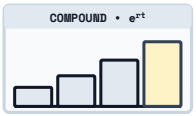
Evaluate growth factor: $(1.05)^{10} =$



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

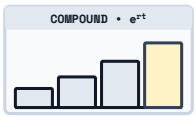
Invest $1,000 at 5% compounded annually for 10 years: $A =$



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

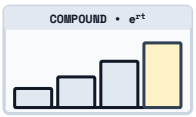
Invest $2,000 at 6% compounded annually for 5 years: $A =$



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

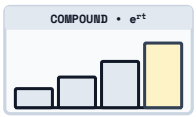
Invest $5,000 at 4% compounded annually for 8 years: $A =$



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

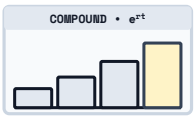
Invest $3,000 at 7% compounded annually for 12 years: $A =$



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

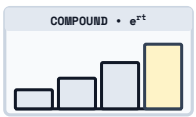
Invest $10,000 at 8% compounded annually for 10 years: $A =$



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

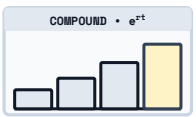
A bacterial culture of 200 cells doubles every hour for 7 hours. Find the population:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

A startup's revenue of $50,000 grows 20% per year for 4 years. Find the revenue:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 05 Practice B — Periodic Compounding (Semiannual, Quarterly & Monthly)

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Invest $1,000 at 6% compounded quarterly (n=4) for 5 years (20 periods).

$r/n = 0.015$; $1000 \times (1.015)^{20} = \$1{,}346.86$.

Verified Answer: $1,346.86

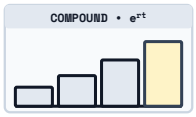
EXAMPLE 2 Invest $2,000 at 8% compounded semiannually (n=2) for 4 years (8 periods).

$r/n = 0.04$; $2000 \times (1.04)^{8} = \$2{,}737.14$.

Verified Answer: $2,737.14

1 Tier 1 • Builder

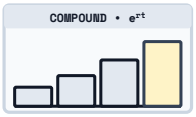
Semiannual compounding: $1,000 at 6% annual rate for 5 years:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

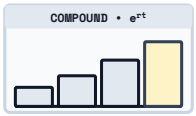
Semiannual compounding: $2,000 at 8% annual rate for 4 years:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

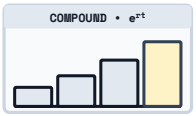
Quarterly compounding: $1,000 at 6% annual rate for 5 years:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

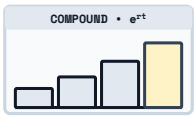
Quarterly compounding: $4,000 at 8% annual rate for 3 years:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

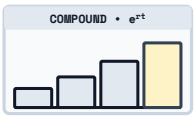
Quarterly compounding: $5,000 at 4% annual rate for 4 years:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

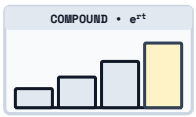
Monthly compounding: $1,000 at 6% annual rate for 5 years:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

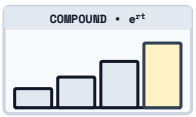
Monthly compounding: $3,000 at 12% annual rate for 2 years:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

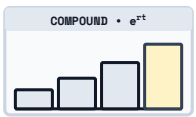
Effective Annual Yield (APY) of 12% compounded monthly:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

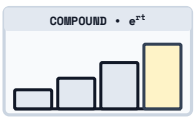
Effective Annual Yield (APY) of 8% compounded quarterly:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Extra interest from monthly ($1,348.85) vs annual ($1,338.23) on $1,000 at 6% for 5 yrs:



Paper Step: _____ | | SC-32 Ans: _____

Unit 05 Practice C — Continuous Compounding with Natural Exponential [e^x]: $A = Pe^{rt}$

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 Invest $1,000 at 5% compounded continuously for 10 years.

$r \times t = 0.5$; $1000 \times e^{0.5} = \$1{,}648.72$. Key: [0.05] [ENTER] [10] [×] [e^x] [1000] [×].

Verified Answer: $1,648.72

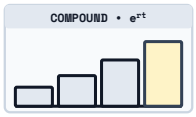
EXAMPLE 2 500 bacteria grow continuously at $r = 0.12$/hr for 8 hours.

$500 \times e^{0.96} = 1{,}305.85$ cells.

Verified Answer: 1305.85

1 Tier 2 • Detective

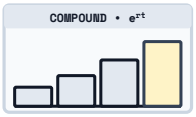
Evaluate Euler's number e^1 on the stack: 1 [e^x] =



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 2 • Detective

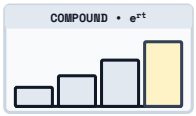
Evaluate natural exponential: e^2 =



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 2 • Detective

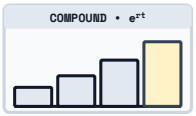
Evaluate natural exponential: $e^{0.5}$ =



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 2 • Detective

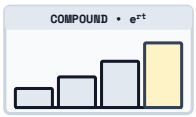
Continuous compounding: $1,000 at 5% for 10 years: A =



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 2 • Detective

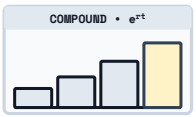
Continuous compounding: $5,000 at 4% for 8 years: A =



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 2 • Detective

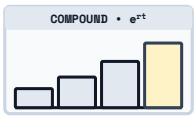
Continuous compounding: $2,000 at 6% for 5 years: A =



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 2 • Detective

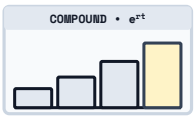
Continuous compounding: $10,000 at 7% for 10 years: A =



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 2 • Detective

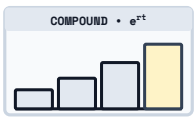
Continuous bacterial growth: 500 cells at $r = 0.12$/hr for 8 hrs:



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 2 • Detective

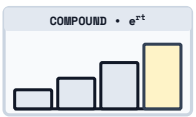
Continuous population growth: 4,000 at $r = 0.03$/yr for 15 yrs:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 2 • Detective

Continuous compounding: $2,500 at 8% for 12.5 years: A =



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 05 Practice D — Exponential Decay, Depreciation & Radioactive Half-Life

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 **A $30,000 car depreciates 15% per year for 4 years.**

$30000 \times (0.85)^4 = \$15{,}660.19$. Key: [30000] [ENTER] [0.85] [ENTER] [4] [y^x] [×].

Verified Answer: $15,660.19

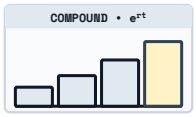
EXAMPLE 2 **100 g of an isotope goes through 3 half-lives.**

$100 \times (0.5)^3 = 12.5$ g.

Verified Answer: 12.5 g

1 Tier 2 • Detective

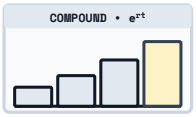
Half-life decay: 80 g after 3 half-lives:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

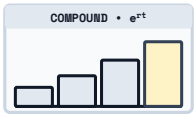
Half-life decay: 100 g after 18 days (with a 6-day half-life):



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

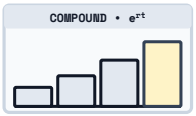
Half-life decay: 240 g after 4 half-lives:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

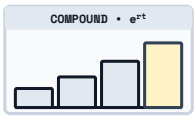
Half-life decay: 500 mg after 2.5 half-lives:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

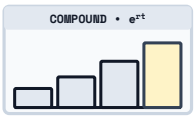
Vehicle depreciation: $30,000 drops 15% per year for 4 years:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

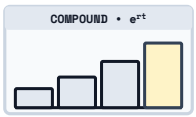
Equipment depreciation: $20,000 drops 20% per year for 5 years:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

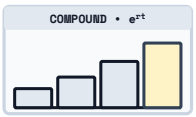
Atmospheric pressure $P = 101.3 \times e^{-0.12 \times h}$ at altitude $h = 5$ km:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

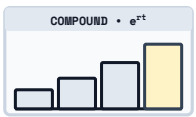
Capacitor discharge $V(t) = 12 \times e^{-0.5 \times t}$ at $t = 4$ seconds:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

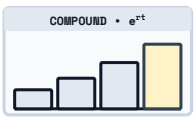
Medicine elimination: 400 mg decays continuously at $k = 0.25$/hr for 6 hrs:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Light intensity through water: 100% drops 10% per meter at 8 m depth:



Paper Step: _____ | | SC-32 Ans: _____

Unit 05 Practice E — Present Value Discounting, Moore's Law & STEM Models

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Find Present Value needed to reach $10,000 in 8 years at 5% annual.**

$PV = 10000 \div (1.05)^{8} = \$6{,}768.39$.

Verified Answer: $6,768.39

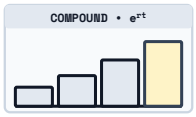
EXAMPLE 2 **Transistor count of 1,000 doubles 8 times under Moore's Law.**

$1000 \times 2^{8} = 256{,}000$.

Verified Answer: 256000

1 Tier 3 • Challenge

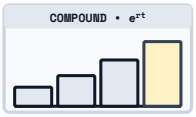
Present Value of $10,000 in 8 years at 5% annual:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

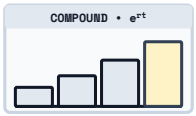
Present Value of $5,000 in 6 years at 6% annual:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

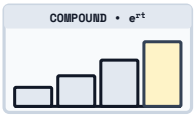
Present Value of $20,000 in 10 years at 5% continuous:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

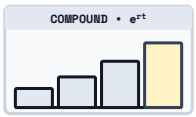
Moore's Law: chip with 2,500 transistors after 6 doublings:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

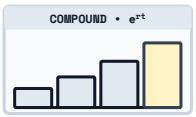
Fiber bandwidth of 10 Gbps grows 50% per year for 4 years:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

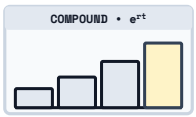
Newton's Law of Cooling: $T(t) = 20 + 70 \times e^{-0.1 \times 5} =$



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

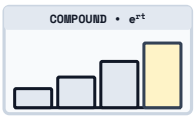
Inflation purchasing power: $1,000 after 10 years of 3% inflation:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

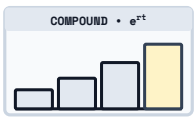
Total interest earned on $4,000 at 5% compounded annually for 6 years:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

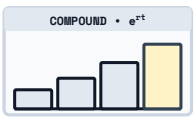
Growth ratio difference of continuous vs annual at 10% for 1 yr:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Spot the Mistake: Student used $r = 5$ instead of 0.05 in $100 \times (1 + r)^2$. Find the true value:



Paper Step: _____ | | SC-32 Ans: _____

Unit 05 Practice F — Multi-Stage Compounding, Logistic Curves & Challenge

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 $1,000 grows at 5% for 4 years, then 8% for 3 years.

$1000 \times (1.05)^4 \times (1.08)^3 = \$1{,}531.02$.

Verified Answer: $1,531.02

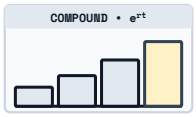
EXAMPLE 2 Evaluate logistic population $L(t) = 1000 \div (1 + 9 \times e^{-0.5 \times 4})$.

$1000 \div (1 + 9 \times e^{-2}) = 450.85$.

Verified Answer: 450.85

1 Tier 3 • Challenge

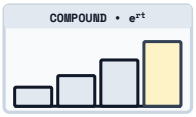
Two-stage investment: $1,000 at 5% for 4 yrs, then 8% for 3 yrs:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

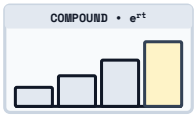
Two-stage investment: $2,000 at 6% for 5 yrs, then 4% for 5 yrs:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

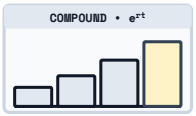
Difference between continuous and annual on $10,000 at 6% for 10 yrs:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

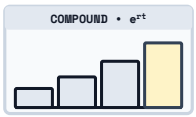
Logistic growth model: $1000 \div (1 + 9 \times e^{-2}) =$



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

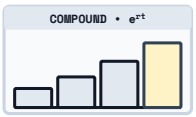
Logistic growth model at $t = 0$: $1000 \div (1 + 9 \times e^{0}) =$



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

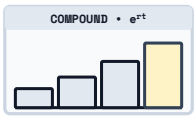
RC circuit charging voltage $V(t) = 10 \times (1 - e^{-1}) =$



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

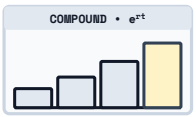
Hyperbolic cosine $\cosh(1) = (e^{1} + e^{-1}) \div 2 =$



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

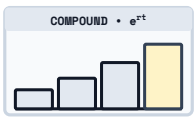
Hyperbolic sine $\sinh(1) = (e^{1} - e^{-1}) \div 2 =$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

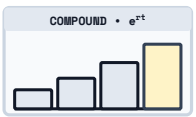
Verify hyperbolic identity $\cosh^2(1) - \sinh^2(1) = 1$:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Continuous Endowment Challenge: Find the value of a $500 grant after 20 years of 5% continuous growth ($A = 500 \cdot e^{0.05 \times 20}$):



Paper Step: _____ | | SC-32 Ans: _____

Unit 05 Mastery Check: Exponential Growth & Decay

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Invest \$1,000 at 5% compounded annually for 10 years: A =

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Invest \$2,000 at 6% compounded annually for 5 years: A =

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Quarterly compounding: \$1,000 at 8% annual rate for 3 years:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

Continuous compounding: \$1,000 at 5% for 10 years: A =

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Continuous compounding: \$5,000 at 4% for 8 years: A =

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Depreciation: A \$30,000 car loses 15% per year for 4 years: A =

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Half-life decay: 160 g of an isotope after 4 half-lives: A =

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

Find Present Value of \$10,000 due in 5 years at 6% annual:

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Evaluate hyperbolic sine $\sinh(1) = (e^{1} - e^{-1}) \div 2$:

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Error Analysis: A student multiplied 1000×1.05 first, then raised 1050^2 instead of $1000 \times (1.05)^2$. True value:

[ ] +10 pts [ ] SC-32 Ans: _____

Unit 05 STEM: Global Fiber Optic Bandwidth Expansion

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: GLOBAL FIBER OPTIC BANDWIDTH EXPANSION

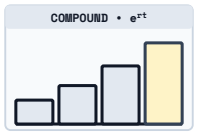
Companion to Vol. II, Ch. 3 & Ch. 11: Network architects and financial engineers model compound traffic growth, TVM cash flows, and optical attenuation using exponential functions.

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: 100-Decade Scaling, Ch. 3: Exponential Growth ([e^x]) & Ch. 11: Fibonacci's [TVM]) • Euler's Base e, Bernoulli's Compounding & Fibonacci's TVM: Jacob Bernoulli discovered $e \approx 2.71828$ by compounding interest every instant (Ch. 3), while Fibonacci's 1202 Liber Abaci launched the time-value-of-money cash-flow equations (Ch. 11)!

Block Studio: Prime Factor Studio & SC-32 Scientific Stack ([y^x], [e^x]) • Stack Shortcut: [1000] [ENTER] [1.05] [ENTER] [10] [y^x] [×] = $1,628.89

CASE INVESTIGATION 1 STEM Modeling

Discrete Backbone Expansion: A 1,000 Gbps node grows 5% annually for 10 years. Find the capacity $A = 1000(1.05)^{10}$:



1. Formula / Stack Plan: _____

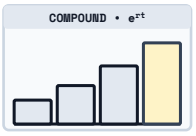
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Continuous Traffic Surge: Model 1,000 units growing continuously at 5% for 10 years ($A = Pe^{rt}$):



1. Formula / Stack Plan: _____

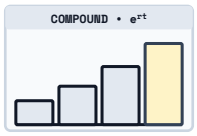
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Optical Laser Attenuation: An 80 mW signal halves every 50 km. Find the power after 3 half-distances:



1. Formula / Stack Plan: _____

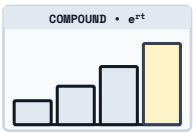
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student enters rate 5% as 5 instead of 0.05. Convert 5% to its decimal rate on the stack:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

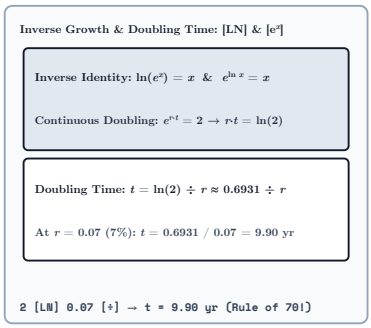
Natural Logarithms, Inverse Exponentials & Doubling Time [LN]

Mathematical Focus: Prime Factor Studio (Logarithmic Height Towers) & SC-32 [LN] • Standards: CCSS.MATH.HSF.BF.B.5, CCSS.MATH.HSF.LE.A.4

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Inverting Exponentials: Solving Doubling Time $t = \ln(2) \div r$):

The Rule of 72 ($72 \div 7 \approx 10.3$ yrs) is an approximation of the exact continuous stack formula $\ln(2)/r$.

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: Slide Rules & Ch. 3: Logarithms, Power Laws & Scaling (Napier 1614, [LN], [LOG])) • John Napier's 20-Year Labor (1614) & Libby's C-14 Clock: John Napier doubled the life of astronomers by turning multiplication into addition via logarithms—the exact principle behind our Prime Factor Log Towers and [LN]!



2. KEY MATHEMATICAL PRINCIPLES: SOLVING FOR UNKNOWN EXPONENTS WITH THE NATURAL LOGARITHM

- The natural logarithm [LN] is the inverse of the natural exponential: $\ln(e^x) = x$ and $e^{\ln x} = x$.
- Logarithmic Block Heights: In the Prime Factor Studio, block height $h(p) = 7.0 \cdot \log_2(p)$ mm, physically turning multiplication into addition!
- To solve $e^{rt} = 2$ for doubling time t: take ln of both sides: $rt = \ln(2) \rightarrow t = \ln(2) \div r$.
- General base change: $\log_b(x) = \ln(x) \div \ln(b)$. On the SC-32, press [x] [LN] [b] [LN] [÷].

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: Place a '2' block (7.0 mm = $7.0 \cdot \log_2(2)$) in the Prime Factor Studio well to model doubling, then solve for continuous doubling time at $r = 0.07$. Question: $\ln(2) \approx 0.69315$. $0.69315 \div 0.07 \approx 9.902$ years. (Prediction: ~9.9 years)

Verification Keystrokes: [2] [LN] [0] [•] [0] [7] [÷]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[2]	2	2 enters Register X
[LN]	0.6931	$\ln(2) = 0.693147$ in Register X
[0] [•] [0] [7]	0.07	Rate r (0.07) enters X; ln(2) lifts to Y
[÷]	9.9024	$\ln(2) \div 0.07 = 9.9024$ years! Exact doubling time

3. PAPER STEPS & SC-32 STACK LEARNING GUIDE & SELF-CHECK

- Mathematical Rule: The natural logarithm [LN] is the inverse of the natural exponential: $\ln(e^x) = x$ and $e^{\ln x} = x$.
- Stack Shortcut & Pattern: [2] [LN] [0.07] [÷] = 9.90 years (Doubling Time at 7%) — General base change: $\log_b(x) = \ln(x) \div \ln(b)$. On the SC-32, press [x] [LN] [b] [LN] [÷].
- 2-Second Paper-to-Screen Audit: Confirm your paper setup and inner values match Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Find the exact doubling time for a population growing at 4% continuously.

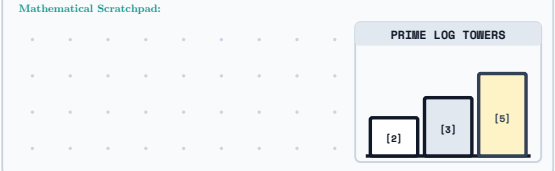
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Evaluating Arbitrary Base Logarithms: $\log_b(x) = \ln(x) \div \ln(b)$

Deepening Concepts • Standards: CCSS.MATH.HSF.BF.B.5

1. EXTENSION CONCEPT & SETUP (Evaluating Arbitrary Base Logarithms: $\log_b(x) = \ln(x) \div \ln(b)$):

The change-of-base theorem converts any base-b logarithm into a simple two-register division on the stack.

Extension Setup: Compare stacking three '2' blocks ($\log_2(8) = 3$) in the Prime Factor Studio with evaluating $\log_{10}(1000)$ via $\ln(1000) \div \ln(10)$. • Question: $\log_{10}(1000) = 3$ because $10^3 = 1000$. $\ln(1000) \div \ln(10)$ must equal exactly 3.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [0] [0] [0] [LN] [1] [0] [LN] [÷]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1] [0] [0] [0]	1000	1000 enters Register X
[LN]	6.9078	ln(1000) = 6.907755 in Register X
[1] [0]	10	10 enters Register X; ln(1000) lifts to Register Y
[LN]	2.3026	ln(10) = 2.302585 in Register X; ln(1000) in Y
[÷]	3	ln(1000) ÷ ln(10) = 3! Base-10 log verified

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1: Believing $\ln(A + B) = \ln(A) + \ln(B)$

Mathematical & Stack Remediation: In the Prime Factor Studio, stacking blocks adds log heights to MULTIPLY values: $\ln(A \times B) = \ln(A) + \ln(B)$, never $\ln(A + B)$.

Misconception 2 — Attempting to Take the Logarithm of a Negative Number: Typing [-5] [LN] produces an error.

Mathematical & Stack Remediation: Exponential functions never output negative numbers, so ln(x) is only defined for $x > 0$.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [2] [LN] [0.07] [÷] = 9.90 years (Doubling Time at 7%)

Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Evaluate $\log_2(64)$ on the stack using natural logarithms.

1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



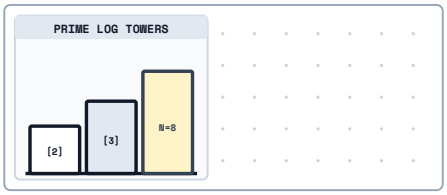
Unit 06 Lab: Natural Logarithms, Inverse Exponentials & Doubling Time [LN]

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the growth/decay curve or timeline in Box 1. (2) Write your exponent/log steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Find the doubling time at 8% continuous interest: $t =$

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

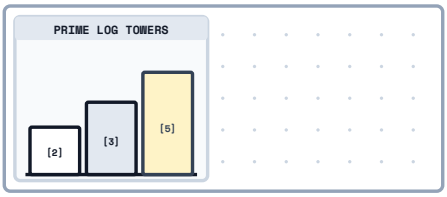
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find the tripling time at 5% continuous growth: $t =$

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

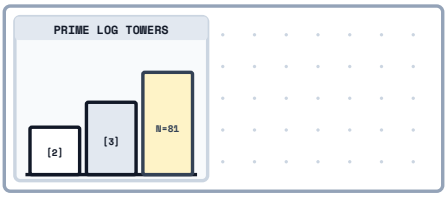
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate $\log_3(81)$ using change of base:

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

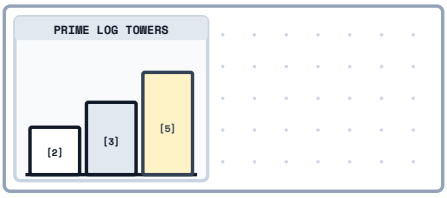
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Solve $e^{0.03\,t} = 5$ for t: $t =$

1. Growth / Decay Timeline Sketch:



2. Exponential / Logarithm Steps:

Exponential / Log Formula:

Exponent or Log Step:

Evaluated Result:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 06 Practice A — John Napier's Logarithms, [LN] & Prime Log Towers

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate $\log_2(32)$ using Prime Factor [2] blocks.

Since $2^5 = 32$ (five [2] blocks stacked), $\log_2(32) = 5$.

Verified Answer: 5

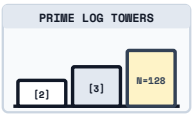
EXAMPLE 2 Evaluate $\log_{10}(1{,}000) + \log_{10}(0.01)$.

$\log_{10}(10^3) + \log_{10}(10^{-2}) = 3 + (-2) = 1$.

Verified Answer: 1

1 Tier 1 • Builder

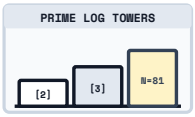
Prime Log Tower: How many [2] blocks stack to reach 128 (find $\log_2(128)$)?



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

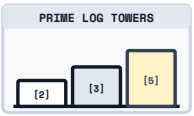
Prime Log Tower: How many [3] blocks stack to reach 81 (find $\log_3(81)$)?



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

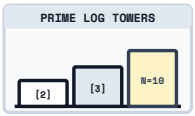
Evaluate natural logarithm on SC-32 (4 dec.): $\ln(2)$ =



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

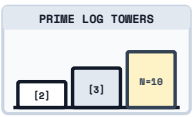
Evaluate natural logarithm on SC-32 (4 dec.): $\ln(10)$ =



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

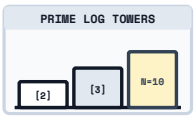
Napier Product Law: Verify $\ln(2) + \ln(5) = \ln(10)$ (4 dec.):



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

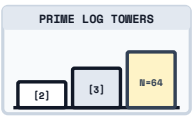
Evaluate base-10 logarithm: $\log_{10}(10{,}000)$ =



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

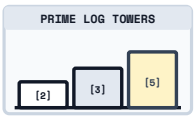
Evaluate base-2 logarithm: $\log_2(64)$ =



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

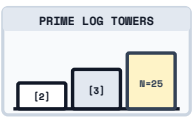
Evaluate natural log identity: $\ln(e^5)$ =



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

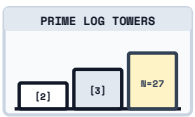
Use log product rule: $\log_{10}(25) + \log_{10}(4)$ =



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Solve the exponential equation: $3^{x-1} = 27$



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Practice B — Logarithmic Inversion, Doubling Time & Half-Life

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the exact continuous tripling time $t = \ln(3) \div r$ at $r = 5\%$ (0.05).

Inside-out: [3] [LN] ≈ 1.0986 → [0.05] [÷] = 21.97 years.

Verified Answer: 21.97 years

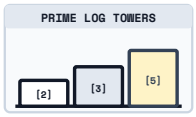
EXAMPLE 2 Find exact continuous doubling time for $r = 10\%$ $(k = 0.10)$.

$t = \ln(2) \div 0.10 \approx 0.6931 \div 0.10 = 6.93$ years.

Verified Answer: 6.93 years

1 Tier 1 • Builder

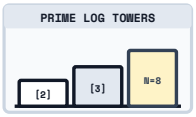
Continuous Doubling Time: At $r = 4\%$ $(k = 0.04)$, find exact $t = \ln(2) \div 0.04$ (2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

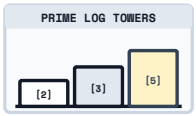
Continuous Doubling Time: At $r = 8\%$ $(k = 0.08)$, find exact $t = \ln(2) \div 0.08$ (2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

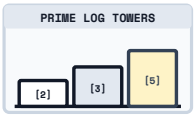
Continuous Tripling Time: At $r = 5\%$ $(k = 0.05)$, find exact $t = \ln(3) \div 0.05$ (2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

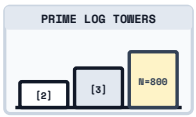
Logarithmic Inversion: Solve $e^{0.05t} = 4$ for $t = \ln(4) \div 0.05$ (2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

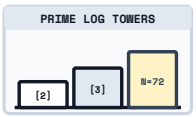
Half-Life Inversion: How many half-lives $n = \log_2(800 \div 100)$ reduce 800 g to 100 g?



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

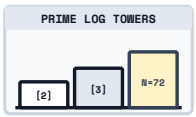
Rule of 72: Estimate doubling time at 6% annual interest:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

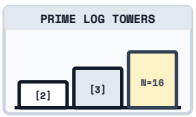
Rule of 72: Estimate doubling time at 9% annual interest:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

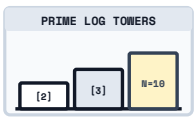
Solve for t using change of base: $2^t = 16$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

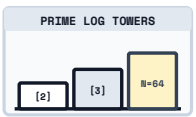
Solve for x using inverse base-10 power: $\log_{10}(x) = 3$



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Change of base: Evaluate $\log_4(64) =$



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Practice C — Real-World Financial, pH & Decibel Applications

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 Sound A is 1,000 times as intense as threshold I_0. Find decibels $dB = 10 \log_{10}(1000)$.

$10 \times \log_{10}(10^3) = 10 \times 3 = 30$ dB.

Verified Answer: 30 dB

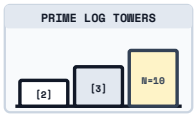
EXAMPLE 2 Spot the Mistake: For $1000 \times (1.05)^2$, Sam multiplied $1000 \times 1.05 = 1050$ and squared 1050.

Exponents come BEFORE multiplication! Square 1.05 first (1.1025), then multiply by 1000.

Verified Answer: 1,102.50

1 Tier 2 • Detective

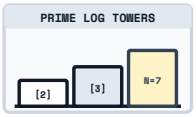
Chemistry pH: Find pH = $-\log_{10}[H^+]$ when hydrogen ion concentration is $[H^+] = 10^{-4}$ M:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

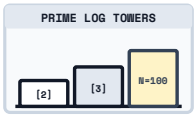
Earthquake Magnitude: How many times more wave amplitude is a Magnitude 7 quake than Magnitude 5?



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

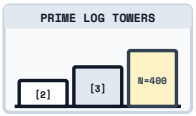
Acoustic Gain: If power increases by a factor of 100, how many decibels does it rise?



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

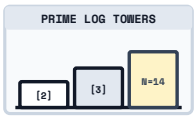
Savings Account: $400 doubles 3 times over a career. What is the final balance?



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

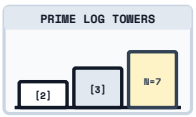
Carbon-14 Decay: A sample starts with 64 mg. How many half-lives until 8 mg remain?



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

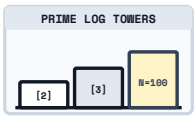
Logarithmic Prime Blocks: Since block [2] has height 7 mm (1U), how tall is a stack representing 16?



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

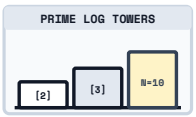
Continuous Growth Initial Value: Evaluate $P(0) = 100 \times e^0$:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

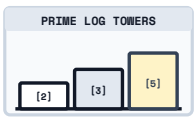
Solve for x: $10^{x+1} = 100{,}000$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

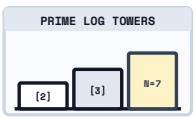
If $\log_{10}(a) = 2$ and $\log_{10}(b) = 3$, what is $\log_{10}(a \times b)$?



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: Leo wrote $\log_{10}(2) + \log_{10}(5) = \log_{10}(7)$. Use the product rule to fix it!



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Practice D — Exponential & Logarithmic Equations and Change of Base

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Solve $4^{x+1} = 64$ by matching bases.

$64 = 4^3 \rightarrow x + 1 = 3 \rightarrow x = 2$.

Verified Answer: $x = 2$

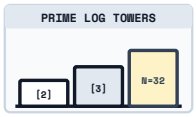
EXAMPLE 2 Solve $\log_2(x) + \log_2(x - 2) = 3$ for $x > 2$.

$\log_2(x(x - 2)) = 3 \rightarrow x^2 - 2x = 8 \rightarrow (x - 4)(x + 2) = 0 \rightarrow x = 4$.

Verified Answer: $x = 4$

1 Tier 2 • Detective

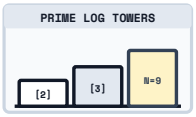
Solve by matching bases or logs: $2^{x+2} = 32$



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

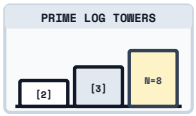
Solve by matching bases or logs: $9^x = 27$



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

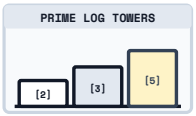
Solve by matching bases or logs: $8^x = 32$



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

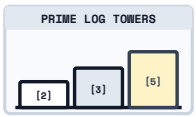
Solve logarithmic equation: $\log_3(x) = 4$



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

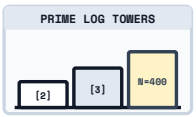
Solve logarithmic equation: $\log_5(2x - 1) = 2$



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

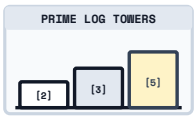
Evaluate using log quotient law: $\log_{10}(400) - \log_{10}(4) =$



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

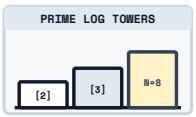
Evaluate using log power law: $3 \times \log_2(4) =$



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

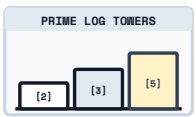
Evaluate telescopic log product: $\log_2(3) \times \log_3(4) \times \log_4(8) =$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

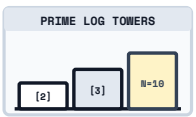
Evaluate inverse identity: $e^{3 \ln 2} =$



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Evaluate inverse identity: $10^{2 \log_10\ 6} =$



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Practice E — Continuous Compounding, Newton's Cooling & Logistic Growth

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate continuous compounding $A = 500\ e^{0.04 \times 5}$ to 2 decimal places.

$0.04 \times 5 = 0.20$; $500 \times e^{0.20} \approx 500 \times 1.22140 = \$610.70$.

Verified Answer: $610.70

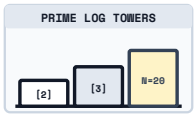
EXAMPLE 2 Newton's Law of Cooling: $T(t) = 20 + 60 \times (0.5)^t$. Find T after $t = 2$ hours.

$20 + 60 \times 0.25 = 20 + 15 = 35°C$.

Verified Answer: 35°C

1 Tier 3 • Challenge

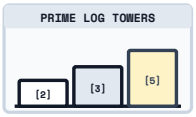
Continuous Growth: Evaluate $1{,}000 \times e^{0.05 \times 20}$ (to 2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

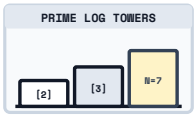
Continuous Doubling Time: At $r = 5\%$ ($k = 0.05$), find the doubling time $t = \ln(2) \div 0.05$ (2 dec):



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

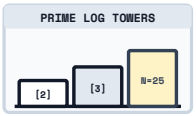
Continuous Doubling Time: At $r = 7\%$ ($k = 0.07$), find the doubling time $t = \ln(2) \div 0.07$ (2 dec):



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

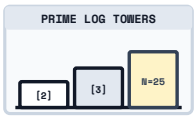
Newton's Cooling: $T(t) = 25 + 80 \times (0.5)^t$. Find $T(3)$:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

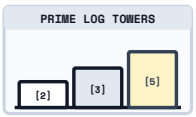
Newton's Cooling Inversion: If $T(t) = 25 + 80 \times (0.5)^t$, after how many hours t does T reach 45°C?



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

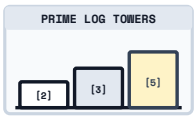
Logistic Initial Population: For $P(t) = 1000 / (1 + 9e^{-t})$, evaluate $P(0) = 1000 \div (1 + 9)$:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

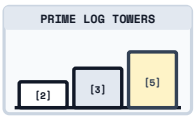
Logistic Inflection Time: For $P(t) = 1000 / (1 + 9e^{-t})$, solve $9e^{-t} = 1$ for $t = \ln(9)$ (2 dec.):



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

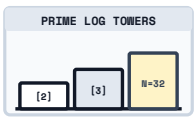
Radioactive Decay: What fraction $(0.5)^4$ of a sample remains after 4 half-lives?



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

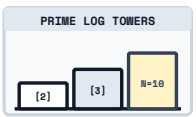
Moore's Law Log Inversion: How many doublings $n = \log_2(32)$ are needed for a 32× transistor increase?



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Half-Life Decay Constant: If half-life is $t_{1/2} = 10$ yr, find $k = \ln(2) \div 10$ (4 dec.):



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Practice F — Scientific Notation, Orders of Magnitude & RPN Pipeline Mastery

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Multiply $(3.0 \times 10^{8}) \times (2.5 \times 10^{-3})$ in scientific notation.

$3.0 \times 2.5 = 7.5$; $10^{8 \cdot -3} = 10^{5} \rightarrow 7.5 \times 10^{5}$.

Verified Answer: 7.5×10^{5}

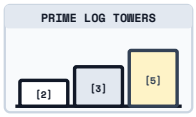
EXAMPLE 2 How many digits does $2^{10} = 1{,}024$ have using $\lfloor \log_{10}(1024) \rfloor + 1$?

$\log_{10}(1024) \approx 3.0103 \rightarrow \lfloor 3.0103 \rfloor + 1 = 4$ digits.

Verified Answer: 4 digits

1 Tier 3 • Challenge

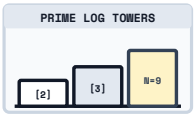
Scientific notation product: $(4 \times 10^{3}) \times (2 \times 10^{4}) =$



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

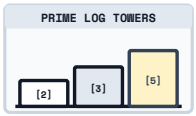
Scientific notation quotient: $(9 \times 10^{8}) \div (3 \times 10^{2}) =$



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

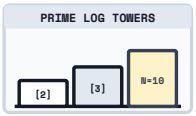
Scientific notation square: $(5 \times 10^{3})^{2} =$



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

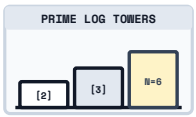
Evaluate $\log_{10}(10^{-9}) =$



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

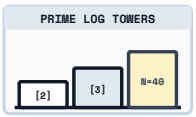
If pH = 3, how many times more acidic is the solution than pH = 6?



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

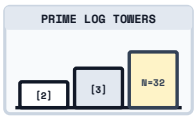
Decibel Difference: Going from 40 dB to 70 dB (+30 dB) multiplies sound power by what factor?



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

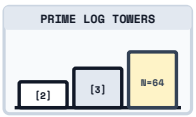
Root-First Rule: Evaluate $32^{3/5} =$



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

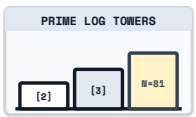
Root-First Rule: Evaluate $64^{2/3} =$



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

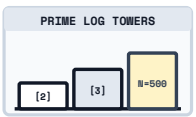
Root-First Rule: Evaluate $81^{3/4} =$



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

RPN Pipeline Check: In $500 \times e^{0.04 \times 5}$, evaluate the exponent 0.04×5 FIRST:



Paper Step: _____ | | SC-32 Ans: _____

Unit 06 Mastery Check: Natural Logarithms & Doubling Time

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Prime Log Tower: How many [2] blocks stack to reach 64 (find $\log_2(64)$)?

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Change of Base on SC-32: Evaluate $\log_3(81) = \ln(81) \div \ln(3)$:

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Evaluate base-10 logarithm: $\log_{10}(1{,}000) =$

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

Evaluate natural logarithm: $\ln(e^4) =$

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Napier Product Law: $\log_2(8) + \log_2(4) =$

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Solve for x: $5^x = 125$

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Logarithmic Doubling Time: A colony of 50 bacteria doubles every hour. How many hours to reach 400?

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

160 mg of medicine halves every 4 hours. How much remains after 8 hours (2 half-lives)?

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

What is the pH = $-\log_{10}[H^+]$ of a solution with $[H^+] = 10^{-7}$ M?

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Error Analysis: Verify $\log(a \cdot b) = \log(a) + \log(b)$ by evaluating $\ln(2) + \ln(3)$ on the stack:

[ ] +10 pts [ ] SC-32 Ans: _____

Unit 06 STEM: Carbon-14 Archeological Dating Laboratory

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: CARBON-14 ARCHEOLOGICAL DATING LABORATORY

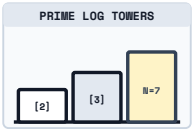
Companion to Vol. II, Ch. 1 & Ch. 3: Geochemists calibrate spectrometer ratios and Willard Libby Carbon-14 decay constants using natural and base-10 logarithms.

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: Slide Rules & Ch. 3: Logarithms, Power Laws & Scaling (Napier 1614, [LN], [LOG])) • John Napier's 20-Year Labor (1614) & Libby's C-14 Clock: John Napier doubled the life of astronomers by turning multiplication into addition via logarithms—the exact principle behind our Prime Factor Log Towers and [LN]!

Mathematical Focus: Prime Factor Studio (Logarithmic Height Towers) & SC-32 [LN] • Stack Shortcut: [2] [LN] [0.07] [÷] = 9.90 years (Doubling Time at 7%)

CASE INVESTIGATION 1 STEM Modeling

Isotope Benchmark Rate: Find the continuous doubling/halving time $t = \ln(2) \div 0.07$ for a 7% rate:



1. Formula / Stack Plan: _____

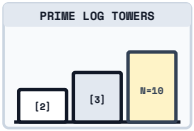
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Decade Ratio Calibration: Evaluate the natural logarithm of a 10× isotope ratio, ln(10):



1. Formula / Stack Plan: _____

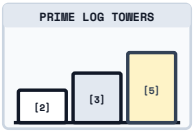
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Spectrometer Base Conversion: Evaluate $\log_{10}(1000)$ using the natural log change-of-base pipeline ln(1000) ÷ ln(10):



1. Formula / Stack Plan: _____

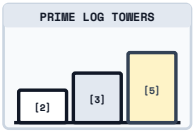
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student tries to calculate ln(-4). Explain the error.



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

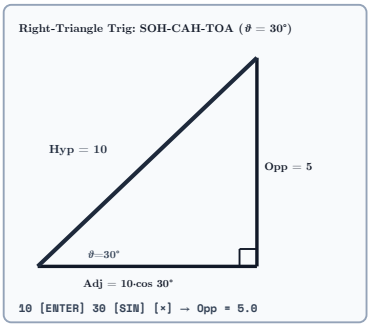
Right-Triangle Trigonometry, Unit Circles & Waveforms

Mathematical Focus: Right-Triangle & Unit-Circle Diagram & SC-32 Trig Keys • Standards: CCSS.MATH.HSG.SRT.C.6, CCSS.MATH.HSG.SRT.C.8

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Finding Right-Triangle Legs with [SIN] and [COS]):

Because [SIN] is a unary postfix operator, it transforms angle 30° into 0.5 in place while 10 waits in Register Y.

Companion Book (Vol. II: Thinking in the Stack — Ch. 4: Circular Trigonometry, Geodetic Triangulation & Hyperbolic Catenaries & Ch. 6: Solid Geometry) • Eratosthenes' Shadow (240 BCE) & The Unit Circle: Using a 7.2° sun shadow at Alexandria, Eratosthenes measured the circumference of the Earth—and right-triangle projections put those exact ratios on the stack!



2. KEY MATHEMATICAL PRINCIPLES: UNDERSTANDING SOH-CAH-TOA ON THE SCIENTIFIC STACK

- Definitions: $\sin(\vartheta)$ = opp / hyp, $\cos(\vartheta)$ = adj / hyp, $\tan(\vartheta)$ = opp / adj.
- To find opposite side: opp = hyp × $\sin(\vartheta)$. On stack: [hyp] [ENTER] [ϑ] [SIN] [×].
- Inverse trig: finding angle ϑ from side ratio: ϑ = atan(opp / adj).
- Degree vs. Radian mode: Ensure calculator is in Degree mode for geometric angles (360° per circle).

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: Sketch a 30°-60°-90° right triangle with hypotenuse 10 m on paper, labeling the vertical opposite leg and horizontal adjacent leg. Question: $\sin(30°) = 0.5$. Opposite side = 10 × $\sin(30°)$ = 5.0 m. (Prediction: 5.0 m)

Verification Keystrokes: [1] [0] [ENTER] [3] [0] [SIN] [×]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[1] [0]	10	Hypotenuse (10) enters Register X
[ENTER]	10	10 lifts to Waiting Shelf Register Y
[3] [0]	30	Angle ϑ (30°) enters Workbench Register X
[SIN]	0.5	$\sin(30°)$ = 0.5000 in Register X
[×]	5	10 × 0.5 = 5.0 m! Opposite side verified

STACK SHORTCUT & PAPER STRATEGY TIP (SC-32):

[10] [ENTER] [30] [SIN] [×] = 5.0 (Opposite Side) • Definitions: $\sin(\vartheta)$ = opp / hyp, $\cos(\vartheta)$ = adj / hyp, $\tan(\vartheta)$ = opp / adj.

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: A 24 m guy wire makes a 30° angle with the ground. Find the tower height.

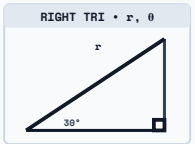
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Inverse Tangent: Finding Angle of Elevation

Deepening Concepts • Standards: CCSS.MATH.HSG.SRT.C.8

1. EXTENSION CONCEPT & SETUP (Inverse Tangent: Finding Angle of Elevation):

Inverse tangent maps the dimensionless slope ratio back into an exact angular elevation.

Extension Setup: Sketch a right triangle on paper with vertical opposite leg = 3 m and horizontal adjacent leg = 4 m to model tan(θ) = opp/adj = 3/4 = 0.75. • Question: Ratio = 3 ÷ 4 = 0.75. Angle θ = atan(0.75) ≈ 36.87°. (Prediction: ~36.9°)

2. EXTENSION STACK SEQUENCE (SC-32): [3] [ENTER] [4] [÷] [↶] [ATAN]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[3]	3	Opposite leg (3) enters Register X
[ENTER]	3	3 lifts to Waiting Shelf Register Y
[4]	4	Adjacent leg (4) enters Workbench Register X
[÷]	0.75	opp ÷ adj = 3 ÷ 4 = 0.75 in Register X
[↶] [ATAN]	36.8699	atan(0.75) = 36.87°! Angle of elevation verified

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1 — Angle Mode Error (Radians vs Degrees): Evaluating sin(30) in Radian mode gives −0.988 instead of 0.5.

Mathematical & Stack Remediation: Verify DEG annunciator is lit on the SC-32 LCD screen.

Misconception 2 — Swapping Opposite and Adjacent Legs in Tangent: Entering adj ÷ opp gives the complement angle.

Mathematical & Stack Remediation: SOH-CAH-TOA mnemonic: TOA is Opposite over Adjacent!

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [10] [ENTER] [30] [SIN] [×] = 5.0 (Opposite Side)

Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: A flagpole 12 m tall casts a 9 m shadow. Find the sun's angle of elevation.

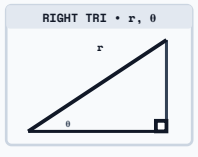
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



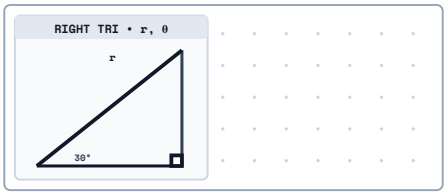
Unit 07 Lab: Right-Triangle Trigonometry, Unit Circles & Waveforms

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the right triangle or coordinate diagram in Box 1. (2) Write your formula steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Find opposite leg: hyp = 20, ϑ = 30°: opp =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

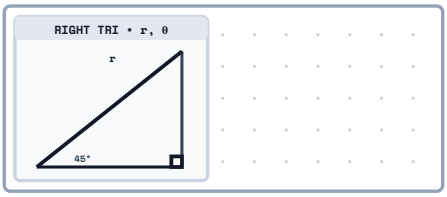
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find adjacent leg: hyp = 50, ϑ = 45°: adj =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

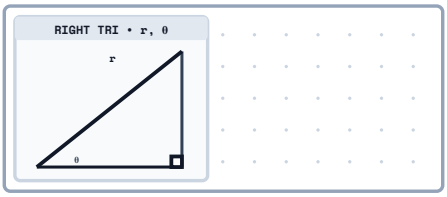
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Find angle ϑ where opp = 5, adj = 12: ϑ =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

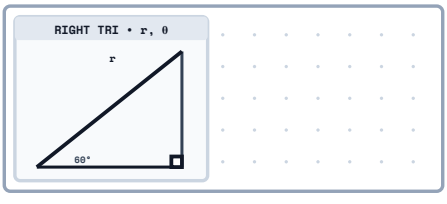
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate tan(60°): tan(60°) =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 07 Practice A — Right-Triangle Trigonometric Ratios [SIN, COS, TAN]

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Evaluate sin(30°) on the SC-32 in Degree mode.

Key: [30] [SIN] = 0.5.

Verified Answer: 0.5

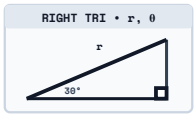
EXAMPLE 2 Evaluate tan(45°) on the SC-32 in Degree mode.

Key: [45] [TAN] = 1.0.

Verified Answer: 1

1 Tier 1 • Builder

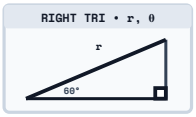
Evaluate exact special angle: sin(30°) =



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

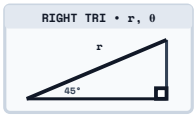
Evaluate exact special angle: cos(60°) =



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

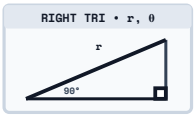
Evaluate exact special angle: tan(45°) =



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

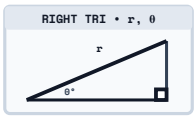
Evaluate special angle: sin(90°) =



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

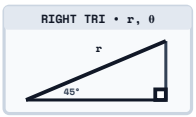
Evaluate special angle: cos(0°) =



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

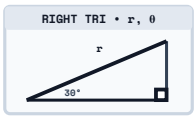
Evaluate to 4 decimal places: sin(45°) =



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

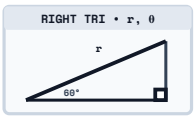
Evaluate to 4 decimal places: cos(30°) =



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

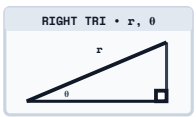
Evaluate to 4 decimal places: tan(60°) =



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

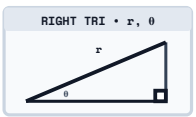
In a 3-4-5 right triangle, find sin(ϑ) for the angle opposite side 3:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

In a 5-12-13 right triangle, find cos(ϑ) for the angle adjacent to side 12:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 07 Practice B — Solving for Unknown Right-Triangle Legs

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Hypotenuse = 20 m, $\vartheta = 30°$. Find opposite leg.**

opp = 20 × sin(30°) = 10 m. Key: [20] [ENTER] [30] [SIN] [×].

Verified Answer: 10 m

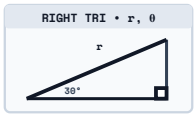
EXAMPLE 2 **Adjacent leg = 15 m, $\vartheta = 35°$. Find opposite leg.**

opp = 15 × tan(35°) = 10.50 m. Key: [15] [ENTER] [35] [TAN] [×].

Verified Answer: 10.5 m

1 Tier 1 • Builder

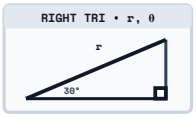
Find opposite leg: hyp = 10, $\vartheta = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

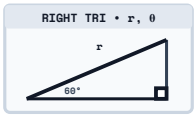
Find opposite leg: hyp = 24, $\vartheta = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

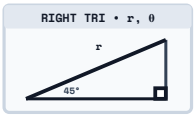
Find adjacent leg: hyp = 40, $\vartheta = 60°$:



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

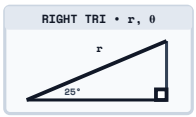
Find adjacent leg: hyp = 50, $\vartheta = 45°$:



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

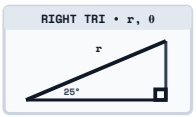
Find opposite leg: hyp = 80, $\vartheta = 25°$:



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

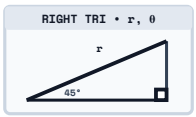
Find adjacent leg: hyp = 80, $\vartheta = 25°$:



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

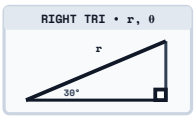
Find opposite leg: adj = 20, $\vartheta = 45°$:



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

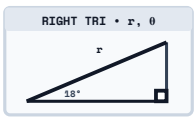
Find opposite leg: adj = 30, $\vartheta = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

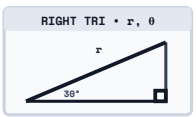
Find opposite leg: adj = 100, $\vartheta = 18°$:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

Find adjacent leg: opp = 15, $\vartheta = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 07 Practice C — Solving for the Hypotenuse & Pythagorean Verification

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 Opposite leg = 12 m, $\vartheta = 30°$. Find hypotenuse.

hyp = 12 ÷ sin(30°) = 24 m. Key: [12] [ENTER] [30] [SIN] [÷].

Verified Answer: 24 m

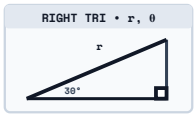
EXAMPLE 2 Adjacent leg = 18 m, $\vartheta = 60°$. Find hypotenuse.

hyp = 18 ÷ cos(60°) = 36 m.

Verified Answer: 36 m

1 Tier 2 • Detective

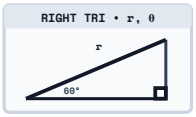
Find hypotenuse: opp = 8, $\vartheta = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 2 • Detective

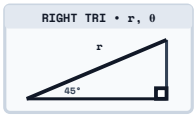
Find hypotenuse: adj = 15, $\vartheta = 60°$:



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 2 • Detective

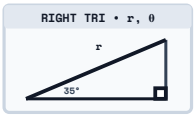
Find hypotenuse: opp = 10, $\vartheta = 45°$:



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 2 • Detective

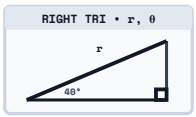
Find hypotenuse: adj = 25, $\vartheta = 35°$:



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 2 • Detective

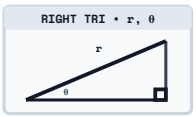
Find hypotenuse: opp = 18, $\vartheta = 40°$:



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 2 • Detective

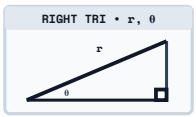
Find the Pythagorean hypotenuse for legs 5 and 12:



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 2 • Detective

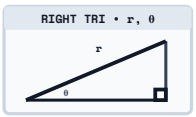
Find the Pythagorean hypotenuse for legs 8 and 15:



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 2 • Detective

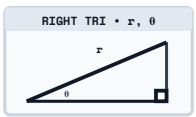
Find the Pythagorean hypotenuse for legs 7 and 24:



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 2 • Detective

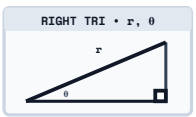
Find the missing Pythagorean leg given hyp = 20 and leg = 16:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 2 • Detective

Find the missing Pythagorean leg given hyp = 26 and leg = 10:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 07 Practice D — Inverse Trigonometry [ASIN, ACOS, ATAN]: Extracting Angles

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Opposite = 3, adjacent = 4. Find angle ϑ.

$\vartheta = \text{atan}(3 \div 4) = 36.87°$. Key: [3] [ENTER] [4] [÷] [ATAN].

Verified Answer: 36.87°

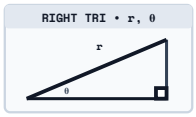
EXAMPLE 2 Opposite = 5, hypotenuse = 10. Find angle ϑ.

$\vartheta = \text{asin}(5 \div 10) = 30°$. Key: [5] [ENTER] [10] [÷] [ASIN].

Verified Answer: 30°

1 Tier 2 • Detective

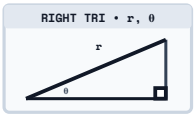
Find angle ϑ where $\sin(\vartheta) = 0.5$:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

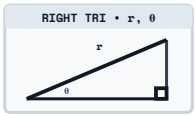
Find angle ϑ where $\cos(\vartheta) = 0.5$:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

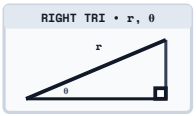
Find angle ϑ where $\tan(\vartheta) = 1$:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

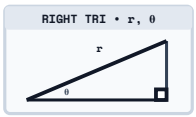
Find angle ϑ in a 3-4-5 right triangle given opp = 3 and adj = 4:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

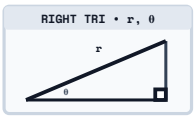
Find the complement angle in a 3-4-5 right triangle given opp = 4 and adj = 3:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

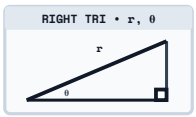
Find angle ϑ in a 5-12-13 right triangle given opp = 5 and adj = 12:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

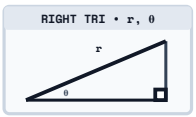
Find angle ϑ in an 8-15-17 right triangle given opp = 8 and hyp = 17:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

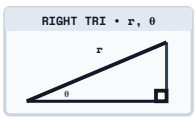
Find angle ϑ given adj = 7 and hyp = 25:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

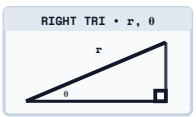
Find angle of elevation for a 12 m flagpole casting a 9 m shadow:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Find the road slope angle ϑ for a 10% grade (rise/run = 0.10):



Paper Step: _____ | | SC-32 Ans: _____

Unit 07 Practice E — Geodetic Surveying, Elevation Angles & Oblique Triangle Area

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 A surveyor stands 200 m from a cliff; angle of elevation is 32°. Cliff height?

200 × tan(32°) = 124.97 m.

Verified Answer: 124.97 m

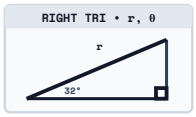
EXAMPLE 2 Find area of triangle with sides $a = 10$, $b = 12$, and included angle $C = 30°$.

Area = 0.5 × 10 × 12 × sin(30°) = 30.

Verified Answer: 30

1 Tier 3 • Challenge

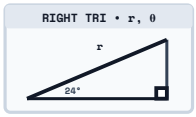
Surveyor 200 m from a tower measures $\vartheta = 32°$. Find the tower height:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

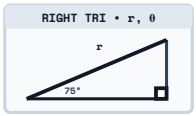
Surveyor 500 m from a mountain peak measures $\vartheta = 24°$ plus 1.5 m eye height:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

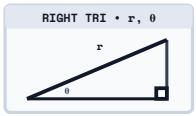
A 10 m ladder leans at 75° to the ground. How high up the wall does it reach?



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

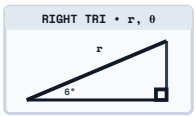
In Problem 3, how far is the base of the 10 m ladder from the wall?



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

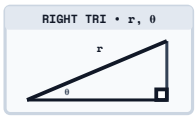
An airplane at altitude 3,000 m descends at a 6° angle. Find the horizontal ground distance:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

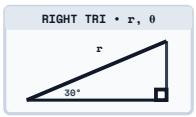
A ramp rises 1.2 m over a 6 m inclined length. Find the ramp angle ϑ:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

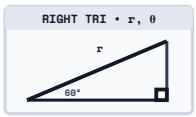
Triangle area $A = 0.5 \times a \times b \times \sin(C)$ for $a = 10$, $b = 12$, $C = 30°$:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

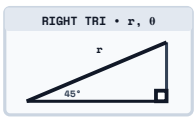
Triangle area $A = 0.5 \times a \times b \times \sin(C)$ for $a = 8$, $b = 15$, $C = 60°$:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

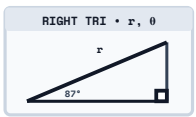
AC voltage $V(t) = 170 \times \sin(45°)$ at phase angle 45°:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Spot the Mistake: Student found complement angle atan(9÷12) = 36.87° instead of atan(12÷9). True ϑ:



Paper Step: _____ | | SC-32 Ans: _____

Unit 07 Practice F — Law of Sines, Law of Cosines & Trigonometric Identities

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Verify Pythagorean identity $\sin^2(37°) + \cos^2(37°) = 1$.

[37] [SIN] [x^2] [37] [COS] [x^2] [+] = 1.0.

Verified Answer: 1

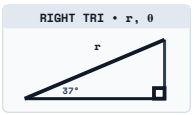
EXAMPLE 2 Law of Cosines: $c = \sqrt{a^2 + b^2 - 2ab\cos C}$ for a=5, b=8, C=60°.

$\sqrt{25 + 64 - 80 \times 0.5} = \sqrt{49} = 7$.

Verified Answer: 7

1 Tier 3 • Challenge

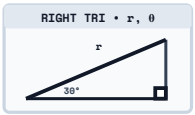
Verify Pythagorean identity: $\sin^2(37°) + \cos^2(37°) =$



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 3 • Challenge

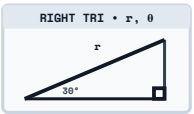
Verify double-angle formula: $2 \times \sin(30°) \times \cos(30°) - \sin(60°) =$



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 3 • Challenge

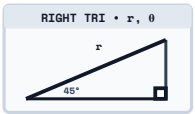
Verify double-angle cosine: $\cos^2(30°) - \sin^2(30°) =$



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 3 • Challenge

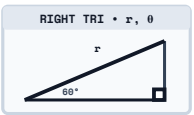
Law of Sines: find $b = a \times \sin(B) \div \sin(A)$ for $a = 12$, $B = 45°$, $A = 30°$:



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 3 • Challenge

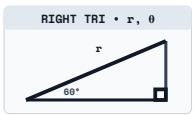
Law of Sines: find $b = a \times \sin(B) \div \sin(A)$ for $a = 20$, $B = 60°$, $A = 45°$:



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 3 • Challenge

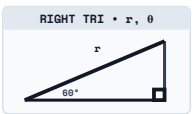
Law of Cosines: $c = \sqrt{a^2 + b^2 - 2ab\cos C}$ for $a = 5$, $b = 8$, C = 60°:



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 3 • Challenge

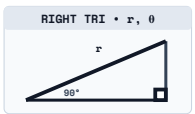
Law of Cosines: $c = \sqrt{a^2 + b^2 - 2ab\cos C}$ for $a = 6$, $b = 10$, C = 60°:



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 3 • Challenge

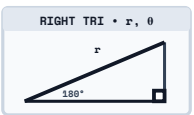
Law of Cosines: $c = \sqrt{a^2 + b^2 - 2ab\cos C}$ for $a = 3$, $b = 4$, C = 90°:



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 3 • Challenge

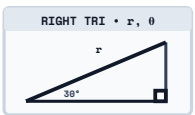
Convert 180° to radians (to 4 decimal places):



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 3 • Challenge

Cofunction Identity Challenge: On the unit circle, evaluate $\sin(30°) + \cos(60°)$:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 07 Mastery Check: Trigonometric Functions [SIN, COS, TAN]

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Evaluate special angle ratio: sin(30°) =

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Evaluate special angle ratio: tan(60°) =

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Find opposite leg: hyp = 20, $\vartheta = 30°$: opp =

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Find adjacent leg: hyp = 50, $\vartheta = 45°$: adj =

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Find hypotenuse: opp = 15, $\vartheta = 30°$: hyp =

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Find angle of elevation ϑ where opp = 3, adj = 4:

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

Find angle ϑ where opp = 5, adj = 12:

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

A surveyor 100 m from a building measures $\vartheta = 40°$. Find the building height:

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Law of Cosines: find side c for $a = 5$, $b = 8$, and included angle $C = 60°$:

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: Verify $\sin^2(25°) + \cos^2(25°)$ on the stack (must equal 1, not sin 50°):

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 07 STEM: Geodetic Mountain Peak Triangulation

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: GEODETIC MOUNTAIN PEAK TRIANGULATION

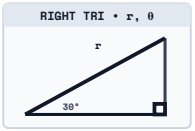
Companion to Vol. II, Ch. 4 & Ch. 6: Topographical surveyors triangulate summit elevations using Eratosthenes sightlines and right-triangle trigonometry.

Companion Book (Vol. II: Thinking in the Stack — Ch. 4: Circular Trigonometry, Geodetic Triangulation & Hyperbolic Catenaries & Ch. 6: Solid Geometry) • Eratosthenes' Shadow (240 BCE) & The Unit Circle: Using a 7.2° sun shadow at Alexandria, Eratosthenes measured the circumference of the Earth—and right-triangle projections put those exact ratios on the stack!

Mathematical Focus: Right-Triangle & Unit-Circle Diagram & SC-32 Trig Keys • Stack Shortcut: [10] [ENTER] [30] [SIN] [×] = 5.0 (Opposite Side)

CASE INVESTIGATION 1 STEM Modeling

Laser Sightline Rise: A 10 m calibration mast is sighted at $\vartheta = 30°$. Find the vertical rise opp = 10·sin(30°):



1. Formula / Stack Plan: _____

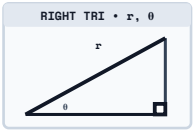
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Ridge Slope Angle: A ridge rises 3 m over a 4 m horizontal run. Find the angle of elevation ϑ = atan(3 ÷ 4):



1. Formula / Stack Plan: _____

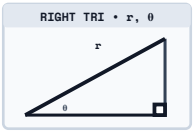
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Direct Line-of-Sight Distance: Survey markers stand 5 km East and 12 km North. Find the hypotenuse distance:



1. Formula / Stack Plan: _____

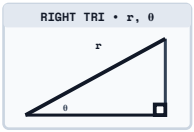
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Calculator in Radian mode gives sin(30) = -0.988.



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

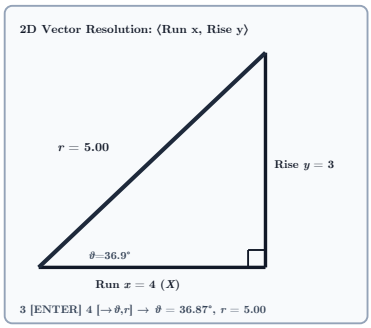
2D Vectors, Polar Conversions & Resultant Forces

Mathematical Focus: 2D Vector Right-Triangle Diagram & SC-32 [→θ,r] • Standards: CCSS.MATH.HSN.VM.A.1, CCSS.MATH.HSN.VM.A.3

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Rectangular-to-Polar Conversion with [→θ,r]):

On the SC-32, [→θ,r] takes y in Register Y and x in Register X, placing magnitude r in X and angle θ in Y.

Companion Book (Vol. II: Thinking in the Stack — Ch. 4: Surveying & Aviation Wind Triangles ([→θ,r], [→y,x]) & Ch. 5: Steinmetz's AC Phasors ([CMPLX])) • Steinmetz's AC Phasor Vectors & The Wind Triangle: Charles Proteus Steinmetz tamed AC power in 1893 by treating waves as 2D vectors—and pressing [→θ,r] resolves (x, y) into magnitude r and angle θ in one tap!



2. KEY MATHEMATICAL PRINCIPLES: DECOMPOSING AND RESOLVING VECTORS ON THE STACK

- Vectors have magnitude r and direction θ: $V = \langle x, y \rangle = \langle r \cos \theta, r \sin \theta \rangle$.
- Pythagorean magnitude: $r = \sqrt{x^2 + y^2}$; heading angle: $\theta = \text{atan}(y / x)$.
- SC-32 Dedicated Converter [→θ,r]: Enter y into Y, x into X, and press [→θ,r] to resolve magnitude r (in X) and angle θ (in Y)!
- Vector addition: add vertical components (Σy) into Y and horizontal components (Σx) into X, then convert to polar resultant.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: Sketch a 2D vector triangle with horizontal East component $x = 60$ m and vertical North component $y = 80$ m; enter $y = 80$ into Waiting Shelf Y and $x = 60$ into Workbench X. Question: $r = \sqrt{60^2 + 80^2} = 100$ m; $\theta = \text{atan}(80/60) \approx 53.13°$.

Verification Keystrokes: [8] [0] [ENTER] [6] [0] [↶] [▸θ,r]

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
[8] [0]	80	Vertical component y (80) enters Register X
[ENTER]	80	$y = 80$ lifts to Waiting Shelf Register Y
[6] [0]	60	Horizontal component x (60) enters Register X
[↶] [▸θ,r]	100	Press [→θ,r]: $r = 100$ m in X, $\theta = 53.13°$ in Y!

3. PAPER STEPS & SC-32 STACK LEARNING GUIDE & SELF-CHECK

- Mathematical Rule: Vectors have magnitude r and direction θ: $V = \langle x, y \rangle = \langle r \cos \theta, r \sin \theta \rangle$.
- Stack Shortcut & Pattern: [4] [ENTER] [3] [→θ,r] → $r = 5.0$ (X), $\theta = 53.13°$ (Y) — Vector addition: add vertical components (Σy) into Y and horizontal components (Σx) into X, then convert to polar resultant.
- 2-Second Paper-to-Screen Audit: Confirm your paper setup and inner values match Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: A surveyor walks 60 m East ($x = 60$) and 80 m North ($y = 80$). Find straight-line distance r and angle θ.

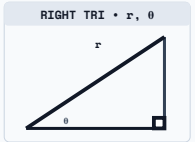
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



Extension: Resolving Resultant Force Vectors: V1 + V2

Deepening Concepts • Standards: CCSS.MATH.HSN.VM.A.3

1. EXTENSION CONCEPT & SETUP (Resolving Resultant Force Vectors: V1 + V2):

Computing Σy first leaves Σy in Register Y and Σx in Register X, ready for [→ϑ,r] with zero register swaps.

Extension Setup: Sketch tip-to-tail vectors $V_1 = \langle 10, 15\rangle$ and $V_2 = \langle 20, 25\rangle$; sum vertical components $\Sigma y = 15 + 25 = 40$ into Waiting Shelf Y and horizontal components $\Sigma x = 10 + 20 = 30$ into Workbench X. • Question: Total $y = 15 + 25 = 40$ N. Total $x = 10 + 20 = 30$ N. Resultant $r = \sqrt{30^2 + 40^2} = 50$ N, $\vartheta \approx 53.13°$.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [5] [ENTER] [2] [5] [+] [1] [0] [ENTER] [2] [0] [+] [↖] [▸ϑ,r]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1] [5]	15	$F1_y$ (15) enters Register X
[ENTER]	15	15 in Y
[2] [5]	25	$F2_y$ (25) in X
[+]	40	$\Sigma y = 40$ N in Register X
[1] [0]	10	$F1_x$ (10) enters; 40 lifts to Register Y
[ENTER]	10	10 in Y; 40 in Z
[2] [0]	20	$F2_x$ (20) in X
[+]	30	$\Sigma x = 30$ N in Register X; $\Sigma y = 40$ waits in Register Y
[↖] [▸ϑ,r]	50	Press [→ϑ,r]: Resultant $r = 50$ N, $\vartheta = 53.13°$!

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1 — Entering Coordinates in Reverse Order for [→ϑ,r]: Entering x before y swaps the angle relative to the axes.

Mathematical & Stack Remediation: Always enter Rise y into Waiting Shelf Y first and Run x into Workbench X.

Misconception 2 — Adding Vector Magnitudes Directly: Claiming $|\langle 3,0\rangle + \langle 0,4\rangle| = 3 + 4 = 7$ instead of 5.

Mathematical & Stack Remediation: Vectors add component-wise (Σx and Σy), not scalar-wise (triangle inequality).

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [4] [ENTER] [3] [→ϑ,r] → $r = 5.0$ (X), $\vartheta = 53.13°$ (Y)

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Add vectors $A = \langle 6, 8\rangle$ and $B = \langle 2, 7\rangle$. Find resultant magnitude and angle.

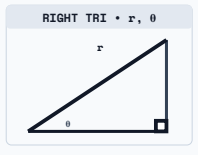
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



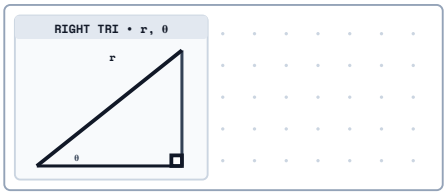
Unit 08 Lab: 2D Vectors, Polar Conversions & Resultant Forces

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch the right triangle or coordinate diagram in Box 1. (2) Write your formula steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Convert ⟨5, 12⟩ to polar: r =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

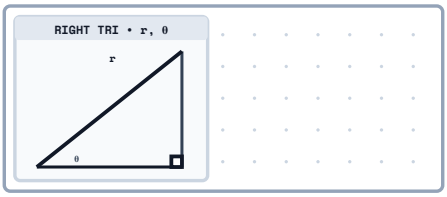
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Find heading angle for ⟨5, 12⟩: ϑ =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

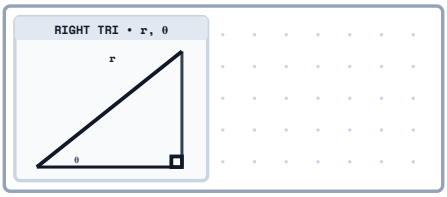
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Convert ⟨8, 6⟩ to polar: r =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

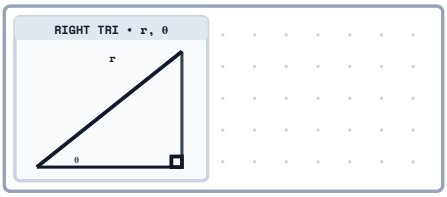
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Add vectors ⟨4, 3⟩ and ⟨8, 6⟩. Find magnitude r =

1. Right Triangle / Vector Sketch:



2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 08 Practice A — Pythagorean Theorem, Distance & 2D Vectors

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the magnitude r of vector ($\Delta x = 3$, $\Delta y = 4$).

$r = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Key: [4] [ENTER] [3] [$\rightarrow \theta, r$].

Verified Answer: 5

EXAMPLE 2 Find the distance between (1, 2) and (7, 10).

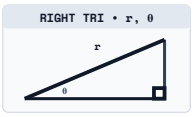
$\Delta x = 6$, $\Delta y = 8 \rightarrow d = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

Verified Answer: 10

1

Tier 1 • Builder

Find the hypotenuse of a right triangle with legs $a = 6$ and $b = 8$:

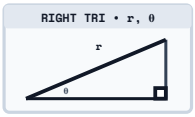


Paper Step: _____ [ ] SC-32 Ans: _____

2

Tier 1 • Builder

Find the hypotenuse of a right triangle with legs $a = 5$ and $b = 12$:

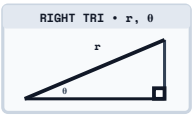


Paper Step: _____ [ ] SC-32 Ans: _____

3

Tier 1 • Builder

Find the hypotenuse of a right triangle with legs $a = 8$ and $b = 15$:

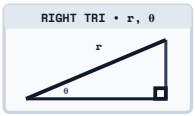


Paper Step: _____ [ ] SC-32 Ans: _____

4

Tier 1 • Builder

Find the missing leg b if leg $a = 9$ and hypotenuse $c = 15$:

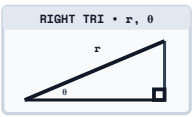


Paper Step: _____ [ ] SC-32 Ans: _____

5

Tier 1 • Builder

Find the missing leg a if leg $b = 24$ and hypotenuse $c = 25$:

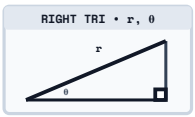


Paper Step: _____ [ ] SC-32 Ans: _____

6

Tier 1 • Builder

Find the distance between (0, 0) and (-9, 12):

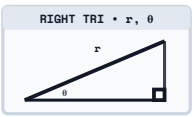


Paper Step: _____ [ ] SC-32 Ans: _____

7

Tier 1 • Builder

Find the distance between (2, -1) and (5, 3):

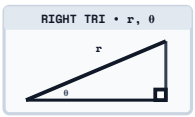


Paper Step: _____ [ ] SC-32 Ans: _____

8

Tier 1 • Builder

Find the midpoint M of the segment connecting (2, 8) and (10, 4):

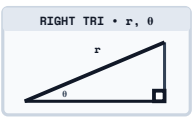


Paper Step: _____ [ ] SC-32 Ans: _____

9

Tier 1 • Builder

Find the slope $m = (y_2 - y_1)/(x_2 - x_1)$ through (1, 3) and (5, 11):

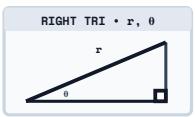


Paper Step: _____ [ ] SC-32 Ans: _____

10

Tier 1 • Builder

Add the 2D vectors $u = \langle 4, -2 \rangle$ and $v = \langle 3, 7 \rangle$:



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 08 Practice B — Polar-Rectangular Conversion [→ϑ,r] & [→y,x] on the Stack

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 Convert polar ($r = 10$, $\vartheta = 60°$) to rectangular (x, y).

$x = 10\cos(60°) = 5.0$; $y = 10\sin(60°) \approx 8.6603$.

Verified Answer: $x = 5$, $y = 8.66$

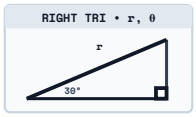
EXAMPLE 2 Convert rectangular vector ⟨3, 4⟩ to polar (r, ϑ) using [4] [ENTER] [3] [→ϑ,r].

Register X shows $r = 5.0$; Register Y holds $\vartheta \approx 53.13°$.

Verified Answer: $r = 5$, $\vartheta = 53.13°$

1 Tier 1 • Builder

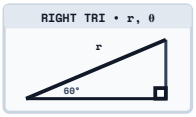
Polar to Rectangular: If $r = 10$ and $\vartheta = 30°$, find $y = r{\cdot}\sin(30°)$:



Paper Step: _____ [ ] SC-32 Ans: _____

2 Tier 1 • Builder

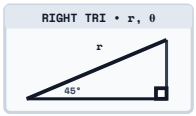
Polar to Rectangular: If $r = 12$ and $\vartheta = 60°$, find $x = r{\cdot}\cos(60°)$:



Paper Step: _____ [ ] SC-32 Ans: _____

3 Tier 1 • Builder

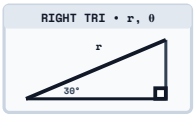
Polar to Rectangular: If $r = 20$ and $\vartheta = 45°$, find $x = r{\cdot}\cos(45°)$ (2 dec.):



Paper Step: _____ [ ] SC-32 Ans: _____

4 Tier 1 • Builder

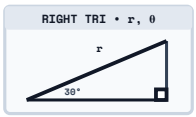
Polar to Rectangular: If $r = 50$ and $\vartheta = 30°$, find $x = r{\cdot}\cos(30°)$ (2 dec.):



Paper Step: _____ [ ] SC-32 Ans: _____

5 Tier 1 • Builder

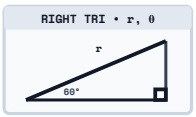
If $r = 20$ and $\vartheta = 30°$, find the vertical component $y = r{\cdot}\sin(30°)$:



Paper Step: _____ [ ] SC-32 Ans: _____

6 Tier 1 • Builder

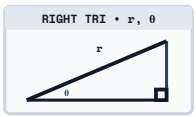
If $r = 14$ and $\vartheta = 60°$, find the horizontal component $x = r{\cdot}\cos(60°)$:



Paper Step: _____ [ ] SC-32 Ans: _____

7 Tier 1 • Builder

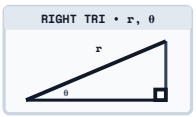
Find the angle ϑ (in degrees) of the vector ($\Delta x = 5$, $\Delta y = 5$):



Paper Step: _____ [ ] SC-32 Ans: _____

8 Tier 1 • Builder

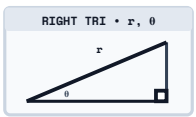
Rectangular to Polar: Find magnitude r of ⟨-5, 12⟩ using [12] [ENTER] [-5] [→ϑ,r]:



Paper Step: _____ [ ] SC-32 Ans: _____

9 Tier 1 • Builder

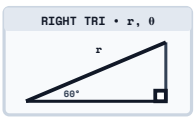
Rectangular to Polar: Find magnitude r of ⟨9, -12⟩ using [-12] [ENTER] [9] [→ϑ,r]:



Paper Step: _____ [ ] SC-32 Ans: _____

10 Tier 1 • Builder

Law of Cosines Resultant: Find c if $a = 8$, $b = 5$, and angle $C = 60°$ ($c^2 = a^2+b^2-2ab\cos 60°$):



Paper Step: _____ [ ] SC-32 Ans: _____

Unit 08 Practice C — Surveying, Navigation Vectors & Mode Error Check

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 A drone flies 12 km East and 5 km North. What is its straight-line distance from base?

$r = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ km.

Verified Answer: 13 km

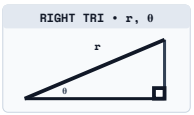
EXAMPLE 2 Spot the Mistake: Noah computed sin(30) and got -0.9880 instead of 0.5000.

His calculator was in RADIAN mode! Switch to DEG mode so sin(30°) = 0.5.

Verified Answer: 0.5 (DEG mode)

1 Tier 2 • Detective

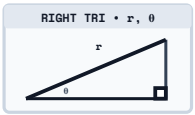
A 10-meter ladder leans against a wall with its base 6 meters from the wall. How high does it reach?



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

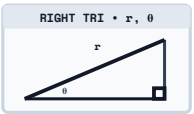
A ramp rises 3 feet over a horizontal run of 4 feet. How long is the inclined ramp surface?



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

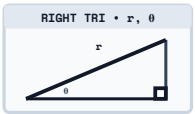
A boat sails 8 miles North, then 6 miles West. How far is it straight-line from the harbor?



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

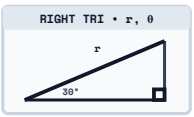
A guy wire is attached 12 m up a tower and anchored 9 m from its base. Length of the wire?



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

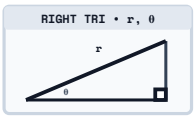
A force of 50 N pulls at $\vartheta = 30°$ above horizontal. What is the vertical lift component?



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

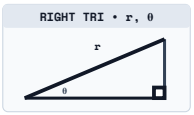
Find the 2D dot product $u \cdot v = x_1x_2 + y_1y_2$ for $u = \langle 3, 4 \rangle$ and $v = \langle 4, -3 \rangle$:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

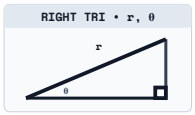
Find the 3D space diagonal $d = \sqrt{l^2 + w^2 + h^2}$ of a box measuring 3 m × 4 m × 12 m:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

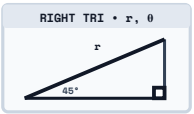
What is the maximum amplitude $R = \sqrt{a^2 + b^2}$ of the wave $y = 5\sin(t) + 12\cos(t)$?



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

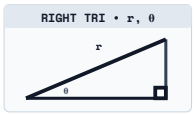
A solar panel tilted at 45° casts a horizontal shadow of 4.2 m. What is the vertical height?



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: To find hypotenuse of legs 3 and 4, Sam computed $\sqrt{3 + 4} = \sqrt{7}$. Fix it!



Paper Step: _____ | | SC-32 Ans: _____

Unit 08 Practice D — Aviation Wind Triangles, Steinmetz Phasors & Vector Superposition

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 A plane flies North at 120 kt in a 50 kt East crosswind. Find ground speed r.

$v = \langle 50, 120\rangle \rightarrow r = \sqrt{50^2 + 120^2} = 130$ kt. Key: [120] [ENTER] [50] [→,θ,r].

Verified Answer: 130 kt

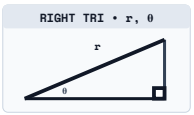
EXAMPLE 2 Steinmetz AC Impedance: Resistance $R = 8\ \Omega$ and Reactance $X = 6\ \Omega$. Find impedance $Z = \sqrt{R^2 + X^2}$.

$Z = \sqrt{8^2 + 6^2} = 10\ \Omega$. Key: [6] [ENTER] [8] [→,θ,r].

Verified Answer: 10 Ω

1 Tier 2 • Detective

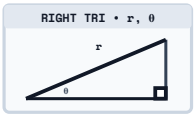
Aviation Wind Triangle: Airspeed 80 kt North, crosswind 60 kt East. Ground speed r?



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

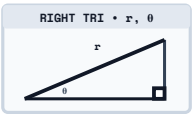
Aviation Wind Triangle: Airspeed 240 kt East, crosswind 70 kt North. Ground speed r?



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

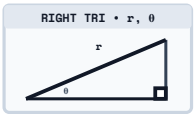
River Crossing: Boat rows East at 4 m/s across a 3 m/s South current. Resultant speed?



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

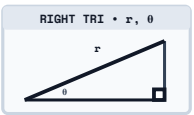
Steinmetz AC Phasor: Resistance $R = 12\ \Omega$, inductive reactance $X_L = 5\ \Omega$. Find Z:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

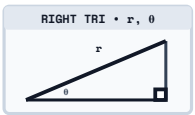
Steinmetz AC Phasor: Resistance $R = 15\ \Omega$, capacitive reactance $X_C = -8\ \Omega$. Find Z:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

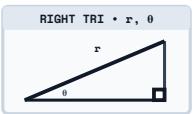
Force Superposition: Combine $F_1 = \langle 2, 6\rangle$ N and $F_2 = \langle 4, 2\rangle$ N and find magnitude $|F_1+F_2|$:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

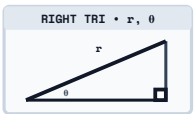
Force Superposition: Combine $F_1 = \langle 10, -3\rangle$ N and $F_2 = \langle 5, 11\rangle$ N and find $|F_1+F_2|$:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

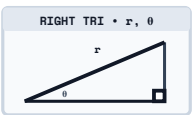
Equilibrant Vector: If net force is $\langle 6, -8\rangle$ N, what is the magnitude of the balancing force?



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

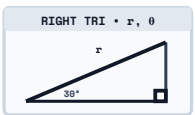
Law of Sines Vector Triangle: If $a = 8$, $\sin(A) = 0.4$, and $\sin(B) = 0.6$, find side b:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Parallelogram Area from Vectors: Find Area $= a{\cdot}b{\cdot}\sin(30°)$ for $|a|=10$, $|b|=12$:



Paper Step: _____ | | SC-32 Ans: _____

Unit 08 Practice E — 2D Vector Projections, Resultant Forces & Dot Product Angles

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the resultant magnitude of $u = \langle 2, 5\rangle$ and $v = \langle 4, 3\rangle$.

$u + v = \langle 6, 8\rangle \rightarrow |u + v| = \sqrt{6^2 + 8^2} = 10$.

Verified Answer: 10

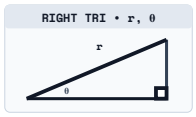
EXAMPLE 2 Find the angle between $u = \langle 3, 0\rangle$ and $v = \langle 2, 2\rangle$.

$\cos\theta = (u \cdot v)/(|u||v|) = 6 / (3 \times 2\sqrt{2}) = 1/\sqrt{2} \rightarrow \theta = 45°$.

Verified Answer: 45°

1 Tier 3 • Challenge

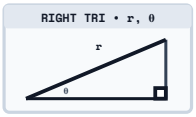
Resultant vector magnitude: $|\langle 1, 2\rangle + \langle 2, 2\rangle| =$



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

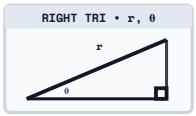
Resultant vector magnitude: $|\langle 3, 7\rangle + \langle 2, 5\rangle| =$



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

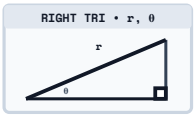
Vector difference magnitude: $|\langle 10, 9\rangle - \langle 2, -6\rangle| =$



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

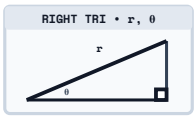
Scalar multiple magnitude: $|3 \times \langle 3, 4\rangle| =$



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

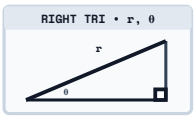
Dot product: $\langle 4, 3\rangle \cdot \langle 5, -2\rangle =$



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

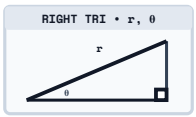
Orthogonal check: Find k so that $\langle 6, 3\rangle$ and $\langle 2, k\rangle$ are perpendicular (dot = 0):



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

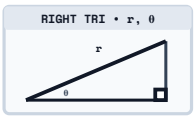
Unit vector: What is the x-component of the unit vector in the direction of $\langle 3, 4\rangle$?



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

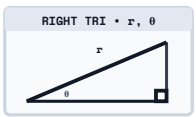
Work Done: Force $\boldsymbol{F} = \langle 10, 4\rangle$ N moves an object $d = \langle 5, 2\rangle$ m. Find Work $\boldsymbol{W} = \boldsymbol{F} \cdot \boldsymbol{d}$:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

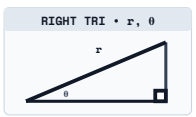
Horizontal Equilibrium: Two forces $\langle 8, 3\rangle$ and $\langle -8, 9\rangle$ act on a point. Net vertical force?



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

3D Vector Magnitude: Find $|\langle 2, 3, 6\rangle| = \sqrt{2^2 + 3^2 + 6^2}$:



Paper Step: _____ | | SC-32 Ans: _____

Unit 08 Practice F — AP Physics / Precalculus Wave Amplitudes & Polar Navigation

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the maximum value (amplitude R) of $f(t) = 8\sin(t) + 15\cos(t)$.

$R = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$. Key: [15] [ENTER] [8] [→ϑ,r].

Verified Answer: 17

EXAMPLE 2 Find the period $T = 360° \div k$ of the wave $y = 4\sin(3x)$.

$T = 360° \div 3 = 120°$ (or $2\pi/3$ radians).

Verified Answer: 120° ($2\pi/3$)

1 Tier 3 • Challenge

Wave Amplitude: Find max value of 6 sin(x) + 8 cos(x):

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Wave Amplitude: Find max value of 7 sin(x) - 24 cos(x):

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

Wave Period: Find the period (in degrees) of $y = \cos(6x)$:

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Equilateral Triangle Area: Find exact area of side $s = 4$ ($A = (\sqrt{3}/4)s^2$):

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Heron's Formula: Find the area of a triangle with sides 3, 4, 5 (semi-perimeter $s = 6$):

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Inscribed Right Triangle: A right triangle with legs 6 and 8 is inscribed in a circle. Radius $R = c/2$?

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Polar Angle: Find the angle ϑ (to 2 dec. places) of vector $\langle 4, 3\rangle$:

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Polar Angle: Find the angle ϑ (to 2 dec. places) of vector $\langle 3, 4\rangle$:

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Angular Velocity: A wheel turns 6 revolutions per second. How many π-radians per second (find k in $k\pi$)?

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Polar Navigation Challenge: Load $\Delta y = 4$ in Register Y and $\Delta x = 3$ in Register X, then convert to polar magnitude r with [→ϑ,r]:

RIGHT TRI • r, θ

Paper Step: _____ | | SC-32 Ans: _____

Unit 08 Mastery Check: 2D Vectors & Polar Coordinates

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Find the hypotenuse for legs $a = 9$ and $b = 12$:

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Find the distance between (1, 1) and (4, 5):

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Resultant magnitude of $\langle 2, 5\rangle + \langle 4, 3\rangle$ using $[\rightarrow \vartheta, r]$:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

If $r = 16$ and $\vartheta = 30°$, find $y = r\cdot\sin(30°)$:

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Find the angle ϑ in degrees when $\Delta x = 10$ and $\Delta y = 10$:

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Evaluate the dot product of $\langle 2, 5\rangle$ and $\langle 3, 4\rangle$:

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

A hiker walks 5 km South and 12 km East. What is the straight-line distance back to camp?

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

A 17-ft beam props up a wall 15 ft high. How far is the base of the beam from the wall?

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Find the maximum value of 3 sin(x) + 4 cos(x):

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Error Analysis: Why is the hypotenuse of a 3-4-5 right triangle shorter than walking $3 + 4 = 7$?

[ ] +10 pts [ ] SC-32 Ans: _____

Unit 08 STEM: Commercial Aircraft Crosswind Navigation

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: COMMERCIAL AIRCRAFT CROSSWIND NAVIGATION

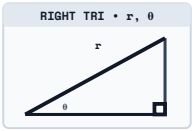
Companion to Vol. II, Ch. 4 & Ch. 5: Flight navigators and electrical engineers compute wind triangles and Steinmetz AC phasors using 2D vector resolution and $[\rightarrow \theta, r]$.

Companion Book (Vol. II: Thinking in the Stack — Ch. 4: Surveying & Aviation Wind Triangles ($[\rightarrow \theta, r]$, $[\rightarrow y, x]$) & Ch. 5: Steinmetz's AC Phasors ([CMPLX])) • Steinmetz's AC Phasor Vectors & The Wind Triangle: Charles Proteus Steinmetz tamed AC power in 1893 by treating waves as 2D vectors—and pressing $[\rightarrow \theta, r]$ resolves (x, y) into magnitude r and angle θ in one tap!

Mathematical Focus: 2D Vector Right-Triangle Diagram & SC-32 $[\rightarrow \theta, r]$ • Stack Shortcut: [4] [ENTER] [3] $[\rightarrow \theta, r] \rightarrow r = 5.0\ (X),\ \theta = 53.13°\ (Y)$

CASE INVESTIGATION 1 STEM Modeling

Drone Test Flight: An aircraft drifts 3 km East (X) and 4 km North (Y). Convert ⟨3, 4⟩ to polar $[\rightarrow \theta, r]$:



1. Formula / Stack Plan: _____

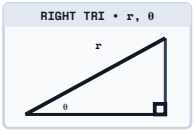
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Jet Crosswind Leg: A jet travels 5 km East (X) and 12 km North (Y). Convert ⟨5, 12⟩ to polar $[\rightarrow \theta, r]$:



1. Formula / Stack Plan: _____

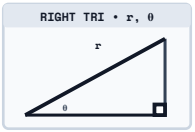
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Combined Thrust & Wind: Add force vectors ⟨10, 15⟩ and ⟨20, 25⟩ and find the resultant magnitude r:



1. Formula / Stack Plan: _____

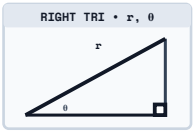
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student adds magnitudes $3 + 4 = 7$ instead of vector sum 5.



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

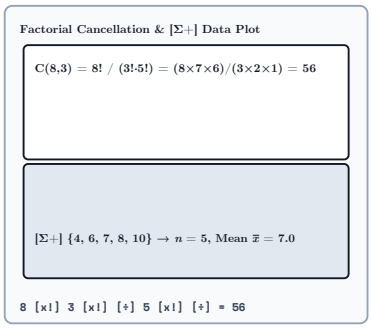
Combinatorics, Probability Ratios & Series Accumulation

Mathematical Focus: Decision Slot Diagram, Statistical Plot & SC-32 [nPr]/[nCr]/[Σ+] • Standards: CCSS.MATH.HSS.CP.B.9

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Combinations vs. Permutations: Subsets & Prime Cancellation):

Combinations nCr = $n!$ / ($r!(n\text{-}r)!$) count unordered selections by dividing ordered permutations by the r! duplicate arrangements of the chosen group. Canceling 5! from 8! leaves (8 × 7 × 6) / 3! = 56.

Companion Book (Vol. II: Thinking in the Stack — Ch. 9: Least Squares, Small Samples & Uncertainty (Gauss's Ceres Recovery, Gosset, [PROB], [Σ+])) • Pascal & Fermat (1654) and Gauss's Recovery of Ceres: Pascal and Fermat founded probability ([nCr], [nPr]) in 1654, and 24-year-old Gauss recovered the lost asteroid Ceres in 1801 using least-squares [Σ+] regression!



2. KEY MATHEMATICAL PRINCIPLES: PERMUTATIONS VS COMBINATIONS ON THE SCIENTIFIC STACK

- Permutations (Order Matters): P(n, r) = $n!$ ÷ $(n - r)!$. Used for ranking, passwords, and assigned positions.
- Combinations (Order Does NOT Matter): C(n, r) = $n!$ ÷ $[r!(n - r)!]$. Used for teams, hands of cards, and committees.
- Always P(n, r) ≥ C(n, r) because combinations divide out the r! internal orderings.
- On the SC-32: Enter n into Y, r into X, then press [nPr] or [nCr].

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Mathematical Setup: Use Prime Factor blocks to model prime factor cancellation in 8! / (3! × 5!) = (8 × 7 × 6) / (3 × 2 × 1) = 8 × 7 = 56. Question: How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! × 5!) = (8 × 7 × 6) / 6 = 56.

Verification Keystrokes: 8 ↶ x! 3 ↶ x! ÷ 5 ↶ x! ÷

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
8	8	Total candidates n = 8
↶ x!	40320	8! = 40,320 in register X
3	3	Subset r = 3 (40320 lifts to Y)
↶ x!	6	3! = 6 in X
÷	6720	40320 / 6 = 6720 in X
5	5	Remaining n - r = 5 (6720 lifts to Y)
↶ x!	120	5! = 120 in X
÷	56	6720 / 120 = 56 combinations!

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

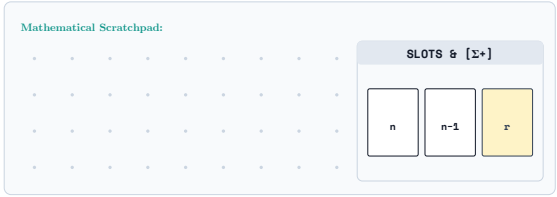
Core Investigation: A lottery draws 4 winning numbers from a pool of 10 (order does not matter). How many winning combinations are possible?

1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)



Extension: Sample Mean & Variance of Discrete Data Sets

Deepening Concepts • Standards: CCSS.MATH.HSS.ID.A.2

1. EXTENSION CONCEPT & SETUP (Sample Mean & Variance of Discrete Data Sets):

Each press of [Σ+] updates the count n, sum Σx, and sum of squares Σx^2 simultaneously.

Extension Setup: Clear statistical registers with [CLΣ], then enter data values {12, 15, 18, 21}. • Question: Count $n = 4$, Sum = 66, Mean $\bar{x} = 66 \div 4 = 16.5$.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [2] [Σ+] [1] [5] [Σ+] [1] [8] [Σ+] [2] [1] [Σ+] [↗] [$\bar{x}$,$\bar{y}$]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1] [2] [Σ+]	1	12 added; $n = 1$
[1] [5] [Σ+]	2	15 added; $n = 2$
[1] [8] [Σ+]	3	18 added; $n = 3$
[2] [1] [Σ+]	4	21 added; $n = 4$
[↗] [$\bar{x}$,$\bar{y}$]	16.5	Sample mean $\bar{x} = 16.5$ in Register X!

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS

Misconception 1: Forgetting to clear the statistical registers ([CLΣ]) before entering a new dataset with [Σ+].

Mathematical & Stack Remediation: Notice that pressing [Σ+] displays the sample count n in Register X; if the first data point shows $n > 1$, press [CLΣ] first so prior data does not corrupt the mean.

Misconception 2: Dividing by r! and then multiplying by $(n - r)!$ when computing C(n, r) = $n!$ / (r!$(n - r)$!).

Mathematical & Stack Remediation: Compute n! into Y, compute r! and $(n - r)$! and multiply them together in X ([×]), then press [÷] once to divide n! by the combined denominator.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [10] [ENTER] [3] [nCr] = 120 (Combinations)

Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Given the four data points {12, 15, 18, 21}, find the sample size n, the total sum of the values, and the sample mean $\bar{x}$.

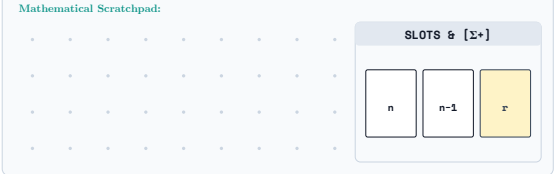
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



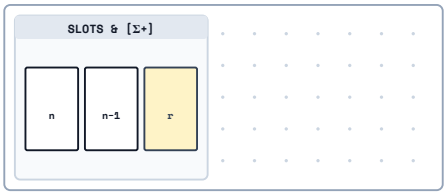
Unit 09 Lab: Combinatorics, Probability Ratios & Series Accumulation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Sketch your slot diagram or data distribution in Box 1. (2) Write your counting/stat steps in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Evaluate 6! (the number of ways to arrange 6 distinct books on a shelf).

1. Slot Diagram / Data Sketch:



2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

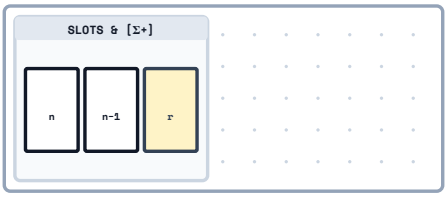
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: How many ways can 5 runners finish in 1st, 2nd, and 3rd place?

1. Slot Diagram / Data Sketch:



2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

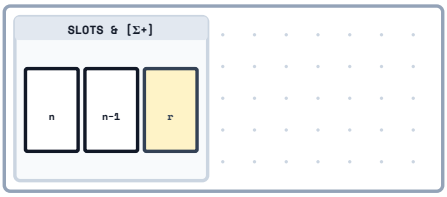
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: How many 2-person partnerships can be formed from 7 students?

1. Slot Diagram / Data Sketch:



2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

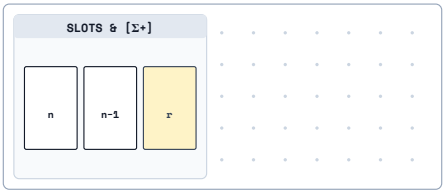
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Find the mean $\bar{x}$ of the two data values 5 and 15.

1. Slot Diagram / Data Sketch:



2. Factorial / Statistical Steps:

Formula (nPr, nCr, or Stat):

Factorial / Sum Expansion:

Computed Value:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did your coordinate, polynomial, or vector diagram confirm the algebraic structure on the SC-32 stack?

Unit 09 Practice A — Factorials, Permutations (nPr) & Combinations (nCr)

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Evaluate 6C2 (choosing 2 items from 6).**

6C2 = (6 × 5) ÷ (2 × 1) = 30 ÷ 2 = 15. Key: [6] [ENTER] [2] [nCr].

Verified Answer: 15

EXAMPLE 2 **How many ways can 4 runners finish 1st, 2nd, and 3rd (4P3)?**

Order matters: 4 × 3 × 2 = 24 ways.

Verified Answer: 24

1 Tier 1 • Builder

Evaluate factorial: 5! =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Simplify factorial ratio: 8! ÷ 6! =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Evaluate permutation: 5P2 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Evaluate permutation: 6P3 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Evaluate combination: 5C2 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Evaluate combination: 7C2 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Evaluate combination: 9C7 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Evaluate combination: 8C3 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Pascal's Row Sum: 4C0 + 4C1 + 4C2 + 4C3 + 4C4 =

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

How many distinct rearrangements can be made from the letters in MOM?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Practice B — Probability Ratios, Binomial Trials & Statistical Summaries

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Roll two 6-sided dice. What is P(sum = 7)?**

6 ways ((1,6)..(6,1)) out of 36 total outcomes → 6/36 = 1/6.

Verified Answer: 1/6

EXAMPLE 2 **Find the Mean and Sample Standard Deviation of {10, 14, 18}.**

Mean = 14. Since {10, 14, 18} are 3 evenly spaced numbers ($d = 4$), Sample SD = 4.

Verified Answer: Mean=14, SD=4

1 Tier 1 • Builder

A bag has 3 red, 5 blue, and 2 green marbles. What is P(blue) in lowest terms?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

If $P(\text{rain}) = 0.35$, what is the complement probability P(no rain)?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Flip 3 fair coins. What is P(all 3 Heads) = $(1/2)^3$?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Flip 4 fair coins. How many ways to get exactly 2 Heads (4C2)?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Flip 4 fair coins. What is P(exactly 2 Heads) = 4C2 ÷ 2^4?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Find the Mean ($\bar{x}$) of the dataset {12, 15, 18, 21, 24}:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Find the Median of the dataset {8, 3, 15, 10, 6} (middle value of sorted {3, 6, 8, 10, 15}):

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Find the Interquartile Range (IQR = Q_3 - Q_1) of {2, 4, 6, 8, 10, 12, 14}:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Find the Sample Standard Deviation of the 3 evenly spaced numbers {5, 10, 15}:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Gauss Series Sum: Evaluate 1 + 2 + 3 + ... + 20 =

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Practice C — Real-World Quality Control, Genetics & Contest Probability

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 A robotics club of 8 members chooses a 3-person repair team. How many teams?

Order does not matter → 8C3 = (8×7×6) ÷ 6 = 56 teams.

Verified Answer: 56 teams

EXAMPLE 2 Spot the Mistake: Before entering a new dataset with [Σ+], Sam forgot to press [CLΣ].

Old data stays in the summation registers! Always press [CLΣ] before a new dataset.

Verified Answer: Press [CLΣ] first

1 Tier 2 • Detective

A security keypad requires a 3-digit code using digits 0–9 (repeats allowed). How many codes?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

How many handshakes occur if 6 people in a meeting each shake hands once with everyone else?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

A pizza shop offers 7 toppings. How many different 2-topping pizzas can you order?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

How many distinct 4-letter arrangements can be made from the letters in BOOK?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

Two independent sensors each have a 0.90 reliability rate. What is P(both work)?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

A spinner has 8 equal sectors (1–8). What is the probability of landing on a prime number?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

After 4 quizzes, Maya's average is 85. What score on Quiz 5 raises her average to 88?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Sum the first 10 odd numbers: 1 + 3 + 5 + ... + 19 =

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

In a clinical study, 5 patients each have a 50% recovery chance. How many total outcome patterns exist?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: For a 3-person committee from 5 people, Eli computed 5P3 = 60. Why use 5C3 instead?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Practice D — Conditional Probability, Expected Value & Hypergeometric Draws

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 A game pays \$10 with probability 0.3 and \$0 with probability 0.7. Expected value E(X)?

$E(X) = 10(0.3) + 0(0.7) = \$3.00$.

Verified Answer: \$3.00

EXAMPLE 2 A bag has 4 Red and 2 Blue marbles. Draw 2 without replacement. Find P(both Red).

$4C2 \div 6C2 = 6 \div 15 = 2/5 = 0.40$.

Verified Answer: 2/5 (0.4)

1 Tier 2 • Detective

Expected Value: Win \$20 with $P = 0.25$ or \$4 with $P = 0.75$. Find E(X):

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

Find the Expected Value of a single fair 6-sided die roll:

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

Expected Value of the sum of two fair 6-sided dice:

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

Without Replacement: Bag has 3 Red and 2 Blue marbles. Draw 2. Find P(both Red):

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

Committee Selection: Choose 2 boys from 4 and 2 girls from 3. How many ways?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Circular Permutations: How many ways can 5 people be seated around a round table?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Polygon Diagonals: How many diagonals does a regular hexagon have?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Polygon Diagonals: How many diagonals does a regular octagon have?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

Independent Failure: If P(defect) = 0.1, what is P(2 items are both defect-free)?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

At Least One: Flip 4 fair coins. What is P(at least 1 Head)?

SLOTS & [Σ+] | n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Practice E — Variance, Z-Scores, Linear Transformations & Regression

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the z-score $z = (x - \mu) / \sigma$ for score $x = 86$ when mean $\mu = 74$ and SD $\sigma = 6$.

$z = (86 - 74) \div 6 = 12 \div 6 = +2.0$.

Verified Answer: $z = 2$

EXAMPLE 2 If every value in a dataset is multiplied by 3 and increased by 5, how do Mean and SD change?

Mean becomes $3\mu + 5$; SD scales only by multiplication → 3σ!

Verified Answer: Mean: $3\mu+5$, SD: 3σ

1 Tier 3 • Challenge

Find the z-score for $x = 90$ when mean $\mu = 75$ and SD $\sigma = 10$:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Find the z-score for $x = 62$ when mean $\mu = 70$ and SD $\sigma = 4$:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

If a student's z-score is $+2.5$ on a test with $\mu = 80$ and $\sigma = 4$, what was their raw score x?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Sample Variance s^2: If Sample Standard Deviation is $s = 6$, what is the sample variance s^2?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Dataset Shift: Adding 10 to every number in {4, 8, 12} changes the Mean from 8 to what?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Dataset Shift: Adding 10 to every number in {4, 8, 12} changes the Standard Deviation from 4 to what?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Dataset Scaling: Multiplying every number in {4, 8, 12} by 2 changes the SD from 4 to what?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Weighted Average: 30% Quiz (score 80) + 70% Exam (score 90) =

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Linear Regression: Given $\hat{y} = 2.5x + 10$, predict $\hat{y}$ when $x = 8$:

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Residual: If actual $y = 33$ and predicted $\hat{y} = 30$, what is the residual $(y - \hat{y})$?

SLOTS & [Σ+] n n-1 r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Practice F — Series Summations (Σk, Σk^2, Σk^3) & Binomial Theorem

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 Find the coefficient of x^2y^3 in $(x + y)^5$.

By the Binomial Theorem, the coefficient is 5C3 = 10.

Verified Answer: 10

EXAMPLE 2 Evaluate the sum of squares $1^2 + 2^2 + 3^2 + 4^2 + 5^2$ using $n(n+1)(2n+1) \div 6$.

$5 \times 6 \times 11 \div 6 = 55$.

Verified Answer: 55

1 Tier 3 • Challenge

Binomial Coefficient: Find the coefficient of x^3 in $(x + 1)^6$:

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Binomial Expansion: Find the coefficient of x^2 in $(x + 2)^4$:

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

Evaluate the sum of squares: $1^2 + 2^2 + 3^2 + 4^2 =$

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Evaluate the sum of squares: $1^2 + 2^2 + \ldots + 6^2 =$

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Evaluate the sum of cubes: $1^3 + 2^3 + 3^3 + 4^3 =$

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

Infinite Geometric Series: Sum $1 + 1/2 + 1/4 + 1/8 + \ldots =$

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Infinite Geometric Series: Sum $6 + 2 + 2/3 + \ldots =$

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Permutations with Repetition: Distinct arrangements of MISSISSIPPI has 11! ÷ (1!4!4!2!). How many I's?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Arrange 5 books on a shelf if 2 specific books must stay together:

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

SC-32 Stats Check: Enter {2, 4, 6, 8, 10} with [Σ+]. What count n is shown in Register X?

SLOTS & [Σ+]: n | n-1 | r

Paper Step: _____ | | SC-32 Ans: _____

Unit 09 Mastery Check: Combinatorics & Probability [nPr, nCr]

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Evaluate: 4! =

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Evaluate: 6P2 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Evaluate: 6C3 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Find the sum of Row 5 of Pascal's Triangle:

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Find the Mean of {14, 18, 22, 26}:

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Find the Sample SD of {6, 12, 18}:

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

How many 2-person lab pairs can be chosen from 9 students?

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

A fair coin is flipped 3 times. What is P(at least 1 Tail)?

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Evaluate the arithmetic series: 1 + 2 + 3 + ... + 15 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: Why is 10C8 equal to 10C2 = 45 without multiplying 8 factors?

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 09 STEM: Cyber Security Password Entropy Cryptography

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: CYBER SECURITY PASSWORD ENTROPY CRYPTOGRAPHY

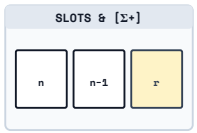
Companion to Vol. II, Ch. 9: Security engineers and statisticians calculate key entropy, combinatorial search spaces, and small-sample distributions using [nPr], [nCr], and [Σ+].

Companion Book (Vol. II: Thinking in the Stack — Ch. 9: Least Squares, Small Samples & Uncertainty (Gauss's Ceres Recovery, Gosset, [PROB], [Σ+])) • Pascal & Fermat (1654) and Gauss's Recovery of Ceres: Pascal and Fermat founded probability ([nCr], [nPr]) in 1654, and 24-year-old Gauss recovered the lost asteroid Ceres in 1801 using least-squares [Σ+] regression!

Mathematical Focus: Decision Slot Diagram, Statistical Plot & SC-32 [nPr]/[nCr]/[Σ+] • Stack Shortcut: [10] [ENTER] [3] [nCr] = 120 (Combinations)

CASE INVESTIGATION 1 STEM Modeling

Ordered PIN Sequence: How many 3-digit ordered PINs can be formed from 10 distinct digits without repetition, P(10, 3)?



1. Formula / Stack Plan: _____

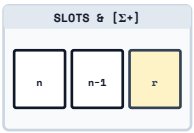
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Unordered Backup Keys: How many unordered 3-key subsets can be chosen from 10 master keys, C(10, 3)?



1. Formula / Stack Plan: _____

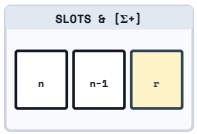
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Cryptographic Token Pool: How many 5-token subsets can be drawn from a 52-token hardware pool, C(52, 5)?



1. Formula / Stack Plan: _____

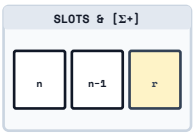
2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student uses P(6, 2) = 30 for a 2-person committee from 6 people. Find true C(6, 2):



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

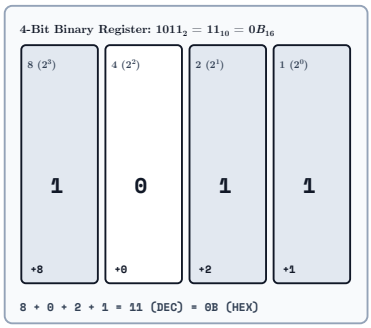
Digital Logic, Base Conversions & Computer Architecture

Hands-On Block Studio: Place Value Studio (Binary [0]/[1] Bit Sockets) & SC-32 [BASE] • Standards: CCSS.MATH.HSN.RN.A.1, CSTA 3A-DA-07

1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Positional Number Systems: Decimal, Binary & Hexadecimal):

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) or 16 instead of powers of 10. Every 4-bit binary group (0000 to 1111) maps to exactly one hexadecimal digit (0 to F).

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: 32,768 Hz Quartz Timebase, Ch. 10: Shannon's Boolean Logic & 36-Bit Word ([BASE]) & Ch. 13: Coupled Systems) • Leibniz's Binary (1703), Shannon (1937) & 32,768 Hz Quartz: Leibniz showed all numbers arise from 1 and 0, Claude Shannon proved Boolean logic maps 1-to-1 onto switches (Ch. 10), and a $2^{15} = 32{,}768$ Hz quartz crystal drives the SC-32 timebase (Ch. 1 & 13)!



2. HANDS-ON BLOCKS & CONCEPT RULES: HOW COMPUTERS REPRESENT DATA IN BINARY AND HEXADECIMAL

- Number Bases: Decimal (Base 10), Hexadecimal (Base 16: 0–9, A–F), Octal (Base 8), Binary (Base 2).
- One Hex Digit = 4 Binary Bits (1 Nybble) in the Place Value Tray sockets: 0xF = [1][1][1][1]$_2$, 0xA = [1][0][1][0]$_2$.
- Bitwise AND masks bits; bitwise OR sets bits; bitwise XOR toggles bits.
- On the SC-32: Press [BASE] to cycle bases; type hex numbers using A–F shift keys.

3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-32

Hands-On Block Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits. Question: Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary $1011_2 = 8 + 0 + 2 + 1 = 11_{10}$. In hexadecimal ($A$=10, B=11), $11_{10} = 0B_{16}$.

Verification Keystrokes: 1 1 ↖ BASE

Button Pressed	Screen (X)	How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X)
1 1	11	Enter decimal integer 11 in DEC mode
↖ BASE	HEX 0B	Press [f:] [BASE]: screen displays Hexadecimal 0B!
↖ BASE	BIN 1011	Press [f:] [BASE]: screen displays Binary 1011!
↖ BASE	OCT 13	Press [f:] [BASE]: screen displays Octal 13!
↖ BASE	DEC 11	Press [f:] [BASE]: returns to Decimal 11

STACK SHORTCUT & BLOCK STUDIO TIP (SC-32):

[255] [ENTER] [BASE] → HEX: 0xFF, BIN: 11111111 • Number Bases: Decimal (Base 10), Hexadecimal (Base 16: 0–9, A–F), Octal (Base 8), Binary (Base 2).

WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Core Investigation: Express decimal 255 as a sum of powers of 2 to write its 8-bit binary and 2-digit hexadecimal representations.

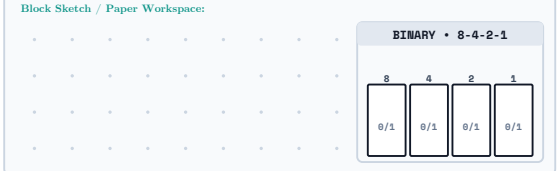
1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



Extension: Bitwise Logic: AND Masking & Column-by-Column Comparison

Deepening Concepts • Standards: CCSS.MATH.PRACTICE.MP5

1. EXTENSION CONCEPT & SETUP (Bitwise Logic: AND Masking & Column-by-Column Comparison):

Bitwise logic operations (AND, OR, XOR) compare corresponding binary place-value columns of two integers in parallel. Placing value A in the top row and mask B in the bottom row lets you inspect each bit column directly.

Hands-On Extension Setup: Row 1 of the Place Value Tray holds nibble A ($[1][1][0][0]_2 = 12$). Row 0 holds mask B ($[1][0][1][0]_2 = 10$). • Question: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: $1000_2 = 8_{10}$.

2. EXTENSION STACK SEQUENCE (SC-32): [1] [2] [ENTER] [1] [0] [AND]

Button Pressed	Screen (X)	Extension Stack Dynamics (T, Z, Y, X)
[1] [2]	12	Enter operand $A = 12$ (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand $B = 10$ (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block Studio Remediation: Model 12 ($[1][1][0][0]_2$) in Row 1 and 10 ($[1][0][1][0]_2$) in Row 0 of the Place Value Tray using [1] and [0] Digit Blocks; only columns where BOTH rows hold [1] remain 1 in 12 AND 10 = 8 ($[1][0][0][0]_2$).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block Studio Remediation: Place [1] in the Sixteens column and [0] in the Ones column on the Place Value Tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-32)

Calculator Shortcut: [255] [ENTER] [BASE] → HEX: 0xFF, BIN: 11111111

Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?

WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

Extension Challenge: Compare 12 and 3 column by column in 4-bit binary to find the result of 12 OR 3 in both binary and decimal.

1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



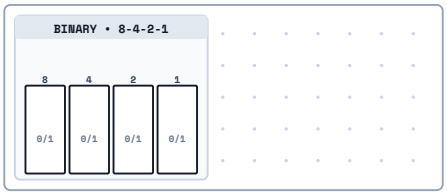
Unit 10 Lab: Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Write the 8-bit / 4-bit place-value table in Box 1. (2) Show your base or bitwise math in Box 2.
(3) Write your Prediction BEFORE pressing keys, then verify your solution on the SC-32!

Investigation 1: Decompose decimal 42 into powers of 2 to write its 6-bit binary representation.

1. Binary Place-Value / Bit Grid:



2. Base Conversion / Bitwise Steps:

Place Weights (8-4-2-1):

Bit Pattern / Logic Mask:

Decimal / Binary / Hex Result:

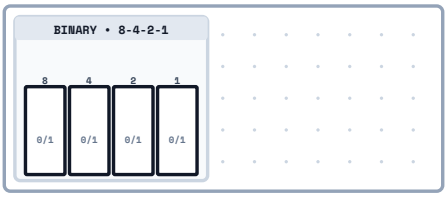
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 2: Convert hexadecimal $2A_{16}$ to decimal using place value.

1. Binary Place-Value / Bit Grid:



2. Base Conversion / Bitwise Steps:

Place Weights (8-4-2-1):

Bit Pattern / Logic Mask:

Decimal / Binary / Hex Result:

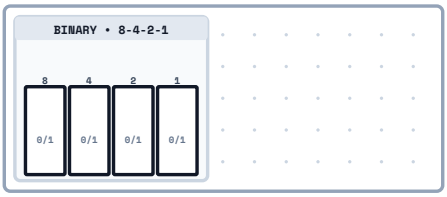
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate the bitwise AND of 15 and 7 in binary and decimal.

1. Binary Place-Value / Bit Grid:



2. Base Conversion / Bitwise Steps:

Place Weights (8-4-2-1):

Bit Pattern / Logic Mask:

Decimal / Binary / Hex Result:

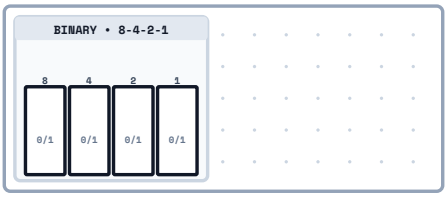
3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate the bitwise XOR of 12 and 10 in binary and decimal.

1. Binary Place-Value / Bit Grid:



2. Base Conversion / Bitwise Steps:

Place Weights (8-4-2-1):

Bit Pattern / Logic Mask:

Decimal / Binary / Hex Result:

3. My Prediction:

Final Verified Answer:

[ ] Verified on SC-32 Stack

EXIT REFLECTION: How did the Prime Factor or Binary Place Value blocks confirm the structure on the SC-32 stack?

Unit 10 Practice A — Binary (Base 2), Hexadecimal (Base 16) & Place Value Bits

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST

EXAMPLE 1 Convert binary 1101_2 to decimal using 8-4-2-1 weights.

$1(8) + 1(4) + 0(2) + 1(1) = 8 + 4 + 1 = 13$.

Verified Answer: 13

EXAMPLE 2 Convert 8-bit binary 10101100_2 to Hexadecimal.

Split into two 4-bit nibbles: $1010_2 = 10 = A$, and $1100_2 = 12 = C \rightarrow AC_{16}$.

Verified Answer: AC_{16}

1 Tier 1 • Builder

Convert 4-bit binary 0101_2 to decimal:



Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Convert 4-bit binary 1011_2 to decimal:



Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Convert 4-bit binary 1111_2 to decimal and hexadecimal:



Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Convert decimal 9 to 4-bit binary:



Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Convert decimal 14 to 4-bit binary and hexadecimal:



Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Convert hexadecimal $1A_{16}$ to decimal:



Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Convert hexadecimal $2F_{16}$ to decimal:



Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Convert hexadecimal FF_{16} to decimal:



Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Convert 8-bit binary 11000111_2 to hexadecimal:



Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

Convert octal 25_8 to decimal:



Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Practice B — Bitwise Logic Gates (AND, OR, XOR, NOT) & Bit Shifts

Name: _____ Date: _____ Score: _____ / 10

TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Evaluate 12 AND 10 using 4-bit binary.**

$12 = 1100_2$, $10 = 1010_2$. Only the 8-bit is 1 in both $\rightarrow 1000_2 = 8$.

Verified Answer: 8

EXAMPLE 2 **Evaluate 12 XOR 10 and 7 << 2.**

12 XOR 10 = differing bits $0110_2 = 6$. Left shift $7 << 2 = 7 \times 2^2 = 28$.

Verified Answer: 6 and 28

1 Tier 1 • Builder

Bitwise AND: Evaluate 6 AND 3 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 1 • Builder

Bitwise OR: Evaluate 6 OR 3 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 1 • Builder

Bitwise XOR: Evaluate 6 XOR 3 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 1 • Builder

Bitwise AND: Evaluate 12 AND 5 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 1 • Builder

Bitwise OR: Evaluate 12 OR 5 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 1 • Builder

Bitwise XOR: Evaluate 15 XOR 9 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 1 • Builder

Self-XOR Identity: For any integer x, what is x XOR x (e.g., 42 XOR 42)?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 1 • Builder

Left Bit Shift: Evaluate 5 << 3 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 1 • Builder

Right Bit Shift: Evaluate 48 >> 2 =

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 1 • Builder

8-Bit Two's Complement: Represent -1 as an 8-bit unsigned hex byte:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Practice C — Computer Architecture, RGB Color Masks & Subnet Logic

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Extract the lower 4 bits of byte 0xB7 (183) using mask 0x0F (15).**

0xB7 AND $0x0F = 0x07 = 7$.

Verified Answer: 7

EXAMPLE 2 **Spot the Mistake: Sam said Hex 10_{16} equals decimal 10.**

In base 16, the '1' is in the Sixteens column: $1 \times 16 + 0 = 16$!

Verified Answer: 16

1 Tier 2 • Detective

How many distinct values can a single 8-bit byte represent?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

What is the largest unsigned integer that fits in a 10-bit register?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

An RGB color is #00FF80. What is the decimal value of the Green channel (FF_{16})?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

In RGB color #00FF80, what is the decimal value of the Blue channel (80_{16})?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

IPv4 Subnet Mask: Evaluate 192 AND 240:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Complementary Bit Masks: Evaluate 170 XOR 85:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Complementary Bit Masks: Evaluate 170 AND 85:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Microcontroller Timer: A 12-bit ADC has max reading 4095. Write 4095 in Hex:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

Binary Addition with Carry: Add $0111_2 + 0001_2$ in binary:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Spot the Mistake: Leo computed 8 AND 4 = 12 by adding. What is 8 AND 4?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Practice D — Binary & Hexadecimal Arithmetic, Carry & Overflow

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

EXAMPLE 1 Add binary 1010_2 (10) + 0110_2 (6) in binary and hex.

$10 + 6 = 16 = 10000_2 = 10_{16}$.

Verified Answer: 10000_2 (10_{16})

EXAMPLE 2 Add hexadecimal $1A_{16} + 16_{16}$ in hexadecimal.

A_{16} (10) + $6_{16} = 16 = 10_{16}$ (0, carry 1); $1 + 1 + 1 = 3 \rightarrow 30_{16}$ (48_{10}).

Verified Answer: 30_{16} (48)

1 Tier 2 • Detective

Binary addition: $0101_2 + 0010_2 =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 2 • Detective

Binary addition: $1100_2 + 0011_2 =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 2 • Detective

Binary addition: $1001_2 + 0111_2 =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 2 • Detective

Binary subtraction: 1110_2 - $0101_2 =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 2 • Detective

Hexadecimal addition: $A_{16} + 5_{16} =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 2 • Detective

Hexadecimal addition: $C_{16} + 4_{16} =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 2 • Detective

Hexadecimal addition: $1F_{16} + 1_{16} =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 2 • Detective

Hexadecimal subtraction: FF_{16} - $F0_{16} =$

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 2 • Detective

Octal to Binary: Convert 75_8 directly to 6-bit binary:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 2 • Detective

Binary to Octal: Convert 101110_2 to octal:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Practice E — Signed 8-Bit Two's Complement & Bit-Field Manipulation

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST

EXAMPLE 1 **Find the 8-bit Two's Complement representation of -5 in binary and hex.**

$256 - 5 = 251 = 11111011_2 = FB_{16}$ (or invert $00000101 \rightarrow 11111010 + 1$).

Verified Answer: 11111011_2 (FB_{16})

EXAMPLE 2 **Extract bits 4..7 (high nibble) of byte 0xA7 (167) by shifting right 4 bits.**

$0xA7 >> 4 = 0x0A = 10$.

Verified Answer: 10 (0x0A)

1 Tier 3 • Challenge

8-Bit Two's Complement: What signed decimal value is 11111111_2 (FF_{16})?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

8-Bit Two's Complement: What signed decimal value is 11111110_2 (FE_{16})?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

8-Bit Two's Complement: Write -4 in 8-bit Hex:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

8-Bit Signed Range: What is the most negative 8-bit Two's Complement integer?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

8-Bit Signed Range: What is the largest positive 8-bit Two's Complement integer?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

High Nibble Extraction: Evaluate (0xC9 >> 4) in decimal:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

Setting a Bit: Use OR with mask 0x04 to set bit 2 of 0x09:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Clearing a Bit: Use AND with mask 0x0E to clear bit 0 of 0x0F:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Toggling Bits: Use XOR with 0xFF to invert all 8 bits of 200:

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

Power-of-2 Test: What is n AND $(n - 1)$ when $n = 16$?

BINARY • 8-4-2-1

8	4	2	1
0/1	0/1	0/1	0/1

Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Practice F — Boolean Algebra Laws, De Morgan's Theorem & Memory Addressing

Name: _____ Date: _____ Score: _____ / 10

TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST

EXAMPLE 1 How many bytes are in 4 Kilobytes (4×2^{10} bytes)?

$4 \times 1{,}024 = 4{,}096$ bytes (0x1000 in Hex).

Verified Answer: 4,096 bytes

EXAMPLE 2 Apply De Morgan's Law to simplify NOT(A OR B) for 4-bit $A = 1100_2$, $B = 1010_2$.

NOT(1100 OR 1010) = NOT(1110_2) = 0001_2 = (NOT A) AND (NOT B).

Verified Answer: 0001_2 (1)

1 Tier 3 • Challenge

Memory Addressing: How many address lines (bits n) are needed to address 1,024 bytes?

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

2 Tier 3 • Challenge

Memory Addressing: How many distinct memory addresses can a 16-bit bus access?

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

3 Tier 3 • Challenge

De Morgan's Law: Verify 4-bit NOT(12 AND 10) by computing 15 XOR (12 AND 10):

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

4 Tier 3 • Challenge

Boolean Identity: Evaluate (25 AND 0) on the stack:

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

5 Tier 3 • Challenge

Boolean Identity: Evaluate (25 OR 0) on the stack:

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

6 Tier 3 • Challenge

XOR Encryption: If message $M = 10$ and key $K = 6$, find cipher C = M XOR K:

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

7 Tier 3 • Challenge

XOR Decryption: Take cipher $C = 12$ and XOR with key $K = 6$ again (12 XOR 6):

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

8 Tier 3 • Challenge

Parity Bit: How many '1' bits (Hamming weight) are in binary 10110101_2?

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

9 Tier 3 • Challenge

Fast Multiply by 10: Compute 7×10 using bit shifts ($7 << 3$) + ($7 << 1$):

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

10 Tier 3 • Challenge

SC-32 Base Check: When switching from [DEC] 42 to [HEX] ($42 = 2{\times}16 + 10$), what appears on Line X?

BINARY • 8-4-2-1 | 8 | 4 | 2 | 1 | 0/1 | 0/1 | 0/1 | 0/1

Paper Step: _____ | | SC-32 Ans: _____

Unit 10 Mastery Check: Digital Logic & Base Conversions

Name: _____ Date: _____ Score: _____ / 100

100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-32 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Convert binary 1101_2 to decimal:

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Convert hexadecimal $3C_{16}$ to decimal:

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Convert binary 11110101_2 to hexadecimal:

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Evaluate: 10 AND 6 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Evaluate: 10 OR 6 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Evaluate: 10 XOR 6 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

Evaluate left shift: 9 << 2 =

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

What is the maximum unsigned decimal value stored in 4 binary bits?

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Extract the low nibble: What is 0x4D AND 0x0F in decimal?

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

Error Analysis: Why does grouping binary into 4-bit nibbles convert directly to Hex?

[ ] +10 pts [ ] SC-32 **Ans:** _____

Unit 10 STEM: Microcontroller Firmware Register Masking

Name: _____ Date: _____ Score: _____ / 4

REAL-WORLD STEM APPLICATION & WORKSHOP: MICROCONTROLLER FIRMWARE REGISTER MASKING

Companion to Vol. II, Ch. 1, Ch. 10 & Ch. 13: Firmware engineers configure GPIO pins on the RP2350 microcontroller using hexadecimal masks and Shannon bitwise AND, OR, and XOR.

Companion Book (Vol. II: Thinking in the Stack — Ch. 1: 32,768 Hz Quartz Timebase, Ch. 10: Shannon's Boolean Logic & 36-Bit Word ([BASE]) & Ch. 13: Coupled Systems) • Leibniz's Binary (1703), Shannon (1937) & 32,768 Hz Quartz: Leibniz showed all numbers arise from 1 and 0, Claude Shannon proved Boolean logic maps 1-to-1 onto switches (Ch. 10), and a $2^{15} = 32{,}768$ Hz quartz crystal drives the SC-32 timebase (Ch. 1 & 13)!

Block Studio: Place Value Studio (Binary [0]/[1] Bit Sockets) & SC-32 [BASE] • Stack Shortcut: [255] [ENTER] [BASE] → HEX: 0xFF, BIN: 11111111

CASE INVESTIGATION 1 STEM Modeling

Full Port Enable: Convert decimal 255 to hexadecimal and binary for an 8-pin GPIO port:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 2 STEM Modeling

Alternating Pin Merge: Combine odd-pin mask 0xAA and even-pin mask 0x55 using bitwise OR:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 3 Applied Calculation

Lower-Nybble Status Mask: Isolate the lower 4 status bits of 0xFF using bitwise AND with 0x0F:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

CASE INVESTIGATION 4 Error Analysis Workshop

Error Analysis: Student adds 0x0F + 1 as 0x10 and claims it is 10 decimal. Find 15 + 1 in decimal:



1. Formula / Stack Plan: _____

2. Algebraic / Paper Steps: _____

3. Final Answer: _____

[ ] Verified on SC-32 Stack (Online Key: stackcalc.io/curriculum/answers)

STEM Applied Lab Summary & Career Connection

Professional scientists, engineers, and quantitative analysts model complex physical and financial systems on paper first before running computational simulations on the SC-32. Verify all calculations online at stackcalc.io/curriculum/answers.

APPENDIX A • STUDENT STACK MASTERY CERTIFICATE & KEYCAP GUIDE

Certificate of Scientific Stem Stack Mastery

Awarded for completing all 10 units of the High School & STEM Lab (840 Problems).

CERTIFICATE OF SCIENTIFIC STEM STACK MASTERY

This certifies that: _____

has completed all 10 Hands-On Math Lab units, leveled practice worksheets (Practice A through F), 100-Point Unit Mastery Checks, and STEM Applied Cases—solving on paper first with physical manipulative models and verifying step-by-step on the 4-level StackCalc SC-32 calculator.

[ ] Scientific STEM Stack Master (Units 01–10 • Grades 9–12 & AP • SC-32)

Portfolio Summary: 10 Lab Units • 840 Verified Problems • 100-Pt Mastery Avg: _____ %

[ ] CRA Manipulative Mastery [ ] Paper-First Predictions [ ] 4-Level Stack Verification

Teacher / Parent Signature: _____ **Date:** _____

ESSENTIAL SC-32 KEYCAP SEQUENCES TO REMEMBER

Evaluating Slope: $m = (y_2 - y_1) \div (x_2 - x_1)$

1 1 ENTER 3 - 6 ENTER 2 - ÷

Coordinate Slope Diagram: Computes Δy in Waiting Shelf Y while computing Δx in Workbench X.

Instant Quadratic Roots with [POLY]: $x^2 + 5x + 6 = 0$

1 ENTER 5 ENTER 6 ↷ POLY

2×2 Area Grid & Prime Studio: Enters coefficients [a, b, c] to solve roots into Y and X.

Rectangular-to-Polar Vector Conversion [→ϑ,r]: (x=3, y=4)

4 ENTER 3 ↖ ▸ϑ,r

2D Vector Diagram: Places Rise y in Register Y and Run x in Register X to resolve r and ϑ.

Continuous Compounding: $A = Pe^{rt}$

0 • 0 5 ENTER 1 0 × e^x 1 0 0 0 ×

Prime Factor Studio: Evaluates exponent product $r{\cdot}t$ first, applies [e^x], then scales by P.

Doubling Time via Natural Logarithm: $t = \ln(2) \div r$

2 LN 0 • 0 7 ÷

Prime Factor Studio: Uses natural log height ln(2) divided by continuous rate r.

CONGRATULATIONS! YOUR NEXT STEP IN THE STACKCALC LIBRARY (AP, CONTESTS & STEM):

- **Timed Contest & AP Speed Training: Book 5 — Stack Speed & Fluency (15 Champion Playbooks & 180 Timed Tests / 3,690 Problems).**
- Companion Trade Book & App: Master Volume II: Thinking in the Stack & the StackCalc 32 iOS/watchOS App at https://stackcalc.io!

High School & STEM Lab Score Tracker & Standards Log

Record your scores for the Lab Investigation, Practice Worksheets A–F, Mastery Checks, and STEM Cases.

Unit & Mathematical Topic	Standards	Lab (/4)	Prac A–F	Quiz (/100)	STEM (/4)
Unit 01: Linear Equations & Slope Grade 9 – 10 (Algebra 1) • pp. 5–15	CCSS.MATH.HSA.CED.A.2 CCSS.MATH.HSF.IF.B.6	____ / 4	__/__/__	____ / 100	____ / 4
Unit 02: Quadratic Factoring & Area Models Grade 9 – 10 (Algebra 1) • pp. 16–26	CCSS.MATH.HSA.SSE.B.3 CCSS.MATH.HSA.REI.B.4	____ / 4	__/__/__	____ / 100	____ / 4
Unit 03: Quadratic Formula & [POLY] Solver Grade 9 – 11 (Algebra 1 & 2) • pp. 27–37	CCSS.MATH.HSA.REI.B.4.B CCSS.MATH.HSN.CN.C.7	____ / 4	__/__/__	____ / 100	____ / 4
Unit 04: Polynomial Operations & Division Grade 10 – 11 (Algebra 2) • pp. 38–48	CCSS.MATH.HSA.APR.A.1 CCSS.MATH.HSA.APR.B.2	____ / 4	__/__/__	____ / 100	____ / 4
Unit 05: Exponential Growth & Decay Grade 10 – 12 (Algebra 2 & Precalculus) • pp. 49–59	CCSS.MATH.HSF.LE.A.1 CCSS.MATH.HSF.LE.A.4	____ / 4	__/__/__	____ / 100	____ / 4
Unit 06: Natural Logarithms & Doubling Time Grade 11 – 12 (Algebra 2 & Precalc) • pp. 60–70	CCSS.MATH.HSF.BF.B.5 CCSS.MATH.HSF.LE.A.4	____ / 4	__/__/__	____ / 100	____ / 4
Unit 07: Trigonometric Functions [SIN, COS, TAN] Grade 10 – 12 (Geometry & Precalculus) • pp. 71–81	CCSS.MATH.HSG.SRT.C.6 CCSS.MATH.HSG.SRT.C.8	____ / 4	__/__/__	____ / 100	____ / 4
Unit 08: 2D Vectors & Polar Coordinates Grade 11 – 12 (Precalc & Physics) • pp. 82–92	CCSS.MATH.HSN.VM.A.1 CCSS.MATH.HSN.VM.A.3	____ / 4	__/__/__	____ / 100	____ / 4
Unit 09: Combinatorics & Probability [nPr, nCr] Grade 11 – 12 & AP (Precalc & AP Stats) • pp. 93–103	CCSS.MATH.HSS.CP.B.9	____ / 4	__/__/__	____ / 100	____ / 4
Unit 10: Digital Logic & Base Conversions Grade 9 – 12 & AP (AP CSP & CS) • pp. 104–114	CCSS.MATH.HSN.RN.A.1 CSTA 3A-DA-07	____ / 4	__/__/__	____ / 100	____ / 4

ONLINE ANSWER KEY & COMPLETE STEP-BY-STEP KEYSTROKE PORTAL

In accordance with modern evidence-based mathematics instruction, complete worked answers are hosted online rather than printed in the student lab book to encourage authentic problem-solving and paper-first predictions.

• Online Portal: https://stackcalc.io/curriculum/answers#high-school

1. PAPER PREDICTION

Solve by hand & write your prediction before touching keys.

2. STACK VERIFICATION

Enter your steps on the SC-32 to check for agreement.

3. ERROR ANALYSIS

If X differs, inspect where your paper step or key diverged.

APPENDIX C (EXAM 1 OF 4) • MID-BOOK BENCHMARK ASSESSMENT (UNITS 01–05 • SC-32)

Cumulative Exam 1: Units 01–05 Benchmark

Name: _____ Date: _____ Score: _____ / 100

100-POINT CUMULATIVE EXAM 1 — UNITS 01–05 BENCHMARK (10 PROBLEMS • 10 PTS EACH)

Covers Linear Slope & Distance, Quadratic Discriminants, [POLY] Roots, Horner's Nested Polynomial Evaluation, and Exponential Growth/Decay. Solve on paper first, verify on the SC-32 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Coordinate Slope on the Stack: Find the slope m through (2, 5) and (6, 21):

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

2D Euclidean Distance: Find the Euclidean distance d between (1, 2) and (7, 10):

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Quadratic Discriminant: Evaluate $\Delta = b^2 - 4ac$ for $2x^2 - 7x + 3 = 0$:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

Quadratic Roots via [POLY]: Find the larger root of $x^2 - 7x + 12 = 0$:

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Horner's Nested Polynomial: Evaluate $P(x) = 2x^3 - 3x^2 + 4x - 5$ at $x = 3$ using Horner's nested method:

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Difference of Cubes Evaluation: Compute $5^3 - 2^3$ on the SC-32 stack:

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Compound Growth: Find the balance A of \$1,000 after 2 years of 5% annual compounding:

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

Exponential Half-Life Decay: Starting with 160 g, how many grams remain after 4 half-lives?

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Parallel Resistor Network: Find the equivalent resistance R of a 6 Ω resistor and a 12 Ω resistor in parallel:

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Error Spotter: A student squared −6 by typing 6 [x^2] [CHS] and got −36. What is the true value of $(-6)^2$?

[ ] +10 pts [ ] SC-32 Ans: _____

Cumulative Exam 2: Units 06–10 Benchmark

Name: _____ Date: _____ Score: _____ / 100

100-POINT CUMULATIVE EXAM 2 — UNITS 06–10 BENCHMARK (10 PROBLEMS • 10 PTS EACH)

Covers Logarithms & Base e, Right-Triangle Trigonometry, 2D Polar Vectors [→ϑ,r], Combinatorics [nCr]/[nPr], Statistics [Σ+], and Digital Logic. Solve on paper first, verify on the SC-32 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Logarithmic Prime Tower Identity: Evaluate $\log_{10}(1000) + \log_{10}(100)$ using [LOG] on the SC-32:

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Continuous Doubling Time: Find the continuous doubling time t (to 4 decimal places) for 5% continuous growth:

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Natural Log & Base-e Inverse: Evaluate $e^{\ln(25)} + 10$ on the SC-32:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

Pythagorean Trig Identity: Evaluate $\sin^2(30°) + \cos^2(30°)$ on the stack:

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

Right-Triangle Ramp Height: A 20 m ramp is inclined at 30°. Find the vertical rise h:

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

2D Vector Magnitude via [→ϑ,r]: Convert rectangular vector (x = 5, y = 12) to polar magnitude r:

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Combinatorics Committee Count: How many 3-person teams can be chosen from 8 engineers, C(8, 3)?

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

Permutation Podium Orderings: In how many ways can 1st, 2nd, and 3rd place be awarded among 6 finalists, P(6, 3)?

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Sample Mean via [Σ+]: Accumulate {14, 18, 22, 26} with [Σ+] and compute $\bar{x}$ with [$\bar{x}$,$\bar{y}$]:

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Bitwise Masking (AP CSP): Apply bitwise AND between 0xF0 and 0x3C (report answer in decimal):

[ ] +10 pts [ ] SC-32 Ans: _____

Cumulative Exam 3: Multi-Step Synthesis

Name: _____ Date: _____ Score: _____ / 100

100-POINT CUMULATIVE EXAM 3 — FULL-CURRICULUM SYNTHESIS (10 PROBLEMS • 10 PTS EACH)

Synthesizes Difference Quotients, Law of Cosines, 3D Vectors, pH & Decibel Scales, Binomial Counts, and Bitwise Register Operations. Solve on paper first, verify on the SC-32 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Difference Quotient (Average Rate of Change): Evaluate $(f(5) - f(2)) \div (5 - 2)$ for $f(x) = x^2$:

[ ] +10 pts [ ] SC-32 Ans: _____

2 Core Fluency • 10 pts

Quadratic Vertex Output: Evaluate $y = 3x^2 - 12x + 17$ at its axis of symmetry $x = 2$:

[ ] +10 pts [ ] SC-32 Ans: _____

3 Core Fluency • 10 pts

Law of Cosines Side c: For a triangle with sides $a = 5$, $b = 8$, and included angle $C = 60°$, find side c:

[ ] +10 pts [ ] SC-32 Ans: _____

4 Core Fluency • 10 pts

3D Space Diagonal: Find the space diagonal d of a $3 \times 4 \times 12$ rectangular prism:

[ ] +10 pts [ ] SC-32 Ans: _____

5 Core Fluency • 10 pts

pH Hydrogen Ion Scale: Calculate the pH of a solution with hydrogen ion concentration $[H^+] = 0.0001$ M:

[ ] +10 pts [ ] SC-32 Ans: _____

6 Core Fluency • 10 pts

Decibel Gain Ratio: Find the decibel gain $dB = 10 \cdot \log_{10}(P_2 / P_1)$ for a 1000× power ratio:

[ ] +10 pts [ ] SC-32 Ans: _____

7 Application • 10 pts

Binomial Probability Count: Multiply $C(6, 2) \times 2^2$ to count weighted 2-success paths:

[ ] +10 pts [ ] SC-32 Ans: _____

8 Application • 10 pts

Factorial Ratio: Evaluate 7! ÷ 5! on the SC-32 stack:

[ ] +10 pts [ ] SC-32 Ans: _____

9 Application • 10 pts

Bitwise OR Register Merge: Combine upper nybble 0xA0 and lower nybble 0x05 using [OR] (report in decimal):

[ ] +10 pts [ ] SC-32 Ans: _____

10 Error Analysis • 10 pts

Bitwise XOR Toggle: Toggle bits of 0xFF with mask 0x0F using [XOR] (report in decimal):

[ ] +10 pts [ ] SC-32 Ans: _____

Cumulative Exam 4: Final Capstone Mastery

Name: _____ Date: _____ Score: _____ / 100

100-POINT CUMULATIVE EXAM 4 — FINAL CAPSTONE & STEM ENGINEERING EXAM (10 PTS EACH)

Final Grades 9–12 & AP mastery assessment integrating Physics Kinematics, Harmonic & RMS Means, Golden Ratio Radicals, and Stack Architecture. Solve on paper first, verify on the SC-32 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

1 Core Fluency • 10 pts

Root-Mean-Square (RMS) Speed: Compute the RMS speed of 6 m/s and 8 m/s on the 4-level stack:

[ ] +10 pts [ ] SC-32 **Ans:** _____

2 Core Fluency • 10 pts

Kinetic Energy Formula: Find the kinetic energy KE (in Joules) for mass $m = 4$ kg and velocity $v = 15$ m/s:

[ ] +10 pts [ ] SC-32 **Ans:** _____

3 Core Fluency • 10 pts

Projectile Range at 45°: Compute $R = (v^2 \cdot \sin(90°)) \div g$ for $v = 28$ m/s and $g = 9.8$ m/s^2:

[ ] +10 pts [ ] SC-32 **Ans:** _____

4 Core Fluency • 10 pts

Continuous Compounding Return: Evaluate $A = 500 \cdot e^{0.05 \times 20}$ (to 2 decimal places) on the SC-32:

[ ] +10 pts [ ] SC-32 **Ans:** _____

5 Core Fluency • 10 pts

Harmonic Mean of Two Rates: Find the average round-trip speed for 30 mph outbound and 60 mph return:

[ ] +10 pts [ ] SC-32 **Ans:** _____

6 Core Fluency • 10 pts

Golden Ratio Quadratic Root: Compute $(1 + \sqrt{5}) \div 2$ to 4 decimal places on the SC-32:

[ ] +10 pts [ ] SC-32 **Ans:** _____

7 Application • 10 pts

Polar-to-Rectangular X Component: For polar magnitude $r = 10$ and angle $\vartheta = 60°$, find the horizontal component x:

[ ] +10 pts [ ] SC-32 **Ans:** _____

8 Application • 10 pts

Acoustics Inverse-Square Ratio: If initial sound intensity is 120 units and drops by a factor of 2 per doubling, find the intensity after 3 doublings:

[ ] +10 pts [ ] SC-32 **Ans:** _____

9 Application • 10 pts

Byte Address Offset: Compute base address 256 + (4 × 16) for a 32-bit register array:

[ ] +10 pts [ ] SC-32 **Ans:** _____

10 Error Analysis • 10 pts

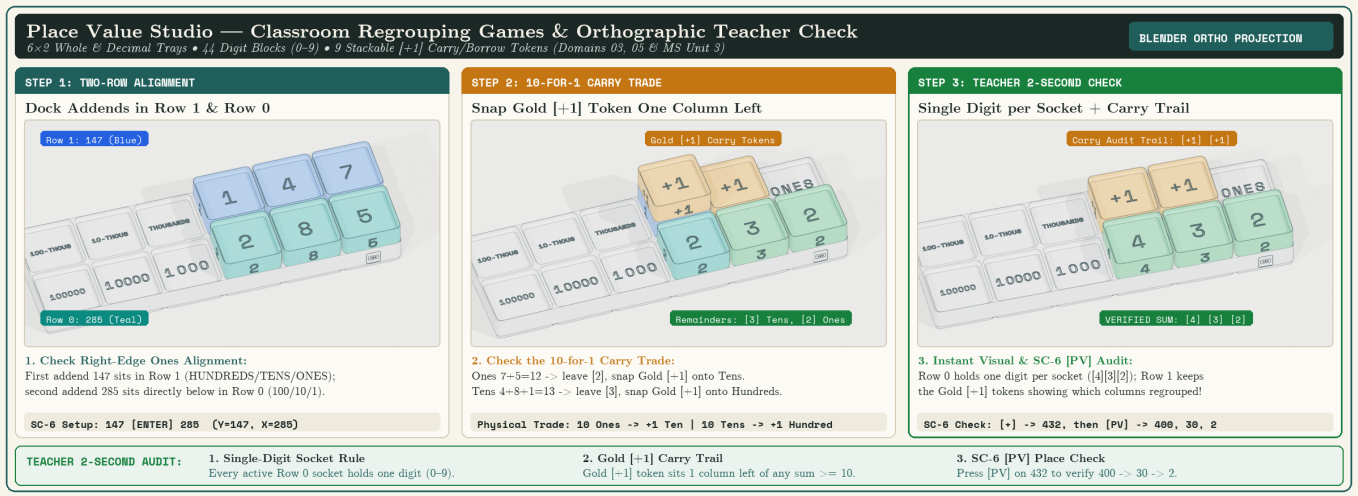
Capstone Error Analysis: To evaluate (20 + 16) ÷ (11 − 2), a student keyed 20 [ENTER] 16 [+] 11 [-] 2 [÷] and got 12.5. Find the true value:

[ ] +10 pts [ ] SC-32 **Ans:** _____

APPENDIX D (PART 1 OF 4) • OFFICIAL 3D BLOCK KIT 1 — PLACE VALUE STUDIO

Place Value Studio: 3D Blocks & Desk Games

Hands-on activities and 2-second visual checks for the 6×2 Whole/Decimal Trays, Digit Blocks (0–9), and Gold [+1] Carry Tokens.



HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

Activity 1: 'Race to 1,000 Carry Relay' (2-Row Tray & Gold [+1] Carry Tokens)

Start at 000 on the 6×2 Whole Number Tray. Place a 3-digit addend in Row 0 (Line X) and add it to the waiting quantity in Row 1 (Line Y). Whenever a column sum reaches 10 or more, leave the ones digit in Row 0 and snap a Gold [+1] Carry Token one column to the left, then verify on the stack!

Stack Check (147 + 285 = 432): `1` `4` `7` `ENTER` `2` `8` `5` `+`

Activity 2: 'The Decimal Point Shifter (×10 & ÷10 Slide)' (Whole & Decimal Trays)

Dock the Whole Number Tray (100,000 to 1) and Decimal Tray (0.1 to 0.000001) side-by-side at the raised decimal point (•). Build 4.75 and multiply by 10 by physically sliding every digit block one column left across the stationary decimal point—seeing why digits shift columns!

Stack Check (4.75 × 10 = 47.5): `4` `•` `7` `5` `ENTER` `1` `0` `×`

Activity 3: 'The Borrowing Bank Vault (Zero-Placeholder Unbundling)'

Set 502 in Row 1 (Line Y) and 168 in Row 0 (Line X). Because Ones (2 − 8) and Tens (0 − 6) cannot subtract directly, unbundle 1 Hundred ([5] → [4]), trade across the zero Tens ([0] → [9]), and snap the Gold [+1] token onto Ones (12 − 8 = 4) to reveal 334.

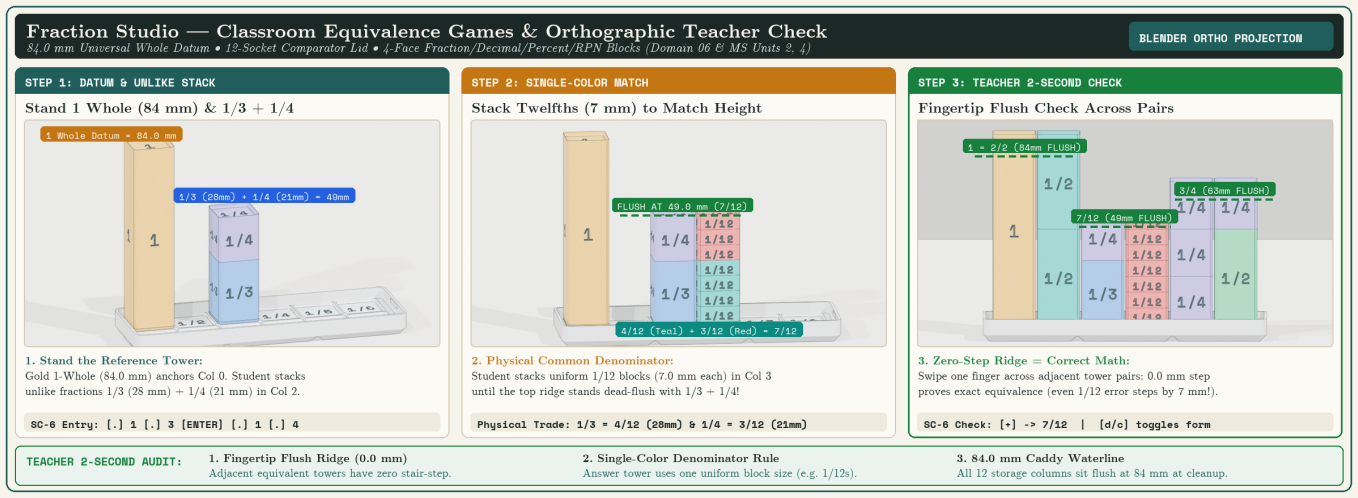
Stack Check (502 − 168 = 334): `5` `0` `2` `ENTER` `1` `6` `8` `-`

WHY PHYSICAL 3D PLACE VALUE BLOCKS BEAT FLAT PAPER (2-SECOND VISUAL CHECK)

- Check 1 — Single-Digit Socket Rule: Each machined socket on the Place Value Tray holds one digit block (0–9). Whenever a column reaches 10, the physical tray forces a 10-for-1 trade into the next column left.
- Check 2 — Tactile Gold [+1] Carry Trail: Every Gold [+1] token sits visibly one column to the left of the column that crossed 10—eliminating forgotten carries.
- Check 3 — Direct 1-to-1 Stack Mapping: Row 1 of the physical tray maps directly to Register Y (Waiting Shelf) and Row 0 maps directly to Register X (Workbench).

Fraction Studio: 84 mm Towers & Flush Checks

Proportional 84.0 mm fraction towers (1 Whole down to 1/12) and the 12-Socket Comparator Lid.



HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

Activity 1: 'The 84.0 mm Flush Tower Challenge (Make 1 Whole)'

Stand the Gold 1-Whole block (84.0 mm) in Column 0 of the 12-Socket Comparator Lid. Build an equal 84.0 mm tower in Column 1 using four blocks from different denominator families: 1/2 (42 mm) + 1/4 (21 mm) + 1/6 (14 mm) + 1/12 (7 mm) = 84.0 mm. Swipe a finger across the top ridge to feel 0.0 mm step!

Stack Check (1/2 + 1/4 + 1/6 + 1/12 = 1): `1` `ENTER` `2` `÷` `1` `ENTER` `4` `÷` `+` `1` `ENTER` `6` `÷` `+` `1` `ENTER` `1` `2` `÷` `+`

Activity 2: 'Two-Dot Common Denominator Duel (1/3 + 1/4 = 7/12)'

Stack 1/3 (28.0 mm) + 1/4 (21.0 mm) = 49.0 mm in Column 2. Next, find a single-color block family (twelfths, 7.0 mm each) and stack seven 1/12 blocks (49.0 mm) in Column 3 so both towers stand 100% flush!

Stack Check (1/3 + 1/4 = 7/12): `1` `ENTER` `3` `÷` `1` `ENTER` `4` `÷` `+`

Activity 3: 'Split & Rebuild Fraction Division (1/2 ÷ 3/4 = 2/3)'

Stand the 1/2 block (42.0 mm) in Column 1. Since it represents 3 fourths of the mystery tower, split 42.0 mm into 3 equal 1/6 blocks (÷ 3 → 14.0 mm is 1 fourth), then stack 4 of those 1/6 blocks in Column 3 (× 4 → 56.0 mm = two 1/3 blocks = 2/3)!

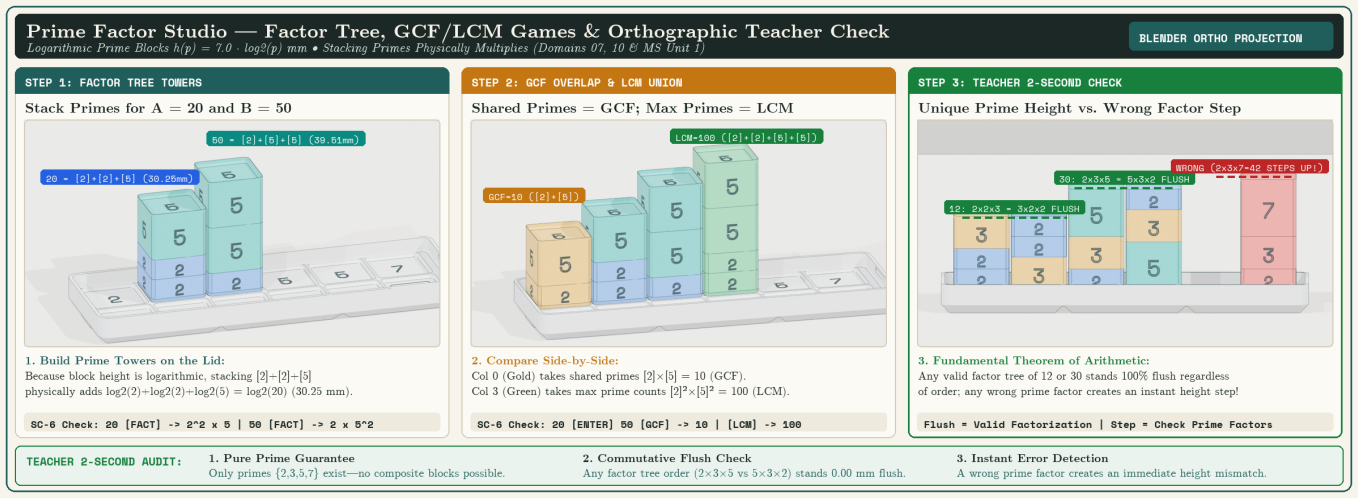
Stack Check (1/2 ÷ 3/4 = 2/3): `1` `ENTER` `2` `÷` `3` `ENTER` `4` `÷` `÷`

WHY 84.0 MM 3D FRACTION TOWERS BEAT FLAT DRAWINGS (2-SECOND FLUSH AUDIT)

- Check 1 — The 0.0 mm Fingertip Flush Test: Because $84 = 2^2 \times 3 \times 7$, halves, thirds, fourths, sixths, sevenths, and twelfths all have exact integer millimeter heights! Even a 1/12 error creates a bold 7.0 mm physical step.
- Check 2 — Single-Color Answer Tower Rule: Adding unlike fractions requires trading into one uniform block color (common denominator) that stands dead-flush with the mixed tower.
- Check 3 — Four-Face Rotational Engraving: Rotate any Fraction Studio block 90° in your hand to read its Fraction (1/4), Decimal (0.25), Percent (25%), and Two-Dot Key (.1.4).

Prime Factor Studio: Logarithmic Prime Blocks

Logarithmic prime factor blocks ($h(p) = 7.0 \cdot \log_2(p)$ mm for primes 2, 3, 5, 7) where multiplication becomes physical height addition.



HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

Activity 1: 'The Commutative Skyscraper Race ($2 \times 3 \times 5 = 30$)'

Factor 30 two different ways: Tower A uses $6 \times 5 \rightarrow [2] + [3] + [5]$ in Column 2, while Tower B uses $10 \times 3 \rightarrow [5] + [2] + [3]$ in Column 3. Because each block's height is proportional to $\log_2(p)$, both towers stand 100% flush at 34.35 mm!

Stack Check ($2 \times 3 \times 5 = 30$): 2 ENTER 3 × 5 ×

Activity 2: 'Fundamental Theorem of Arithmetic Challenge ($12 = 2 \times 2 \times 3$)'

Stack the prime tower for 12 ([2] + [2] + [3] = 25.10 mm) in Column 1, then try to match its exact height in Column 2 using different primes (e.g., [5] + [2] = 10 = 23.25 mm or [2] + [7] = 14 = 26.65 mm). No other prime combination ever stands flush!

Stack Check ($2 \times 2 \times 3 = 12$): 2 ENTER 2 × 3 ×

Activity 3: 'Side-by-Side GCF Overlap & LCM Union Duel (20 & 50)'

Build Tower $A = 20$ ([2]+[2]+[5]) and Tower $B = 50$ ([2]+[5]+[5]). Copy the shared prime blocks ([2]+[5] = 10) to build GCF(20, 50) = 10 on the left, and stack the tallest count of each prime ([2]+[2]+[5]+[5] = 100) to build LCM(20, 50) = 100 on the right!

Stack Check (GCF = 10, LCM = 100 → $10 \times 100 = 1000$): 2 ENTER 5 × 2 ENTER 2 × 5 × 5 × ×

HOW LOGARITHMIC 3D PRIME BLOCKS MAKE FACTORING & LOGARITHMS VISIBLE

- Check 1 — Pure Prime Guarantee: Because the Prime Factor Studio contains only prime blocks ($\{2, 3, 5, 7\}$), you can never stop a factor tree early at a composite number like 4, 6, 8, or 9.
- Check 2 — Physical Logarithm Law: Stacking blocks adds their heights $h(a) + h(b) = 7.0 \cdot \log_2(a) + 7.0 \cdot \log_2(b) = 7.0 \cdot \log_2(a \cdot b)$, turning abstract logarithms into a 3D tower!
- Check 3 — GCF + LCM Height Symmetry: On the comparator lid, $h(\text{GCF}) + h(\text{LCM}) = h(A) + h(B)$, proving visually why GCF(A, B) × LCM(A, B) = $A \times B$.

Official StackCalc Block Studios, Calculators & Library Guide

Order official 3D Manipulative Block Studios, distraction-free calculators, and companion books at https://stackcalc.io.

1. THE OFFICIAL STACKCALC HANDS-ON HARDWARE & 3D BLOCK STUDIO ECOSYSTEM (STACKCALC.IO)

STUDIO 1	**Place Value Studio**	Binary Kit • K–12

6×2 Whole & Decimal Trays, 40 Digit Blocks (0–9), Gold [+1] Carry Tokens & Base-10 Set

STUDIO 2	**Fraction Studio**	Grades 2–8

78 Proportional Fraction Towers (84.0 mm Whole down to 1/12) & 12-Socket Comparator Lid

STUDIO 3	**Prime Factor Studio**	Core Kit • 3–12

Logarithmic Prime Blocks ($h(p) = 7.0 \cdot \log2(p)$ mm for 2, 3, 5, 7) & Comparator Lid

SC-6	**Elementary RPN Calculator**	Grades K–8

4-Register Visible Stack (T, Z, Y, X), Two-Dot Fractions (.a.b), [PV], [FACT], [GCF] + Web Sim

SC-32	**Scientific STEM Calculator**	Core Calc • 8–12

32-Key Scientific RPN: [POLY], [→θ,r], [SIN/COS/TAN], [LN/LOG], [BASE] + iOS/watchOS App

2. COMPANION TRADE BOOKS & WORKBOOKS IN THE STACKCALC LIBRARY (STACKCALC.IO)

Volume I & II • Illustrated Trade Books:	Vol. I: From Blocks to the Stack (Grades K–6) • Vol. II: Thinking in the Stack (Grades 7–12 & AP)
Book 1 • Teaching with the Stack:	Master K–12 Teacher & Homeschool Guide, Word-for-Word Scripts & Misconception Matrix (46 pp.)
Book 2A & 2B • Elementary Math Labs:	Book 2A: Early Elementary (K–2) • Book 2B: Upper Elementary (3–5) (10 Units + 4 Exams • 840 Probs)
Book 3 • Middle School Math Lab:	Grades 6–8 Two-Method Block & SC-6 Workbook (10 Units, 45 Sections & 10 Quizzes • 1,000 Problems)
Book 4 & 5 • High School & Speed Labs:	Book 4: High School & STEM Lab (SC-32) • Book 5: Stack Speed & Fluency (180 Timed Tests)

3. ONLINE ANSWER KEY, FREE SIMULATOR & HARDWARE ORDERING PORTAL

- Online Worked Solutions & Keystrokes (840 Verified Problems): https://stackcalc.io/curriculum/answers#high-school
- Order Physical 3D Block Studios & Distraction-Free SC-32 Calculators: https://stackcalc.io
- Free Web Simulator & iOS/watchOS App: Practice on the interactive SC-6 / SC-32 simulator or download StackCalc 32 on the App Store!